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Review

A review of advanced SOFCs and SOECs: Materials, innovative synthesis, functional mechanisms, and system integration

Peng Feng[a], Kuan Yang[a], Xuanyou Liu[a], Jiujun Zhang[b,*], Zhi-Peng Li[a,*]

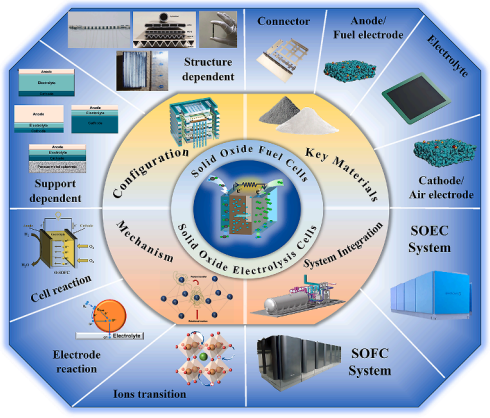
[a] Frontier Science Centre for Flexible Electronics, Xi'an Institute of Flexible Electronics, Northwestern Polytechnical University, 127 Youyi West Road, Xi'an 710072, China

[b] College of Materials Science and Engineering, Fuzhou University, Fuzhou 350108, China

HIGHLIGHTS

- Pioneering SOFC/SOEC review spanning materials design to system integration, delivering the first comprehensive whole-chain perspective.
- Integrates cutting-edge design paradigms—high-entropy doping, machine learning—into a systematic materials-research framework.
- Synthesizes system-integration advances to establish reverse-guided research demands for materials and stacks.
- Provides actionable R&D strategies and forward-looking recommendations.

GRAPHICAL ABSTRACT



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ABSTRACT

Solid oxide fuel cells (SOFCs) and solid oxide electrolysis cells (SOECs) are next-generation energy conversion technologies that have attracted widespread attention due to their high efficiency, fuel flexibility, and environmental friendliness. The reversible reaction processes of the two can achieve power generation and energy storage in one device. This paper provides an extensive overview of the latest developments in the field of SOFCs and SOECs, including types, material synthesis, mechanism research, and system integration. First, we introduce the classification of current SOFCs/SOECs according to their different supports and conducting ions. Then, we summarize the synthesis methods and optimization strategies for key materials, including the latest developments in electrolytes, electrodes, and interconnects. Subsequently, the electrochemical mechanisms, including ion transport, electron conduction, electrochemical reaction kinetics, and interfacial phenomena, are analyzed in depth. This paper also outlines challenges and strategies for system integration, such as thermal management, fluid dynamics, and mechanical stress control. Through comprehensive analysis, this review aims

* Corresponding authors.
E-mail addresses: jiujun.zhang@fzu.edu.cn (J. Zhang), iamzpli@nwpu.edu.cn (Z.-P. Li).

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to provide researchers with a holistic perspective and guidance for the future development of SOFCs and SOECs. We close by discussing the main challenges and future research directions for further promoting the commercialization and large-scale development of these technologies.

1. Introduction

With the continuous growth of the global energy demand and our increased awareness of the need to halt environmental degradation, the development of efficient, clean, and sustainable energy conversion and storage technologies has become a hot topic in the energy research field. Solid oxide fuel cells (SOFCs) and solid oxide electrolysis cells (SOECs) are outstanding representatives of the next generation of energy conversion technologies and have attracted widespread attention due to their high energy conversion efficiency, fuel flexibility, and environmental friendliness. The reversible reaction processes of the two have achieved the dual functions of power generation and energy storage, offering innovative solutions for maximizing energy utilization.

The SOFC is an efficient device that directly converts chemical energy into electrical and thermal energy. SOFCs have been widely developed for several decades. In general, a SOFC has a high energy conversion efficiency ($\geq$ 55%) and can utilize a variety of fuels, including hydrogen, methane, methanol, etc., demonstrating great fuel flexibility [1]. Furthermore, depending on the fuels used, a SOFC is environmentally friendly and does not produce significant harmful gas emissions during the working process. These advantages give it a wide-ranging application potential in the fields of distributed power generation, combined heat and power (CHP), and transportation. In contrast, the SOEC is a device that converts electrical or thermal energy into chemical energy for storage. With the increased popularity of renewable energy, SOECs have emerged as a promising technology to produce clean fuels such as hydrogen and chemical feedstocks like carbon monoxide. This is achieved through the electrolysis of water or carbon dioxide, offering new solutions for renewable energy storage and economic development. Additionally, SOECs can perform CO_2/H_2O co-electrolysis, directly converting carbon dioxide and water into valuable chemical feedstocks such as syngas, methanol, and methane within the same cell [2]. A search on *Web of Science* for studies on "SOFC" and "SOEC" from 2006 to 2024, limited to "article" publications, revealed that research on SOFCs remains popular, and research on SOECs has experienced rapid growth, as shown in Fig. 1; since relatively few studies have looked at both SOFCs and SOECs, they have not been counted here (see Fig. 2).

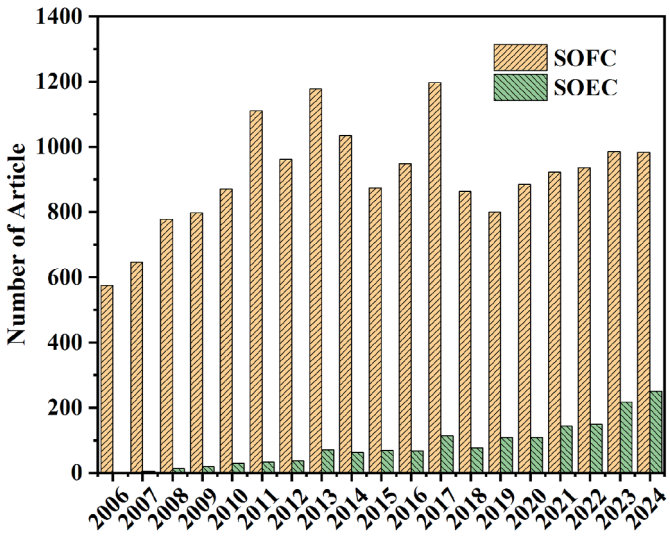

Fig. 1. Annual publication volume of SOFC and SOEC research, 2006–2024.

To further advance SOFC and SOEC technology, researchers in recent years have carried out numerous studies in the fields of material synthesis, electrochemical mechanisms, and system integration, making significant progress. In the field of material synthesis, new types of electrolytes, electrodes, and interconnect materials are continuously emerging, contributing to better battery performance and lower costs. In the field of electrochemical mechanisms, researchers have explored in detail the microscopic mechanisms of ion transport, electron conduction, and electrochemical reaction kinetics, providing a theoretical basis for optimizing battery structure and improving reaction efficiency. In the field of system integration, researchers are committed to solving key issues such as thermal management, fluid dynamics, and mechanical stress control, striving to achieve efficient and reliable system integration. However, the development of SOFC and SOEC technology has not been smooth sailing. High-temperature working environments necessitate very stringent requirements for the thermal stability and chemical stability of materials, and urgent issues include material aging, corrosion, and performance degradation. In addition, the complex electrochemical reaction mechanisms and system integration strategies are also challenges. Current researchers face several obstacles, including how to optimize the performance of electrode materials, improve the ionic conductivity of the electrolyte, reduce the system operating costs, and achieve stable long-term operation.

To date, numerous reviews on SOFCs and SOECs have been published, but most of these have focused on analyzing and summarizing either SOFCs or SOECs. In addition, the majority have concentrated on the synthesis methods, properties, and mechanisms of key materials, with only a few discussing the configuration and performance of single cells. Table 1 presents the typical reviews, showing that a comprehensive review covering both SOFCs and SOECs, from materials to systems, is currently lacking. Furthermore, with recent novel research methods such as high-entropy doping and machine learning, there have been significant advancements in the development of high-performance key materials for SOFCs/SOECs. These advancements need to be summarized so researchers have an updated overview of current progress. This paper therefore aims to present a comprehensive and detailed overview of SOFCs and SOECs, including the latest research progress.

We begin by introducing the classification of SOFCs/SOECs and then delve into their electrochemical mechanisms, focusing on ion transport, electronic conduction, electrochemical reaction kinetics, and interfacial phenomena. Next, we summarize the core challenges and recent advances in key materials such as SOFC/SOEC anodes, cathodes, electrolytes, and interconnects. This is followed by an exploration of the challenges and strategies for system integration, including thermal management, fluid dynamics, and mechanical stress control, as well as ways to achieve efficient and reliable system integration. This comprehensive analysis aims to provide a thorough research perspective and guide for future technological advancements in the field of SOFCs and SOECs. Readers can thereby gain insights into the current state of research and identify potential areas for further exploration and innovation.

2. Principles and mechanisms

2.1. Operating principles of SOFCs and SOECs

SOFCs are high-efficiency devices that convert the chemical energy in fuels directly into electrical and thermal energy through electrochemical reactions. They consist of porous cathodes and anodes separated by a dense solid electrolyte. The anode, where the fuel gas is

introduced, and the cathode, where air is introduced, can also serve as the mixed conductors for ions and electrons. The electrolyte situated between them acts as a medium for ion transport. In a SOFC, the electrolyte is typically a dense solid oxide that facilitates ionic conduction and separates the air side from the fuel side during operation [14]. The anode is the site of the fuel oxidation reaction, and hydrogen-rich compounds such as methane (CH_4) and methanol (CH_3OH) are commonly used as fuels due to their abundance, low cost, and ease of storage and transportation, making them the most industrially viable options for SOFCs; the cathode is where oxygen reduction occurs. Both electrodes contain catalysts that accelerate the electrochemical reactions taking place at the electrodes [15]. Depending on the type of ion transfer, SOFCs can be categorized into oxygen ion-conducting solid oxide fuel cells (O–SOFCs) and proton-conducting solid oxide fuel cells (H–SOFCs), each of which has different working principles.

Using H_2 as a fuel, the working process of an O–SOFC is as follows: H_2 is continuously introduced to the anode side. The anode is a porously structured material with catalytic properties that facilitate the adsorption of fuel gas molecules and their diffusion toward the anode–electrolyte interface. The catalytic surface of the anode promotes the

Table 1
A summary of recent reviews on SOFCs or SOECs.

Review	Related fields	Subject	Ref.
1	SOFC	Synthesis process of key materials	[3]
2	SOFC	Electrolyte	[4]
3	SOFC	Key materials and mechanisms	[5]
4	SOFC	Single-cell configuration	[6]
5	SOFC	Key materials	[7]
6	SOFC	Performance and preparation methods of key materials and single cells	[8]
7	SOEC	CO_2/H_2O electrolysis performance	[9]
8	SOEC	Performance of key materials for oxygen ion conduction SOECs	[10]
9	SOEC	Electrolyte and electrophoresis materials and methods for preparing the same	[11]
10	SOEC	Proton-conducting oxide electrolyte materials, modeling, and current leakage	[12]
11	SOEC	Comparison of SOECs with other electrolysis technologies	[13]
This review	**SOFC/ SOEC**	**Full coverage, from key materials to system integration**	–

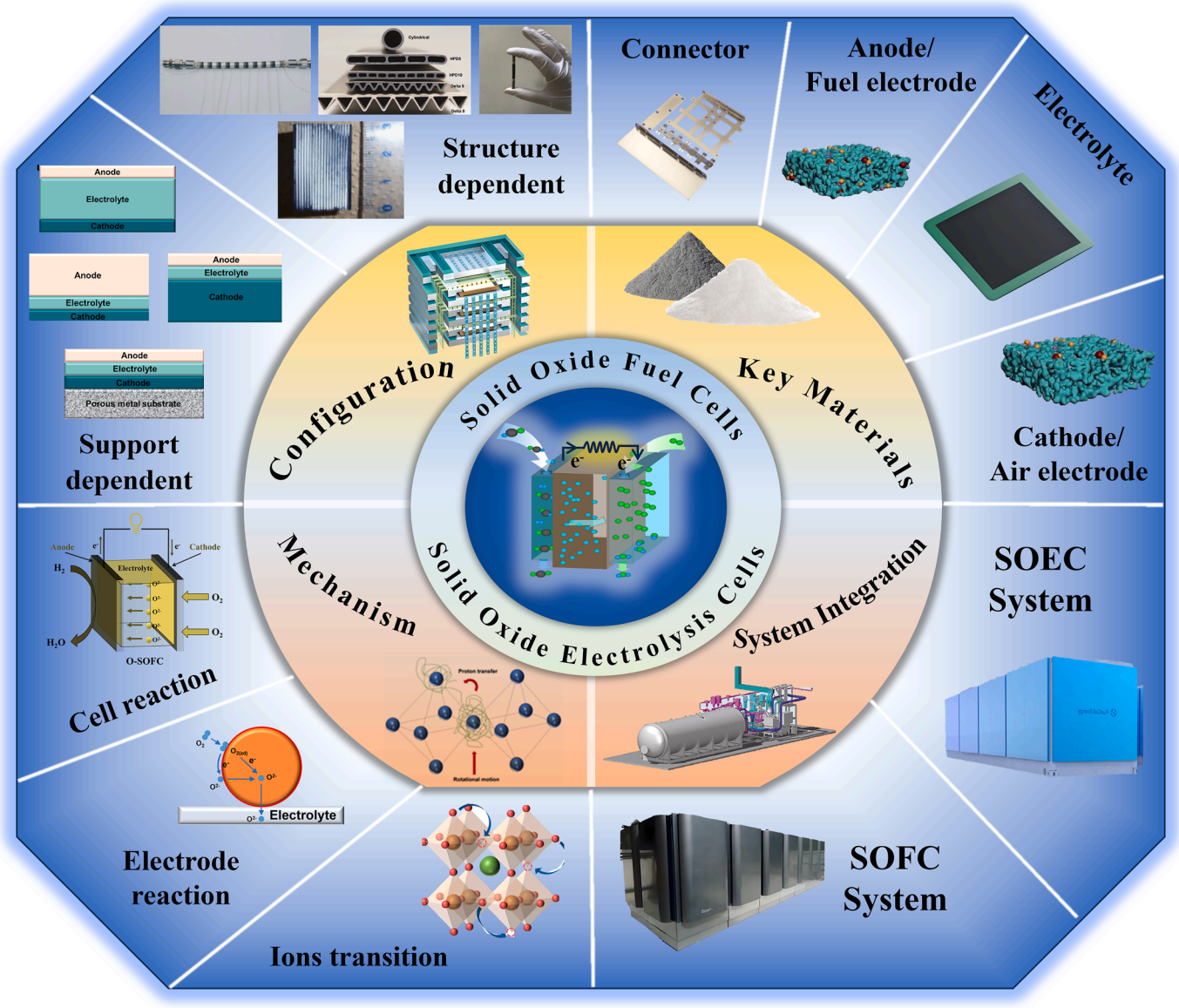

Fig. 2. SOFC/SOEC synthesis framework design. Reproduced with permission [299]. Copyright 2015, Mitsubishi Heavy Industries Technical Review.

oxidation of the fuel; hydrogen molecules are oxidized, releasing electrons in the process. Concurrently, on the cathode side, the oxidant (typically air) is continuously fed in. Oxygen molecules are adsorbed on the surface of the porous structure of the cathode, capture electrons, and are converted into oxygen ions (O^{2-}). These oxygen ions, under the influence of chemical potential, enter the solid-state oxygen ion conductor, which is the electrolyte. Driven by concentration gradients and kinetic diffusion, the oxygen ions ultimately reach the interface between the solid electrolyte and the anode, where they react with the fuel gas to form H_2O. During this process, the fuel gas loses electrons, and the electrons flow back to the cathode through the external circuit, completing the circuit and generating an electric current (Fig. 3a). The reactions that occur throughout this process are as follows:

cathode reaction: $O_2 + 4e^- \rightarrow 2O^{2-}$ (1)

anode reaction: $H_2 + O^{2-} \rightarrow H_2O + 2e^-$ (2)

total reaction: $2H_2 + O_2 \rightarrow 2H_2O$ (3)

During operation of a H–SOFC, H_2 is introduced into the anode compartment of the cell. Under the catalytic effect of the anode, hydrogen molecules are oxidized into hydrogen ions, also known as protons (H^+), which then diffuse through the proton-conducting electrolyte to the interface with the cathode. Concurrently, O_2 is introduced into the cathode compartment, where adsorbed oxygen molecules are reduced to O^{2-} and react with the protons that have diffused to the cathode–electrolyte interface to form H_2O. Throughout this process, the fuel gas loses electrons, which are then returned to the cathode through an external circuit, completing the circuit and generating an electric current (Fig. 3b). The reactions that take place during this process are:

cathode reaction: $4H^+ + O_2 + 4e^- \rightarrow 2H_2O$ (4)

anode reaction: $2H_2 \rightarrow 4H^+ + 4e^-$ (5)

total reaction: $2H_2 + O_2 \rightarrow 2H_2O$ (6)

It is evident that the O–SOFC and the H–SOFC introduce gases to the anode and cathode compartments identically; however, the interfaces at which the electrochemical reactions transpire, the direction of ion conduction, and the location of product formation are reversed. In the O–SOFC, the reactions occur at the interface of the anode and the electrolyte, with the resultant products being generated on the anode side. In contrast, within the H–SOFC, the reactions take place at the cathode–electrolyte interface, with the products formed on the cathode side. This distinction results in divergent reaction mechanisms at the electrode sites for each cell type, which will be thoroughly examined in what follows.

The SOEC operates on the converse principle of the SOFC, converting electrical or thermal energy into chemical energy. The SOEC is distinguished by its capacity to efficiently generate H_2 or CO, and its potential to produce synthetic gas through the process of co-electrolysis; this gas can subsequently be transformed into a diverse array of chemical

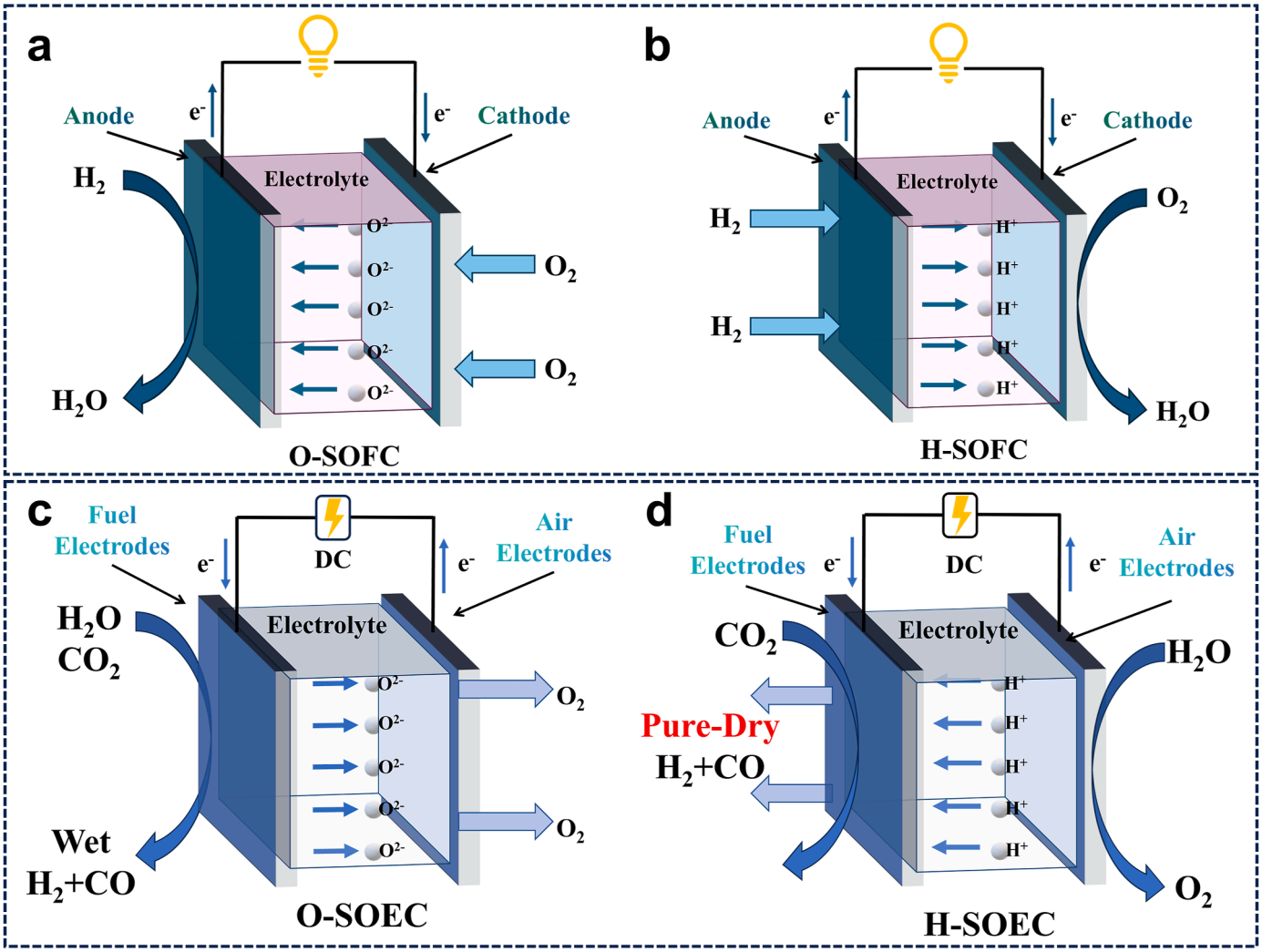

Fig. 3. Operating principle of (a) O–SOFC, (b) H–SOFC, (c) O-SOEC, and (d) H-SOEC.

products, yielding extensive application prospects. Analogous to the SOFC, the SOEC is composed of catalytic porous cathodes and anodes, coupled with a dense electrolyte that facilitates ion transport. It is further classified into two types, based on the ion conduction mechanism: the oxygen ion-conducting (O-SOEC) and the proton-conducting (H-SOEC) variants [16].

During the electrolysis of H_2O in an O-SOEC, $H_2O(g)$ is introduced to the fuel electrode side. Subject to an external direct current (DC) voltage, the water vapor accepts a pair of electrons via the external circuit and undergoes a reduction reaction, resulting in the formation of H_2 and O^{2-}. The O^{2-} are then transported through the electrolyte to the air electrode side, where they release electrons and are further reduced to generate O_2, which subsequently desorbs from the anode surface. The electrons, drawn to the positive charge at the fuel electrode, traverse the external circuit to reach the fuel electrode side, where they combine with H^+ to form H_2 [17]. The principle of operation is shown in Fig. 3c.

reaction at the fuel electrode: $H_2O + 2e^- \rightarrow H_2 + O^{2-}$ (7)

reaction at the air electrode: $2O^{2-} \rightarrow O_2 + 4e^-$ (8)

total reaction: $2H_2O \rightarrow 2H_2 + O_2$ (9)

During the operation of a H-SOEC, high-temperature water vapor is introduced at the air electrode side. Subject to the application of DC electricity, it undergoes an electrochemical reaction that results in decomposition into O_2 and H^+. These protons then migrate across the dense electrolyte layer and towards the fuel electrode side, where they are rapidly reduced to form H_2. The proton-conducting electrolyte, being impermeable to oxide ions and molecular gases, permits the passage of hydrogen ions exclusively, ensuring that the product generated at the fuel electrode side is pure, dry H_2 (Fig. 3d). The reactions at the two electrodes, and the overall cell reaction, are as follows:

reaction at the fuel electrode: $2H^+ + 2e^- \rightarrow H_2$ (10)

reaction at the air electrode: $2H_2O \rightarrow 4H^+ + O_2 + 4e^-$ (11)

total reaction: $2H_2O \rightarrow O_2 + 2H_2$ (12)

In addition to hydrogen production through water electrolysis, SOECs can also decompose CO_2 into CO and O_2, and co-electrolyze with H_2O to produce synthetic gas. This synthetic gas can be further converted into a variety of chemicals through catalytic reactions that vary in the ratio of H_2 to CO. Moreover, the technology for synthesizing NH_3 using SOECs is under development; this involves introducing air and steam into the electrolysis cell. In the SOFC mode, the synthesized ammonia can be used as a fuel for marine applications, but the current yield of SOEC ammonia synthesis remains low. This versatility makes SOECs superior to other electrolysis modes [18]. Similarly, in O-SOECs

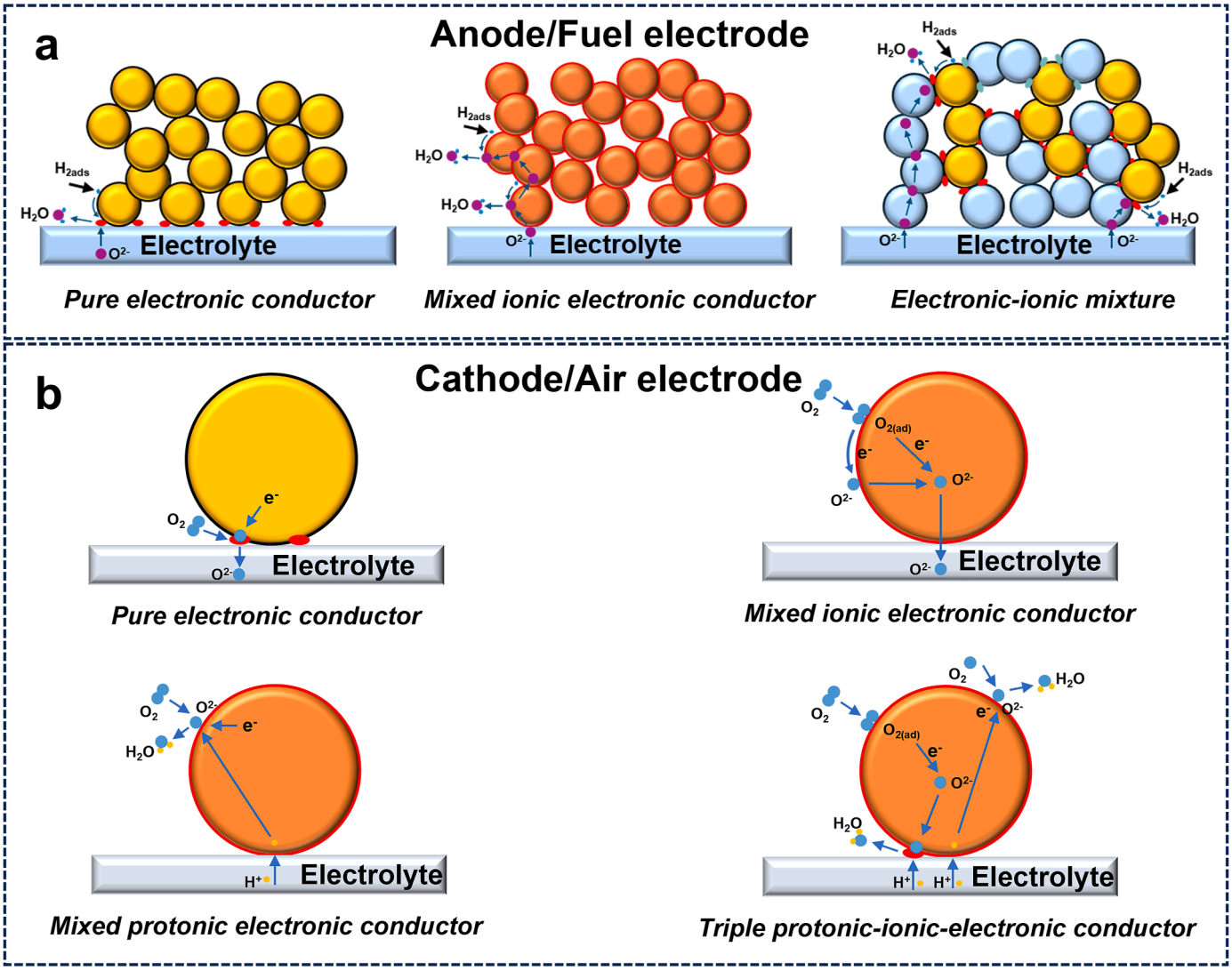

Fig. 4. Reaction mechanism of (a) anode/fuel electrode; (b) cathode/air electrode.

and H-SOECs, a situation can arise where the reaction interface side and the product generation side are opposite: the former is supplied with H_2O(g) at the fuel electrode, generating H_2 on the same side; in contrast, the latter is supplied with H_2O(g) at the air electrode, with H_2 being produced at the fuel electrode side. The H_2 produced by O-SOECs is mixed with H_2O(g), necessitating additional drying steps, whereas the H_2 produced by H-SOECs is completely pure and dry, making H-SOECs more practical than O-SOECs in terms of application.

2.2. Mechanisms of SOFCs/SOECs

2.2.1. Electrode reaction mechanisms

The electrode reaction mechanisms of SOFCs and SOECs involve multiple steps, including the adsorption/dissociation of gas molecules, electron transfer, ion migration, and chemical reactions. The choices of electrode materials, interfacial characteristics, and operating conditions have a significant impact on the efficiency of the electrode reactions [19]. Electrode materials for SOFCs and SOECs can be categorized based on their conductive properties. Anode/fuel electrode materials typically include pure electronic conductors (PECs), mixed ionic–electronic conductors (MIECs), and composites of electrons and ions (Fig. 4a). On the other hand, cathode and air electrode materials are comprised of PEC, mixed proton–electron conductors (MPECs), MIECs, and triple conductors that exhibit mixed ionic–electronic conduction (THOECs), as depicted in Fig. 4b. These classifications reflect the diverse roles and functionalities of electrode materials in the context of energy conversion and storage technologies [20]. As previously mentioned, whether it involves O^{2-} conduction or proton conduction, the anode of SOFCs or the fuel electrode of SOECs undergoes hydrogen/carbon oxidation–reduction reactions, while the cathode/air electrode undergoes oxygen evolution reaction (OER) or reduction (ORR) processes, accompanied by the generation or decomposition of H_2O and CO_2. All these processes are accomplished through the diffusion of ions within the bulk of the electrode material, adsorption and dissociation at the surface, and charge transfer. The most critical factor affecting the electrode reaction is the triple phase boundary (TPB), where the gas phase, ionic conductor, and electronic conductor come into contact. The length of the TPB directly influences the rate of the electrode reaction [21]. The effective electrochemical reaction zones (ERZs) differ among electrode materials with distinct charge carriers. For instance, in the case of an anode in an O–SOFC fabricated from a PEC, the ERZ is predominantly localized at the bi-phase boundary between the electrolyte and the anode. Conversely, when employing a MIEC for the anode, the reducing environment facilitates a pronounced enlargement of the ERZ, extending it into a three-dimensional volume reaction zone. Nonetheless, within this expanded zone, electrochemical reactions are confined to the interfacial regions where the oxygen ion conductor, the electronic conductor phase, and the gas phase intersect (as depicted in Fig. 4b). It is imperative to ensure that the triple phase boundaries (TPBs) are three-dimensionally integrated with electrons, ions, and gases to yield a substantial decrease in activation polarization and to enhance electrical efficiency. The electrochemical activity of TPBs, denoted as electrochemically active triple phase boundaries (EATPBs), is contingent upon this three-dimensional coupling [22]. The electrochemical activity at a TPB is contingent upon the successful execution of several sequential steps: (a) the migration of ions from the electrolyte to the reaction sites, (b) the delivery of gas molecules to these sites, and (c) the extraction of electrons from the reaction sites. The integrity of the TPB's electrochemical function is predicated on an uninterrupted progression through these steps. Should any of these steps be compromised, the electrochemical reaction is impeded, thereby preventing the reaction from occurring [23].

2.2.2. Ion transport mechanisms

The preceding discussion has established that ions of varying conductive properties adhere to unique transport mechanisms. In pursuit of a more profound understanding of the modalities of ionic transport, a plethora of theoretical models have been conceptualized to delineate the dynamics of ion migration. Presently, the three predominant models that encapsulate these mechanisms in electrolytes are the lattice diffusion model, the Grotthuss mechanism, and the vehicle mechanism.

i. Lattice diffusion model

The lattice diffusion model is one of the most commonly used mechanistic models for oxygen ion-conducting electrolyte materials in SOFCs/SOECs. It is predicated on the diffusion behavior of oxygen ions within a solid lattice. In this model, the electrolyte material typically possesses a cubic lattice structure, such as yttria-stabilized zirconia (YSZ). Within this structure, Zr^{4+} ions are replaced by Y^{3+} ions, creating oxygen ion vacancies. The doped Y^{3+} ions, with their lower charge, induce the formation of oxygen ion vacancies to maintain electrical neutrality. O^{2-} migrates through the lattice via these vacancies [24]. At elevated temperatures, oxygen ions acquire energy, hopping from one vacancy to an adjacent one, a process known as hopping diffusion, as depicted in Fig. 5a. The migration of oxygen ions involves an exchange of energy; during the hopping process, they interact with the cations in the lattice, either absorbing or releasing electrons, which facilitates their movement within the lattice. This migration is generally influenced by factors such as lattice structure, temperature, and doping levels [25]. The electrical conductivity (σ) is related to the mobility of the oxygen ions (D), which can be described by the Nernst–Einstein equation:

$$\sigma = nqD$$

where n represents the concentration of oxygen ions, q is the charge of the oxygen ion, and D is the diffusion coefficient of the oxygen ions.

The mobility of ions is temperature dependent and typically follows the Arrhenius equation, which can be expressed as:

$$D = D_0 \exp\left(-\frac{Ea}{\kappa_B T}\right)$$

The pre-exponential term D_0 in the equation denotes a constant, while Ea signifies the activation energy required for the migration process. The Boltzmann constant is represented by k_B, and T refers to the thermodynamic temperature measured in absolute units (Kelvin). Lattice imperfections, including vacancies and dopant ions, exert a pronounced influence on the ionic conductivity. An optimal level of doping can augment the concentration of oxygen ion vacancies, thus elevating the ionic conductivity. Conversely, an excessive doping concentration may induce lattice distortion, adversely affecting the conductivity. Characterization techniques such as impedance spectroscopy and neutron diffraction, complemented by computational approaches grounded in density functional theory (DFT), are employed to ascertain and model the diffusion trajectories and energetics of oxygen ions within the lattice. These methodologies are instrumental in corroborating the congruence between the lattice diffusion model and empirical findings.

ii. Grotthuss mechanism

The Grotthuss mechanism is a proton conduction mechanism initially used to explain proton transfer in acidic solutions. Later, this concept was extended to solid materials, especially in proton-conducting electrolytes involving hydrated protons or hydroxyl groups. The Grotthuss mechanism is particularly important in proton-conducting electrolyte materials for SOFCs/SOECs, especially in those electrolytes capable of water molecule adsorption and dissociation. In the Grotthuss mechanism, protons are not transferred as independent particles within the electrolyte but are conveyed through a hopping process between adjacent oxygen atoms or water molecules (Fig. 5b). This hopping involves the transfer of a proton from a negatively charged oxygen atom of one molecule to another. In a hydrated environment, protons (H^+) typically exist in the form of hydronium ions (H_3O^+). Water molecules form a hydration sphere around the proton, enabling its migration through the electrolyte. The proton hopping is accompanied by a local

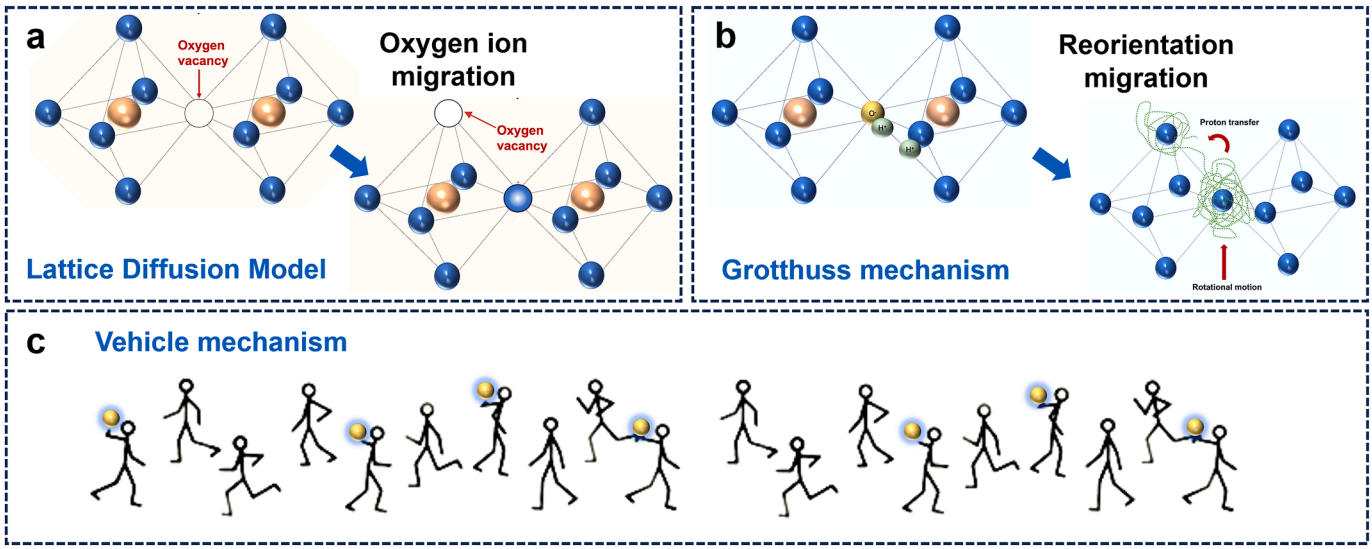

Fig. 5. Mechanistic models for ion transport. (a) lattice diffusion model; (b) Grotthuss mechanism. Reproduced with permission [26]. Copyright 2023, Springer; (c) vehicle model. Reproduced with permission [29]. Copyright 2017, American Chemical Society.

rearrangement of the surrounding oxygen atoms. When a water molecule loses a proton, it becomes an oxygen anion, while another water molecule transforms into a hydronium ion by gaining a proton. Under specific conditions, the Grotthuss mechanism entails the synergistic transport of electrons and protons, where proton hopping in certain oxide electrolytes may be concurrent with the transference of electrons, thereby achieving comprehensive charge conduction [26]. For instance, within cerium-doped ceria, the redox transformations of cerium ions (Ce^{4+}/Ce^{3+}) can facilitate proton hopping [27]. The activation energy associated with proton hopping is generally low, which allows the Grotthuss mechanism to operate efficiently at lower temperatures. Proton conductivity typically rises with an increase in temperature; however, this temperature dependence may be less significant compared to that of conventional oxygen ion conductivity. In certain solid-state materials with particular lattice configurations, such as perovskite oxides, the arrangement of oxygen atoms within the lattice may establish conduits for proton conduction, thereby enhancing the likelihood of the Grotthuss mechanism. Proton hopping behavior under the Grotthuss mechanism can be scrutinized using a range of experimental techniques, including Raman spectroscopy, infrared spectroscopy, nuclear magnetic resonance (NMR), and neutron scattering. First-principles calculations and molecular dynamics simulations are instrumental for investigating proton hopping dynamics and the interactions between protons and the surrounding atoms within the Grotthuss mechanism. Elucidating the Grotthuss mechanism is beneficial for designing novel proton-conducting materials. By modulating the chemical composition, lattice structure, and water molecule adsorption capabilities of these materials, their proton-conducting properties can be optimized. However, the explanatory power of the Grotthuss mechanism falters in some circumstances, such as the potential absence of proton hopping in anhydrous environments, or the compromised stability of water molecules at elevated temperatures.

iii. Vehicle mechanism

In contrast to the Grotthuss mechanism, the vehicle mechanism proposed by Kreuer et al. posits that protons do not migrate as isolated H^+ ions, but rather are combined with a "carrier" molecule in the form of OH^-, NH^{2-}, etc., such as H_2O, NH_3, and so on. This combination results in the formation of a composite ion, for example, H_3O^+ or NH^{2-}. Within this mechanism, proton migration is achieved through the movement of these composite ions. When a proton transfers from an oxygen ion to a carrier molecule, the carrier molecule, now bearing the proton, moves to another oxygen ion and subsequently releases the proton, completing the transfer process (Fig. 5c). During this process, the empty carrier not bearing a proton will move in the opposite direction to maintain charge balance. The diffusion coefficient of the carrier corresponds to the proton conductivity. This complies with the Nernst–Einstein relationship, which states that the diffusion coefficient is directly proportional to the ion's migration rate [28]. In a crystalline milieu, carriers demonstrate the attributes of a Brønsted base, so they are capable of accepting protons, a trait pivotal to their efficacy in proton conduction [29]. The vehicle mechanism model elucidates the substantial temperature factors detected in X-ray and neutron diffraction studies, indicating a correlation between proton conduction and the carriers' translational movements, independent of their rotational motions. Despite providing novel perspectives on the comprehension of rapid proton conductors, the vehicle mechanism model encounters challenges when it comes to practical application, notably its inability to explain the phenomenon of proton hopping in anhydrous settings.

The diffusion of charge carriers such as oxide ions or protons at grain boundaries/interfaces plays a crucial role in the overall performance of solid oxide fuel cells/solid oxide electrolysis cells. These diffusion processes not only impact the kinetics of electrode reactions but also determine the efficiency and stability of the cells. Grain boundaries are important microstructural features characterized by relatively irregular atomic arrangements and a higher concentration of oxygen vacancies and defects, which provide enhanced pathways and lower energy barriers for the migration of oxide ions. For example, research has shown that in YSZ with a mixed grain boundary structure at a temperature of 2300 K, the diffusion coefficient (D_o) of oxide ions at the core of the grain boundary is 4.85×10^{-10} m^2/s. In comparison, the diffusion coefficient of oxide ions in the (21–1) grain is 1.24×10^{-9} m^2/s, nearly double that of the (−511) grain at 8.69×10^{-10} m^2/s. These findings demonstrate that grain orientation significantly influences oxide ion diffusion, with the diffusion rate at grain boundaries surpassing that within the grains [30].

For protons, conduction behavior can vary significantly depending on the grain boundary structure. The presence of defects at grain boundaries can create more pathways and lower energy barriers for proton migration, resulting in faster proton diffusion at grain boundaries compared to within the grain. An example of this is observed in nanocrystalline BZY ceramics, where the plasmonic conductivity at the grain boundaries is nearly three orders of magnitude higher than in microcrystalline BZY [31]. This difference is attributed to the grain boundaries' high sensitivity to the mobility and concentration of defects that

promote proton conduction. In certain materials, the presence of water molecules at grain boundaries can enhance proton conduction by providing additional pathways for conduction. For instance, in hydrogen-bonded organic framework (HOF) materials, water molecule adsorption at grain boundaries can improve proton conduction properties [32].

On the other hand, the blocking effect at grain boundaries can impede proton conduction. Guo et al. studied the sintering properties and conductivity of CuO-modified BZY. In their research, they observed that the relative density and linear shrinkage of BZY increased with higher CuO content at the same temperature, but the electrolyte resistance also increased, leading to a decrease in conductivity. However, the addition of CuO did not decrease the conductivity of BZY. The authors noted that excess CuO not only introduced electronic conductivity, which reduced the open-circuit voltage of the cell, but also displaced Y in BZY, causing Y polarization in the grain boundaries and further obstructing proton conduction. In some materials, structural instability at grain boundaries can also hinder proton conduction [33]. The irregular arrangement of atoms at grain boundaries may induce structural instability, thereby increasing the energy barrier for proton conduction.

3. Structures of SOFCs/SOECs

3.1. SOFCs/SOECs with different supports

SOFCs and SOECs can be distinguished by their structural support mechanisms, which include anode/fuel electrode-supported, electrolyte-supported, cathode/air electrode-supported, and metal-supported designs (Fig. 6a). Each of these configurations possesses unique benefits and drawbacks. Consequently, it is imperative to investigate the

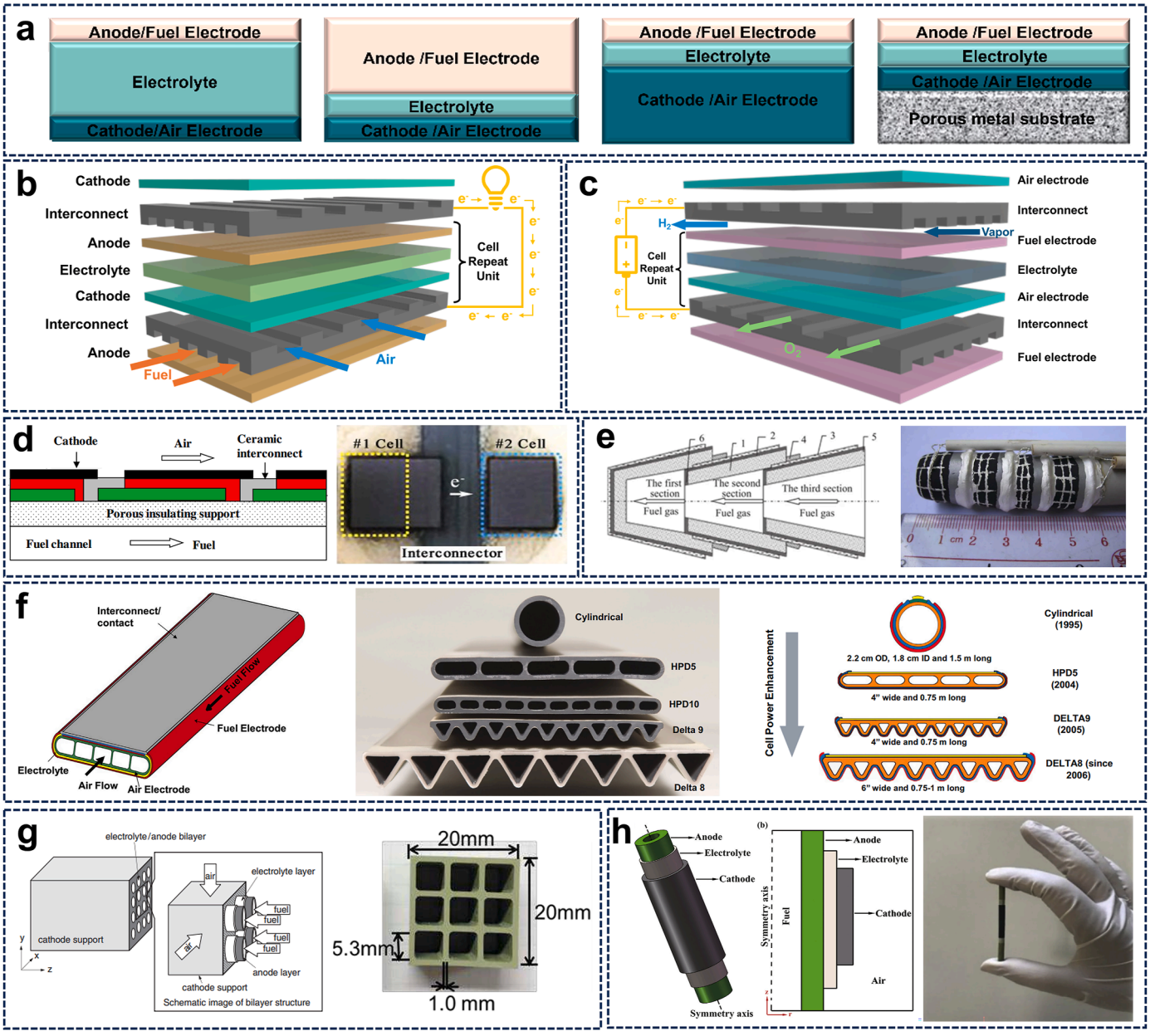

Fig. 6. (a) Different support configurations for SOFCs/SOECs. (b) Flat-panel SOFC. (c) Flat-panel SOEC. (d) Segmented-in-series type. Reproduced with permission [49]. Copyright 2013, Elsevier. (e) Conical tubular type. Reproduced with permission [51]. Copyright 2008, American Ceramic Society. (f) Flat-tubular type [49]. (g) Honeycomb type. Reproduced with permission [53]. Copyright 2021, IOPScience. (h) Micro-tubular type. Reproduced with permission [54]. Copyright 2018, Elsevier.

most effective configuration for specific SOFCs/SOECs to optimize their performance. This involves a comprehensive evaluation of the structural integrity, thermal management, electrical conductivity, and overall efficiency of each design to determine the one that best aligns with the desired operational characteristics and cell longevity.

i. Anode-supported SOFCs and fuel-electrode-supported SOECs

In anode-supported SOFCs and fuel-electrode-supported SOECs, the anode/fuel electrode performs a dual role as the locus for electrochemical reactions and the structural backbone of the cell. The anode/fuel electrode support, commonly fabricated from porous ceramic materials, constitutes a principal element of the cell. This porous architecture is instrumental in enabling the permeation and distribution of the fuel gas, concurrently imparting adequate mechanical rigidity to sustain the cell's integrity. Atop the anode/fuel electrode support, a slender functional layer is present, which, while composed of an analogous material, boasts a more refined microstructure to augment the reactivity at the reaction interface. This functional layer is ensheathed by a tenuous electrolyte membrane, characterized by a compact microstructure that facilitates ionic conduction and precludes interfacial contact between the gases at opposing electrodes. The cathode/air electrode is generally affixed to the electrolyte surface in the form of a thin film. Cells that are anode/fuel electrode-supported often exhibit superior power densities, attributable to the architecture that permits a thinner electrolyte layer, consequently diminishing the cell's ohmic resistance and operational temperature. Such design innovations are conducive to streamlining the cell configuration, curtailing costs, and bolstering the cell's power density and mechanical robustness [34–37].

ii. Electrolyte-supported SOFCs and SOECs

In electrolyte-supported SOFCs and SOECs, the electrolyte constitutes the main supportive element of the cell's architecture, with the thickness of the layer commonly falling between 150 and 300 μm. The electrolyte-supported configuration tends to be more robust, owing to the superior chemical and thermal stability inherent in electrolyte materials, which facilitate the preservation of the cell's structural integrity. The stability of the electrolyte materials endows these structured cells with enhanced durability over extended periods, especially under conditions of thermal cycling and mechanical stress. Nonetheless, the increased thickness of the electrolyte layer can result in diminished ionic conductivity, consequently reducing the power density, so operational temperatures exceeding 800 °C are necessary. Such elevated temperatures demand superior thermal insulation qualities and the employment of costly materials resistant to high temperatures, while simultaneously introducing the potential for material degradation due to prolonged exposure to high thermal conditions [38–40].

iii. Cathode-supported SOFCs and air-electrode-supported SOECs

In cathode-supported SOFCs and air electrode-supported SOECs, the cathode/air electrode is commonly fabricated from porous ceramic materials that have excellent electrical conductivity and adequate mechanical strength to sustain the entire cell structure. The electrolyte layer is typically deposited onto the cathode, with the anode/fuel electrode positioned on the opposite side of the electrolyte layer. The cathode/air electrode-supported configuration offers the advantage of reducing the electrode thickness on the fuel side, shortening the fuel diffusion pathway, and thus enhancing cell performance. Nevertheless, this configuration has significant drawbacks. Its lower fabrication temperature in comparison to the anode/fuel electrode and electrolyte makes it more prone to thermal degradation, which can adversely impact the electrochemical performance of SOFCs/SOECs. Consequently, research on this particular configuration remains limited [41–43].

iv. Metal-supported SOFCs and SOECs

Metal-supported SOFCs and SOECs are underpinned by the same design paradigm, wherein a slender ceramic electrode/electrolyte layer is undergirded by an economical and sturdy stainless-steel substrate. Such an architectural approach not only obviates material expenditure but also substantially augments the cell's mechanical resilience, reducing the likelihood of the brittle fractures prevalent in conventional SOFC/SOEC configurations. These cells exhibit an enhanced capacity to tolerate significant thermal gradients and can endure swift thermal cycling events. In stark contrast to traditional SOFCs/SOECs, which necessitate a protracted heating regimen to counteract thermal shock, metal-supported SOFCs/SOECs can achieve activation within a mere 10 s. This rapid response is of paramount importance for applications involving changeable load demands. The technology demonstrates superior performance stability when subjected to redox cycling conditions, effectively surmounting the electrolytic ruptures inherent to legacy SOFC and SOEC systems. It mitigates structural decay and material translocation by reducing catalytic processing temperatures, refining catalytic particle dimensions, and optimizing electrode architectural designs, while the excision of the buffer layer augments operational efficacy and steadfastness. Collectively, these advancements are encouraging for the implementation of such technologies across pivotal sectors, including renewable energy sequestration and aerospace applications [44–46].

3.2. SOFCs and SOECs in different configurations

Depending on the construction of single cells, the architectural paradigms of SOFCs and SOECs are broadly classified into (1) the planar configuration and (2) the tubular configuration. These distinct structural approaches possess their own unique merits and demerits with respect to mechanical characteristics and electrochemical efficacy.

i. Planar

The planar configuration has a simple geometric structure, consisting of an anode/fuel-electrode material, an electrolyte material, and a cathode/air-electrode material integrated into one device (Figs. 6b, c). Flanked by slotted interconnects on either side, contiguous cathodes and anodes are electrically connected, with gas channels facilitated on both peripheries, concurrently segregating the oxidizing gases from the reducing gases. The fabrication and architectural methodology of planar SOFC/SOEC devices are devoid of complexity, leading to a marked decrease in the costs of production. Moreover, the planar design has abbreviated current pathways and diminished ohmic polarization impedance, yielding better electrical performance and higher power output density. However, the peripheries of planar SOFC/SOEC cell assemblies must be sealed to prevent the commingling of oxygen and fuel gases, a procedural challenge of significant magnitude. The material selection for bipolar current collectors is constrained by the need for thermal compatibility with the electrode materials, robust resistance to oxidation at elevated temperatures, and sustained electrical conductivity to prevent lower performance or inadequate resistance to thermal cycling [47,48].

ii. Tubular

Research on planar SOFCs and SOECs concentrates predominantly on augmenting the current by enlarging the planar reaction surface area, instead of elevating the voltage through the serial connection of multiple individual cells to assemble a cell stack. For enhancing voltage, the tubular design is better than the planar. Within a tubular single cell, the innermost layer comprises a porous ceramic tube that functions as the cell anode and underpins the cell's structural integrity; this is succeeded by layers of electrolyte and cathode materials. Fuel is introduced into the tube's core, whereas air is delivered through the tube's outer shell. Tubular SOFC/SOEC single cells exhibit greater freedom, diminishing the likelihood of fracture; they leverage porous ceramics as a supportive framework and thereby are robust and long-lasting; cell assembly is relatively straightforward, permitting facile aggregation into high-power cell stacks via parallel and series interconnections. Nonetheless, the electrolyte in tubular SOFCs/SOECs is comparatively thick, resulting in greater ohmic resistance and consequently lowering the cell's power density. Tubular SOFCs/SOECs also face the challenge of current aggregation around the circumference, which can precipitate substantial ohmic losses and attenuate the power density of individual tube cells.

Consequently, the enhancement of output power density through geometrical design innovations on the tubular SOFC/SOEC platform has emerged as a principal avenue of investigation. Presently, several configurations have gained prominence, including segmented serial tubular, conical tubular, flat tubular, corrugated tubular, honeycomb, and micro-tubular designs.

① Segmented-in-series type

In response to the challenges previously outlined, Rolls-Royce UK has engineered a novel configuration termed the segmented-in-series SOFC/SOEC (Fig. 6d). This design harnesses the conventional tubular framework and introduces a series of discrete cell segments through the artificial segmentation of the contiguous electrode structure, establishing a cohesive internal architecture. Subsequently, these discrete cell segments are connected in tandem via electrical conduits, culminating in a "striped tubular" series assembly. The segmented series design facilitates a more expedient pathway for current collection, mitigating the ohmic losses inherent to the circumferential current distribution in traditional tubular designs. This configuration not only enhances the power output at the segmental level but also, through serial alignment, augments the cumulative voltage output without necessitating the employment of enlarged cell dimensions. This innovative strategy effectively ameliorates several constraints associated with conventional tubular cell designs while also preserving the structural resilience and integrity essential for robust cell performance [49,50].

② Conical tubular type

The conical tubular SOFC/SOEC employs the segmented-in-series design philosophy described above, assembling conical segments to create a self-supporting tubular structure, with the interconnector fulfilling both sealing and current collection roles. Such a design offers a compact and flexible stack configuration capable of generating higher power densities in a reduced form factor, coupled with better resilience to thermal shocks, as depicted in Fig. 6e. Nevertheless, the intricate structural design and the fabrication challenges associated with the interconnector are hurdles that require resolution [51,52].

③ Flat-tubular type

The flat-tubular SOFC/SOEC represents a high-power density (HPD) configuration, engineered by Siemens-Westinghouse Electric Corporation, expanding upon the traditional tubular design. This structure is distinguished by its amalgamation of the advantageous attributes inherent to both tubular and planar configurations. It circumvents the necessity for high-temperature sealing materials and concurrently achieves a marked enhancement in power density on a per unit length basis, as well as in volumetric power density, as depicted in Fig. 6f [49].

④ Corrugated type

In pursuit of a reduction in the cost of the cell stack, Siemens-Westinghouse Electric Corporation has engineered a novel generation of HPD batteries, advancing from the flat-tubular design to a corrugated form factor. This innovative geometry is capable of augmenting the active surface area by a surplus of 40% within an equivalent length (Fig. 6f). The corrugated architecture has also been instrumental in achieving a more condensed cell stack configuration. Unfortunately, the intricacy of the structural design introduces complexities in the fabrication process, thereby increasing production costs [49].

⑤ Honeycomb type

Beyond the flat-tubular and conical designs, the honeycomb configuration has emerged as a distinctive design paradigm that amalgamates the merits of both tubular and planar architectures. This configuration is distinguished by its superior modularity and mechanical robustness, endowing it with enhanced volumetric power density and endurance, as depicted in Fig. 6g [53]. The honeycomb design harnesses the inherent benefits of structural integrity and efficient space utilization, making it a promising candidate for applications where high power output and long-term stability are paramount. The intricate geometry of the honeycomb facilitates heat management and the optimal distribution of mechanical loads, further augmenting the overall performance and reliability of the energy conversion system.

⑥ Micro-tubular type

The micro-tubular SOFC/SOEC integrates the merits of both tubular and planar configurations. Characterized by a diminutive diameter of less than 10 mm, batteries such as those depicted in Fig. 6h are called micro-tubular [54]. In contrast to other forms, micro-tubular batteries exhibit superior volumetric power density and rapid initiation, making them suitable for the swift activation of supplementary power sources in backup and mobile applications. They are able to rapidly ascend to operational temperatures without incurring damage. However, their diminutive dimensions present challenges for electricity collection, rendering this form particularly fitting for medium-to low-power portable applications.

3.3. Challenges for SOFCs/SOECs

SOFCs and SOECs have a host of advantages but are not devoid of challenges. On the systems level, their all-solid-state construction and the need for high-temperature operation can engender response latency, rendering such systems unsuitable for applications requiring frequent cycling between start-up and shutdown. Additionally, the exigencies of high-temperature operation necessitate materials with enhanced thermal and mechanical stability, given the potential for issues related to thermal stress and vibration. At the material level, efficient conduction at elevated temperatures is a prerequisite for both electronic and ionic conduction within SOFCs and SOECs. Despite research on various electrode and electrolyte materials, their high-temperature conductivity and chemical stability remain suboptimal. During operational deployment, electrodes and interconnect materials may be subject to carbonaceous deposition and contamination by impurities, which can curtail the performance of SOFCs and SOECs. Consequently, further investigation and advancement in the domain of electrode and electrolyte materials is needed to augment their electrochemical performance and stability at high temperatures, which are essential for the evolution of SOFC and SOEC technologies.

4. Key materials

4.1. SOFC cathodes and SOEC air electrodes

In the context of SOFCs, the cathode must take into consideration material chemical stability, ionic conductivity, electronic conductivity, and electrolyte compatibility. Furthermore, their structural design plays an integral role in facilitating gas diffusion and enhancing reaction efficiency. For SOECs, the air electrode is tasked with the diffusion/migration of ions, the transfer of e^-, and the formation and dissociation of O_2. Consequently, the air electrode should exhibit a porous structure for gas transport, coupled with high electronic and ionic conductivity, as well as robust catalytic activity for the ORR/OER. Additionally, the material comprising the air electrode must demonstrate chemical and thermal compatibility with the adjacent electrolyte to ensure sustained cell performance. The spectrum of cathode/air electrode materials is notably diverse, ranging from single perovskite and double perovskite to Ruddlesden–Popper (RP) oxides, among others. The judicious selection and optimization of these materials are pivotal to the performance and durability of both SOFCs and SOECs.

4.1.1. Single perovskite

The single perovskite structure is denoted by the chemical formula ABO_3 and exhibits a diversity of lattice configurations, ranging from cubic to tetragonal and orthorhombic symmetries. A distinctive architectural aspect is the coordination of the A-site cations at the cell's vertices with a dodecahedral array of oxygen ions, whereas the smaller B-site cations are positioned in a central octahedral coordination within the unit cell (Fig. 7a) [55–57]. Within the context of SOFC/SOEC applications, the A-site is commonly occupied by low-valence rare earth elements (e.g., La^{3+}, Gd^{3+}, Pr^{3+}), alkali metals (e.g., Li^+, Na^+, K^+), and

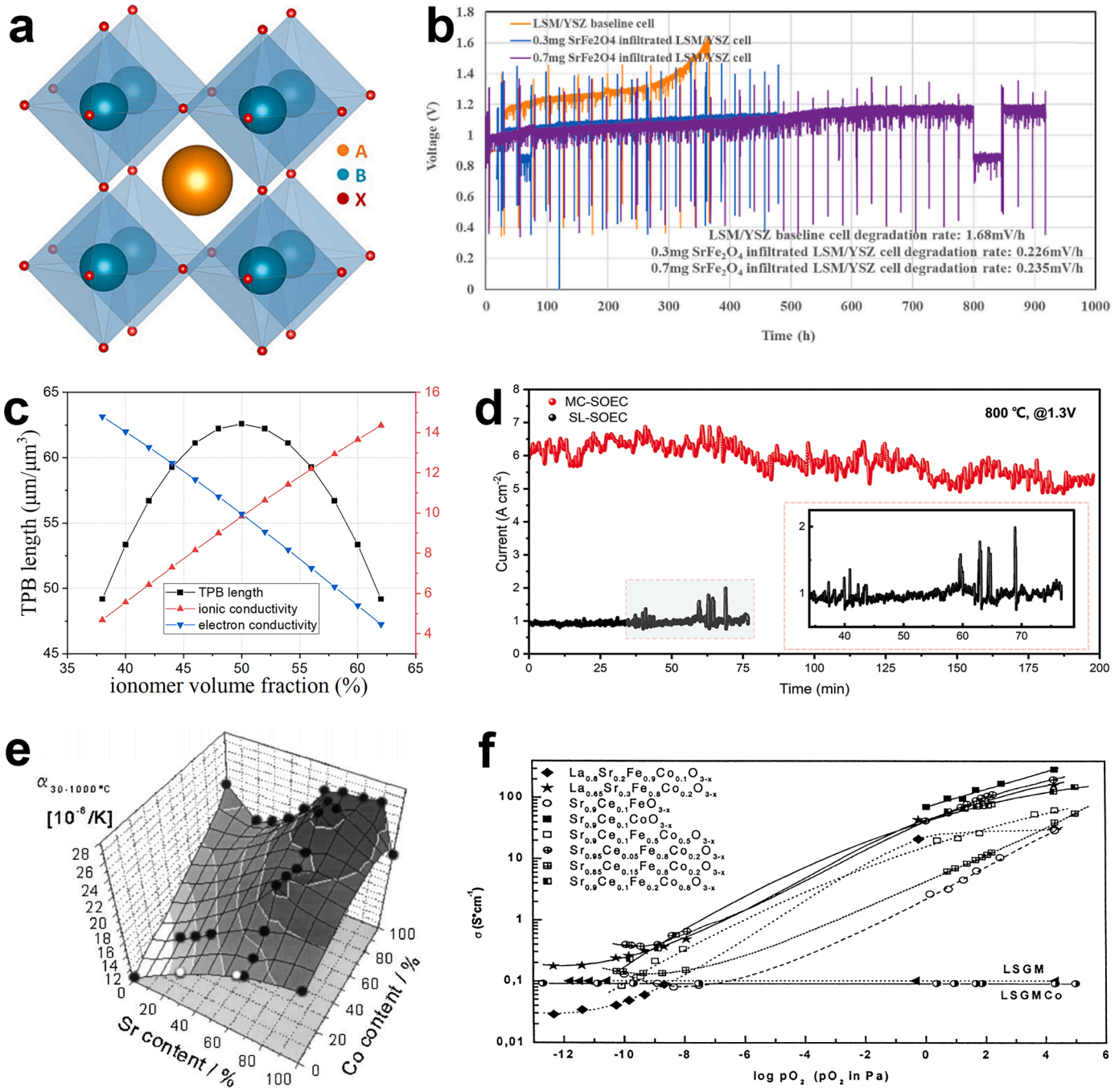

Fig. 7. (a) Crystal structure of single perovskite. (b) Stability of LSM-YSZ composite air pole. Reproduced with permission [67]. Copyright 2023, Elsevier. (c) TPB modulation of composite electrodes. Reproduced with permission [69]. Copyright 2021, Wiley-VCH. (d) Current density of micro-channel and spongelike SOECs at 800 °C, 1.3V. Reproduced with permission [81]. Copyright 2025, Elsevier. (e) Changes in TEC of Co-based single perovskite. Reproduced with permission [82]. Copyright 2000, Elsevier. (f) Effect of Fe on the TEC of Co-based single perovskite. Reproduced with permission [85]. Copyright 2000, Elsevier.

alkaline earth metals (e.g., Ca^{2+}, Sr^{2+}, Ba^{2+}). The B-site, conversely, is inhabited by smaller transition metal ions (e.g., Fe^{3+}, Co^{3+}, Ni^{3+}), chosen for their propensity to confer enhanced ionic conductivity and structural robustness [58,59].

$LaMnO_3$ has a quintessential single perovskite structure, renowned for its exceptional stability [60,61]. The partial substitution of the rare earth cation La^{3+} with the divalent Sr^{2+} yields $La_{1-x}Sr_xMnO_3$ (LSM), a material frequently employed as a cathode/air electrode due to its elevated conductivity, robust catalytic activity, and favorable thermal and chemical compatibility with electrolytes [62,63]. Nonetheless, the oxygen ion conductivity of LSM at 800 °C is only 10^{-8}~10^{-1} S/cm; this restricts the ORR/OER reactions to the LSM/YSZ interface, culminating in suboptimal cell performance [64]. Moreover, the diminished O^{2-} conductivity can precipitate interfacial delamination between LSM and YSZ [65]. The formation of a composite air electrode through the mechanical blending of LSM with YSZ results in a synergistic enhancement of the OER/ORR activity. This augmentation is attributed to the expansion of the TPB and the concomitant improvement in stability, as the ion-conducting YSZ phase is amalgamated with the p-type electronic conductor LSM [66]. In addition, the LSM-YSZ composite air electrode demonstrates enhanced compatibility with the YSZ electrolyte, facilitating stable operation over an extended period spanning thousands of

hours (Fig. 7b) [67]. Despite these merits, the LSM-YSZ composite cathode/air electrode has yet to satisfy the stringent requirements for high-performance SOFCs and SOECs. The particle size of the LSM/YSZ interface is a critical parameter that directly influences the length of the TPB, which in turn dictates the electrochemical performance of the material. Consequently, efforts have been directed towards the fabrication of nanoparticles to augment the TPB length, as depicted in Fig. 7c [68,69]. However, while nanostructured materials may exhibit enhanced initial performance, the relatively inferior catalytic properties of LSM/YSZ necessitate higher operational temperatures. This temperature requirement leads to a swift coarsening of the nanostructures to micrometer-scale dimensions, consequently diminishing the performance gains [70,71]. Moreover, akin to LSM electrodes, the catalytic activity of the LSM-YSZ composite is amplified post-polarization but loses this enhancement under SOEC operating conditions [72]. In light of these challenges, the research community has progressively shifted its focus towards the development of cathode/air electrode materials that exhibit high O^{2-} conductivity and ORR/OER activity at medium and low temperatures, characterized by MIEC properties.

The elevated O^{2-} conductivity and accelerated kinetics of oxygen reactions in MIEC materials stem from the doping of low-valence cations at the A-site and/or B-site, which augments the concentration of oxygen vacancies. As a result, active sites are not confined to the substrate but rather are dispersed across the entire surface of the electrode [73]. At present, exemplary MIEC air electrode materials include $SrCo_{0.8}Nb_{0.2}O_{3-\delta}$, $La_{0.8}Sr_{0.2}CoO_{3-\delta}$ (LSC), $(La,Sr)(Co,Fe)O_{3-\delta}$ (LSCF), $SrNb_{0.1}Co_{0.7}Fe_{0.2}O_{3-\delta}$, $SrSc_{0.175}Nb_{0.025}Co_{0.8}O_{3-\delta}$, and $(Ba,Sr)(Co,Fe)O_{3-\delta}$ (BSCF), among others [74–79]. In contrast to materials such as LSM, MIEC materials extend the TPB reaction zone to the two-dimensional dual-phase boundary (DPB) at the interface between the ion/electron mixed conductor and the gas phase, thereby demonstrating enhanced electrochemical performance. Micro-nanostructured LSC fuel electrodes have achieved a current density of 5.96 A/cm^2 at an electrolysis potential of 1.3 V, with an associated hydrogen production rate of 2.5 L/cm^2·h (Fig. 7d) [80,81]. However, these Co-based materials exhibit a higher thermal expansion coefficient (TEC). For instance, the TEC of LSC can reach up to 23×10^{-6} K^{-1} (30–1000 °C) due to the high-temperature spin state transition of Co^{3+}, which is substantially higher than the TEC values of conventional electrolyte materials, resulting in a thermal expansion mismatch between Co-containing electrodes and electrolytes (Fig. 7e) [82,83]. Strategies to reduce the TEC involve the substitution of La^{3+} with ions of smaller radii, such as Pr^{3+} ($Pr_{0.4}Sr_{0.6}Co_{0.8}Fe_{0.2}O_{3-\delta}$) or Gd^{3+} ($Gd_{0.4}Sr_{0.6}Co_{0.8}Fe_{0.2}O_{3-\delta}$), among others [84]. Furthermore, the replacement of Co with Fe can effectively reduce the TEC of LSC, yet this reduction in TEC comes at the expense of suppressing the material's electrical conductivity. The ionic conductivity of $La_{0.8}Sr_{0.2}Co_{0.8}Fe_{0.2}O_{3-\delta}$ (4.0×10^{-2} S/cm, 800 °C) is an order of magnitude greater than that of $La_{0.8}Sr_{0.2}Co_{0.2}Fe_{0.8}O_{3-\delta}$ (2.3×10^{-3} S/cm, 800 °C) (Fig. 7f) [85]. This dichotomy creates an inherent balance between the TEC and ionic conductivity for Co-based perovskite materials.

Unfortunately, at operational temperatures ranging from 600 to 800 °C for SOFCs/SOECs, cobalt-based materials are susceptible to reactions with electrolytes such as YSZ, leading to the formation of transitional phases like $SrZrO_3$. To counteract this detrimental effect, a buffer layer of doped CeO_2 is often utilized between the electrode and the YSZ electrolyte [86]. Therefore, cobalt-lean or cobalt-free perovskites, including $SrTi_{0.3}Fe_{0.63}Co_{0.07}O_{3-\delta}$ and $SrTi_{0.3}Fe_{0.7}O_{3-\delta}$, are garnering more interest [87,88]. Furthermore, MIEC perovskite materials containing alkaline earth metals (such as Ba and Sr) also encounter challenges associated with elemental segregation, chromium poisoning, and carbon dioxide poisoning, which can compromise the durability of the air electrode. Recently, there has been a growing focus on novel alkali-free and metal-free electrodes, where the A-site is commonly occupied by lanthanide elements, and the B-site is populated by transition metal elements (Fe, Cu, Ni, etc.). Among these electrodes, candidates such as $LaNi_{1-x}Cu_xO_{3-\delta}$ and $LaNi_{1-x}Co_xO_{3-\delta}$ have started to draw significant attention [89,90].

4.1.2. Double perovskite

The chemical formulation of double perovskite compounds is commonly denoted as $A_2B'B''X_6$ or $A'A''B_2X_6$, wherein A, A', A", and B, B', B" represent distinct metallic cations, and X corresponds to an anionic species. This structure is comprised of two perovskite units, which together constitute a more intricate three-dimensional layered framework. The diverse metal cations occupying the A and B sites are instrumental in dictating the material's physical and chemical attributes, including electrical conductivity and oxygen affinity. In comparison to their single perovskite counterparts, double perovskites provide an enhanced repertoire of lattice sites and more extensive opportunities for microstructural control, thus endowing them with superior electronic conduction and enhanced oxygen surface exchange and diffusive capabilities. These characteristics make double perovskites promising candidates for applications in SOFC cathodes and SOEC air electrodes.

$Sr_2Fe_{1.5}Mo_{0.5}O_{6-\delta}$ (SFM) is an $A_2B'B''X_6$-type MIEC double perovskite oxide that has attracted considerable interest on account of its remarkable redox stability and elevated conductivity under both oxidative and reductive conditions (50% reduction in polarization resistance (Rp)) (Fig. 8a) [91,92]. SFM's application as an electrode in symmetric cells for SOFCs and SOECs has yielded exceptional performance metrics. For example, an SOEC utilizing SFM electrodes realized a hydrogen generation efficiency of 380 mL/cm^2·h throughout the sustained electrolysis of a 40% H_2–60% H_2O mixture for a period of 100 h at 900 °C, with a current density of 0.88 A/cm^2 and a cell voltage of 1.3 V (Fig. 8b) [93]. The electrochemical attributes of this material can be notably augmented through the incorporation of 10% Cu into the lattice, resulting in $Sr_2Fe_{1.4}Cu_{0.1}Mo_{0.5}O_{6-\delta}$ (SFCM), which exhibits a total conductivity of 49.3 S/cm at 450 °C and a marked decrease in interfacial resistance (with a surface resistance of 0.26 Ω/cm^2 at 800 °C). However, beyond a 10% Cu doping threshold, an expansion of the material's unit cell is observed, potentially impeding the diffusive motion of oxygen ions. A composite electrode fabricated by integrating SFCM with $Sm_{0.2}Ce_{0.8}O_{1.9}$ (SDC), akin to the approach taken with LSM-YSZ composite electrodes, manifests an interfacial impedance of 0.15 Ω/cm^2, underscoring its superior electrochemical performance (Fig. 8c) [94].

Lanthanide-based $LnBaCo_2O_{5+\delta}$, belonging to the $A'A''B_2X_6$-type double perovskite oxides, exhibits an ordered arrangement of Ba and Ln ions within alternating layers (001) [95]. The oxygen content within the material varies with the alteration of the Ln element (La→Lu), adhering to the gadolinium break effect, wherein the non-stoichiometric ratio δ of oxygen is minimal in $GdBaCo_2O_{5+\delta}$ (0.698). This phenomenon can be ascribed to the half-filled f^7 electronic configuration of the +3 valence Gd ions, which imparts a significant electron shielding effect and a relatively diminished effective nuclear charge. Consequently, this leads to a reduced stability constant for the complex, culminating in decreased oxygen content [96]. The layered perovskite materials are advantageous due to their enhanced ionic conductivity at lower temperatures, a result of the weakened oxygen bonds within the [AO] layers and the emergence of crystallographic channels that facilitate ion transport. Moreover, the variable valency of Co ions diminishes the hopping energy for O^{2-}, optimizing the processes of oxygen adsorption and dissociation [97].

$PrBaCo_2O_{5+\delta}$ (PBC), recognized for its superior electrochemical performance among the $LnBaCo_2O_{5+\delta}$ series, exhibits an electrical conductivity of approximately 682 S/cm at 800 °C [98]. Specifically, the Ba-deficient variant $PrBa_{0.94}Co_2O_{5+\delta}$ (PBxCO, with x = 0.90–1.0) manifests enhanced performance relative to LSCF. At x = 0.94, $PrBa_{0.94}Co_2O_{5+\delta}$ achieves the most pronounced electrocatalytic activity, featuring a Rp as low as 42 mΩ/cm^2 at 600 °C; this surpasses alternative cathode/air electrode materials, including cobalt-free perovskite $SrNb_{0.1}Fe_{0.9}O_{3-\delta}$ (SNF) and cobalt-containing perovskite BSCF. A SOFC

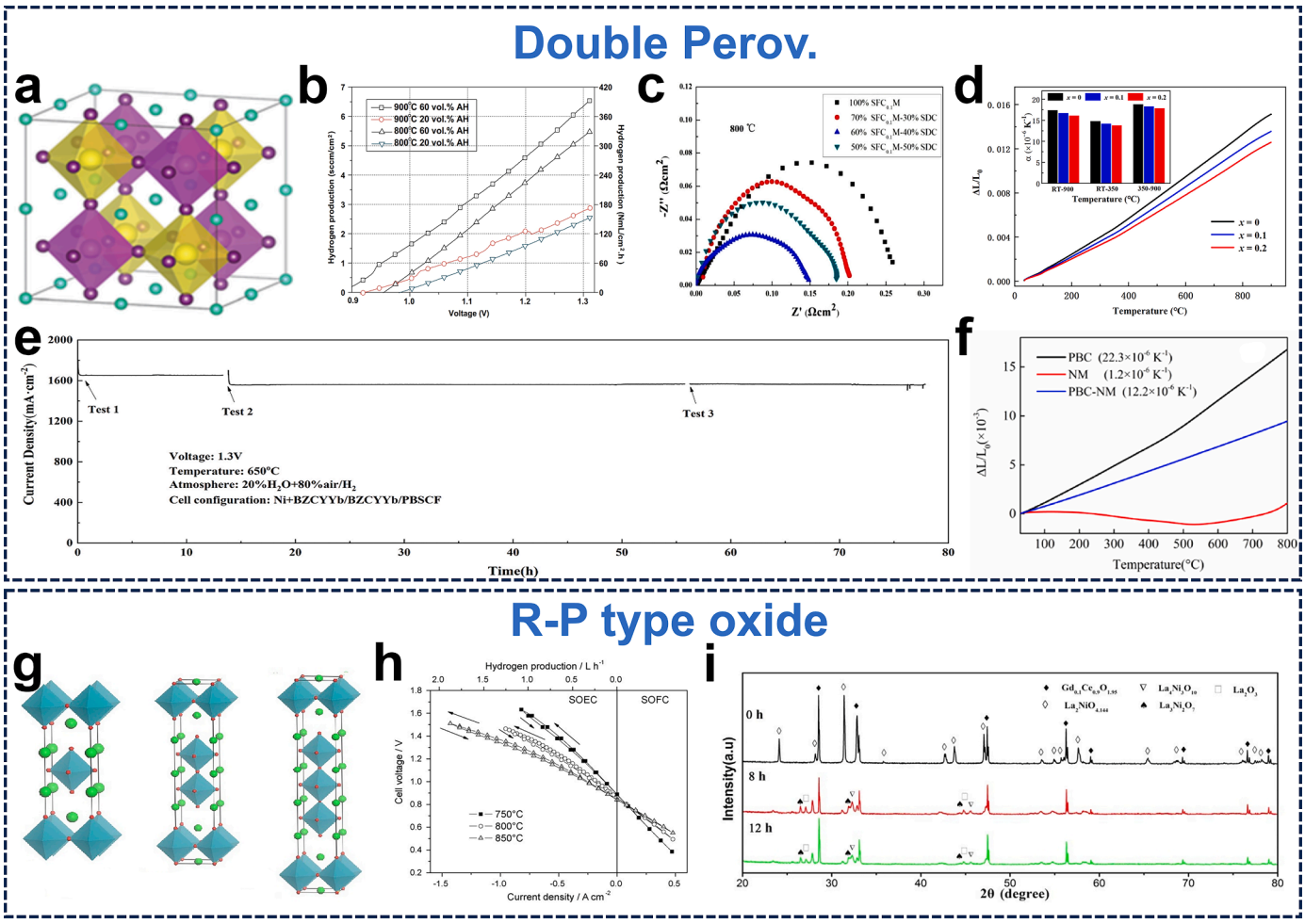

Fig. 8. (a) Crystal structure of double-perovskite. (b) Hydrogen production curves for SOEC using SFM electrodes. Reproduced with permission [93]. Copyright 2010, Elsevier. (c) Changes in conductivity and interfacial resistance of Cu-doped SFCMs. Reproduced with permission [94]. Copyright 2021, Springer. (d) Conductivity and TEC properties of $PrBa_{1-x}Ca_xCoCuO_{5+\delta}$. Reproduced with permission [100]. Copyright 2022, Elsevier. (e) Stability of PBSCF at 650 °C, 1.3 V. Reproduced with permission [102]. Copyright 2023, Elsevier. (f) TEC of PBC with $NdMnO_{3-\delta}$ composite electrode. Reproduced with permission [98]. Copyright 2022, Elsevier. (g) Crystal structures of R–P type oxides. (h) Ionic conductivity and hydrogen production properties of NNO. Reproduced with permission [112]. Copyright 2021, Elsevier. (i) LNO decomposes at 750 °C to $La_3Ni_2O_7$ and $La_4Ni_3O_{10}$. Reproduced with permission [119]. Copyright 2017, Elsevier.

with this cathode yielded a peak power density of 1.0 W/cm^2 at 700 °C, underscoring PBC's potential for electrochemical device applications [99]. However, akin to other Co-based perovskite materials, these materials are characterized by a significant TEC of approximately 17.5×10^{-6} K^{-1}, as depicted in Fig. 8d [100]. Consequently, a strategy to mitigate the TEC involves the substitution of Co with Fe, akin to the modification approach for LSC, to reduce the material's TEC. One material of particular interest in this context is $PrBa_{0.5}Sr_{0.5}Co_{1.5}Fe_{0.5}O_{5+\delta}$ (PBSCF), which boasts exceptional oxygen surface exchange coefficients, elevated O^{2-} conductivity, robust OER/ORR catalytic activity, and commendable structural stability [101]. Under operating conditions of 1.3 V and 650 °C, a SOEC equipped with a PBSCF air electrode sustains a current density of 1.56 A/cm^2 after 78 h of operation, with no discernible delamination (Fig. 8e) [102]. Furthermore, the superior OER activity of PBSCF has been substantiated in SOECs equipped with Ni-YSZ anode electrodes [103]. A secondary strategy to diminish the TEC encompasses the integration of materials exhibiting negative thermal expansion (NTE), including $NdMnO_{3-\delta}$ and $Y_2Mo_3O_{12}$. These NTE materials are capable of significantly reducing the TEC of PBSCF, consequently alleviating thermal stress and bolstering the thermal cycling stability of the cell. Specifically, by forming a composite with $NdMnO_{3-\delta}$, the TEC of PBSCF is reduced from 22.3×10^{-6} K^{-1} to 12.2×10^{-6} K^{-1} when the mass ratio of PBSCF to $NdMnO_{3-\delta}$ in the composite electrode is 1:1 (refer to Fig. 8f) [98]. Beyond SFM and PBSCF, other double perovskite oxides, such as $NdBa_{0.5}Ca_{0.5}Co_{1.5}Fe_{0.5}O_{5+\delta}$, $PrBaFe_{1.9}Zr_{0.1}O_{5+\delta}$, and $SmBa_{0.75}Ca_{0.25}CoFeO_{5+\delta}$, have also been documented for their application as cathodes in SOFCs and air electrodes in SOECs, yielding commendable performance outcomes [104–106].

4.1.3. *R–P type oxides*

R–P oxides, characterized by the chemical formula $A_{n+1}B_nO_{3n+1}$ with n = 1, 2, 3, are a category of materials noted for their pronounced oxygen ion conductivity and oxygen exchange capabilities. Owing to their distinctive mixed ionic-electronic conductive properties, they play a significant role in the investigation of materials for SOFC cathodes and SOEC air electrodes. The structural framework of these materials is defined by A-sites and B-sites; the A-sites are commonly occupied by rare earth elements, including La, Nd, Pr, Sr, and Ga, whereas the B-sites are populated by transition metals such as Ni, Co, and Cu. In R–P oxides, an excess of O^{2-} is introduced within the [AO] layers at the A-site, creating electron holes around the B-site to preserve the structural integrity. This architecture promotes the conduction of O^{2-} within the material and provides active sites that catalyze the ORR and OER, as depicted in Fig. 8g [107]. In comparison to conventional perovskite oxides, R–P oxides demonstrate enhanced oxygen diffusivity and surface exchange

kinetics, consequently leading to a marked improvement in their performance within SOFC/SOEC applications. Research indicates that the concentration of oxygen vacancies is not the sole determinant of the catalytic activity for OER/ORR reactions; rather, it is intricately linked to the concentration and mobility of interstitial oxygen ions. The electrochemical performance and structural stability of R–P type materials can be fine-tuned through doping at the A-site or B-site. For instance, the incorporation of low-valence elements at the A-site can diminish the concentration of interstitial oxygen ions within the AO layers, thereby inhibiting the reactivity of lattice oxygen in the perovskite oxide layers and enhancing the material's ionic conductivity [108].

Lower-order Ruddlesden–Popper (R–P) nickelates, with the notation n = 1, such as $Nd_2NiO_{4+\delta}$(NNO), $Pr_2NiO_{4+\delta}$ (PNO), and $La_2NiO_{4+\delta}$(LNO), are regarded as the most promising electrode materials for SOFCs and SOECs [109–111]. NNO demonstrates superior ionic conductivity compared to LSCF at 800 °C, and an SOEC utilizing NNO as the air electrode has achieved a current density of 0.76 A/cm^2 at 1.2 V and 800 °C (refer to Fig. 8h) [112]. In contrast to NNO, PNO is characterized by enhanced diffusion rates, reduced diffusion activation energy, and improved ionic-electronic conductivity, particularly within the intermediate temperature range of 600–700 °C [113]. Higher-order R–P nickelates, including $Ln_3Ni_2O_{7\pm\delta}$ (n = 2) and $Ln_4Ni_3O_{10\pm\delta}$ (n = 3), are structured by the linkage of NiO_6 layers along the c-axis of the crystal lattice. These higher-order nickelates present a relatively higher total conductivity compared to $Ln_2NiO_{4+\delta}$ [114]. However, the thermal stability of R–P oxides continues to pose a significant challenge. PNO undergoes decomposition into PrOx and $Pr_4Ni_3O_{4\pm\delta}$ at elevated temperatures between 800 and 950 °C and further decomposes into Pr_6O_{11}, $PrNiO_{3-\delta}$, and $Pr_4Ni_3O_{10+\delta}$ within the intermediate temperature range of 600–800 °C [115]. LNO, after being subjected to operating conditions of 500 mA/cm^2 and 750 °C for 48 h, experiences decomposition into $La_3Ni_2O_7$ and $La_4Ni_3O_{10}$ (refer to Fig. 8i) [116]. Moreover, R–P oxides with elevated cobalt concentrations are also prone to rapid degradation.

In pursuit of better performance and stability for R–P oxides, a plethora of approaches have been investigated by researchers, with ionic doping being a prominent strategy. For example, the incorporation of low-valence elements such as Sr or Ca at the A-site, or the introduction of other transition metal ions like Co, Fe, or Cu at the B-site, can enhance the electronic and ionic conductivity of the materials. Atomic simulation outcomes have demonstrated that the substitution of Ni^{2+} at the B-site with Co^{3+} or Fe^{3+} results in an increased concentration of oxygen vacancies, albeit with potential detrimental effects on the oxygen ion diffusion coefficient [117]. An alternative strategy to bolster material properties is the fabrication of mixed or composite nickelate electrodes, which have exhibited superior performance and more favorable thermal expansion coefficient matching. For instance, a composite electrode $La_4Ni_3O_{10-\delta}$ has demonstrated an electrical conductivity of approximately 440 S/cm at 700 °C, and the conductivity and electrochemical performance of electrodes based on the $La_{2-x}Pr_xNiO_{4+\delta}$ system (with $0 \leq x \leq 2.0$) have likewise been notably enhanced [118]. Additional R–P oxides, including $La_{1.5-x}Bi_xSr_{0.5}Ni_{0.5}Mn_{0.5}O_{4+\delta}$, $Pr_2Ni_{1-x}Co_xO_{4+\delta}$, $La_{1.5}Sr_{0.5}Ni_{1-x}Mn_xO_{4+\delta}$, and $Nd_{0.8}Sr_{1.2}CoO_{4+\delta}$, have emerged as prospective candidates for applications in SOFC cathodes and SOEC air electrodes [119–122].

4.1.4. Non-perovskite oxides

Non-perovskite oxide cathode and air electrode materials are instrumental in the research and development of SOFCs and SOECs. $YBa_2Cu_3O_{7-\delta}$ (YBCO) type oxides, while exhibiting robust oxygen reduction kinetics at intermediate temperatures, face limitations due to their elevated area-specific resistance, which can precipitate performance decline [123]. Spinel-structured materials and layered materials have garnered attention as potential candidates owing to their distinctive crystal lattice structures and electronic/ionic conduction properties. Lithium nickel manganese oxide, $LiNi_xMn_{2-x}O_4$ (LNMO), renowned for its low cost and plentiful supply as a cathode material in high-voltage lithium-ion batteries, has also been probed for its applicability as a SOFC cathode [124]. $NiMn_2O_4$ (NMO), synthesized via the sol-gel method, manifests a stable spinel structure and has been extensively investigated for its use as a cathode in intermediate-temperature solid oxide fuel cells (IT-SOFCs), showcasing mixed ionic-electronic conductivity [125]. The spinel material $MnCo_2O_4$, employed as an air electrode in SOECs, demonstrates a Rp of 1.43 Ω cm^2 under open-circuit voltage (OCV) conditions at 800 °C. At an electrolysis voltage of 1.5 V, it achieves a current density of 0.75 A/cm^2 and maintains stable operation for 84 h under a constant current load of 0.2 A/cm^2, with a minimal degradation rate of 7.23 × 10^{-4} V/h. These findings suggest that the $MnCo_2O_4$ spinel material not only possesses superior electrochemical performance but also exhibits commendable long-term stability [126]. Nevertheless, it is not without shortcomings, including a propensity for phase transitions under high voltages, which can result in micro-cracks and severe interfacial side reactions. Issues such as structural degradation or compatibility challenges with electrolyte materials may arise during prolonged high-temperature operation, potentially compromising long-term stability.

4.2. SOFC anodes and SOEC fuel electrodes

4.2.1. Metal

Metallic anode/fuel electrode materials, predominantly comprising Ni and its alloys, have garnered widespread research attention for their superior catalytic activity. These nickel-based anode/fuel electrode materials retain significant strength at high temperatures, potentially exceeding 1200 °C, and demonstrate elevated catalytic activity for electrode reactions, realizing a power density of 1079 mW/cm^2 at 700 °C (Fig. 9a) [127]. However, the TEC discrepancy between nickel-based materials and electrolyte materials, such as YSZ and gadolinium-doped ceria (GDC), may precipitate interfacial cracking and contribute to electrolyte layer delamination after prolonged use (Figs. 9b, c) [128]. To surmount these challenges, researchers have investigated an array of alloys and composite materials. For example, Ni–Fe alloys, utilized as support structures, have achieved a peak power density of 1.4 W/cm^2 at 800 °C, notwithstanding their susceptibility to oxidation and thermal expansion coefficient mismatch, through refined design and coating technologies (Fig. 9d) [129]. The integration of Ni–Fe alloys has also reduced costs and partially alleviated the thermal expansion incompatibility between pure Ni and the electrolyte, yet issues of carbon deposition and the deleterious effects of sulfur on cell performance persist [130].

Porous Ni and its alloyed derivatives are considered promising candidates for fuel electrode materials that exhibit enhanced performance during the electrolysis of carbon dioxide in SOECs. The addition of alloying elements, such as Fe and Cr, to these materials can mitigate against coke formation and Ni particle sintering, consequently improving the electrode's stability during operation. Pt serves as a direct fuel electrode material for the electrolytic reduction of CO_2, circumventing the need to introduce a reducing gas to counteract oxidation. The fervor for researching single-phase metal fuel electrodes has diminished, though, due to their deficient ionic conductivity and the constraints associated with the TPB, which plays an integral role in high-efficiency electrochemical reactions.

4.2.2. Composite electrode

The inherent limitations of metallic electrodes, specifically in terms of ionic conductivity and TPB, have prompted a shift towards the investigation of composite electrodes within academic research. Of particular interest are metal–ceramic composite electrodes, which effectively combine the catalytic prowess of metals with the ionic conductivity of ceramic oxides, while maintaining compatibility with electrolytes. These unique characteristics position them as promising materials for employment in SOFC anodes and SOEC fuel electrodes. Notably, metal–ceramic composite

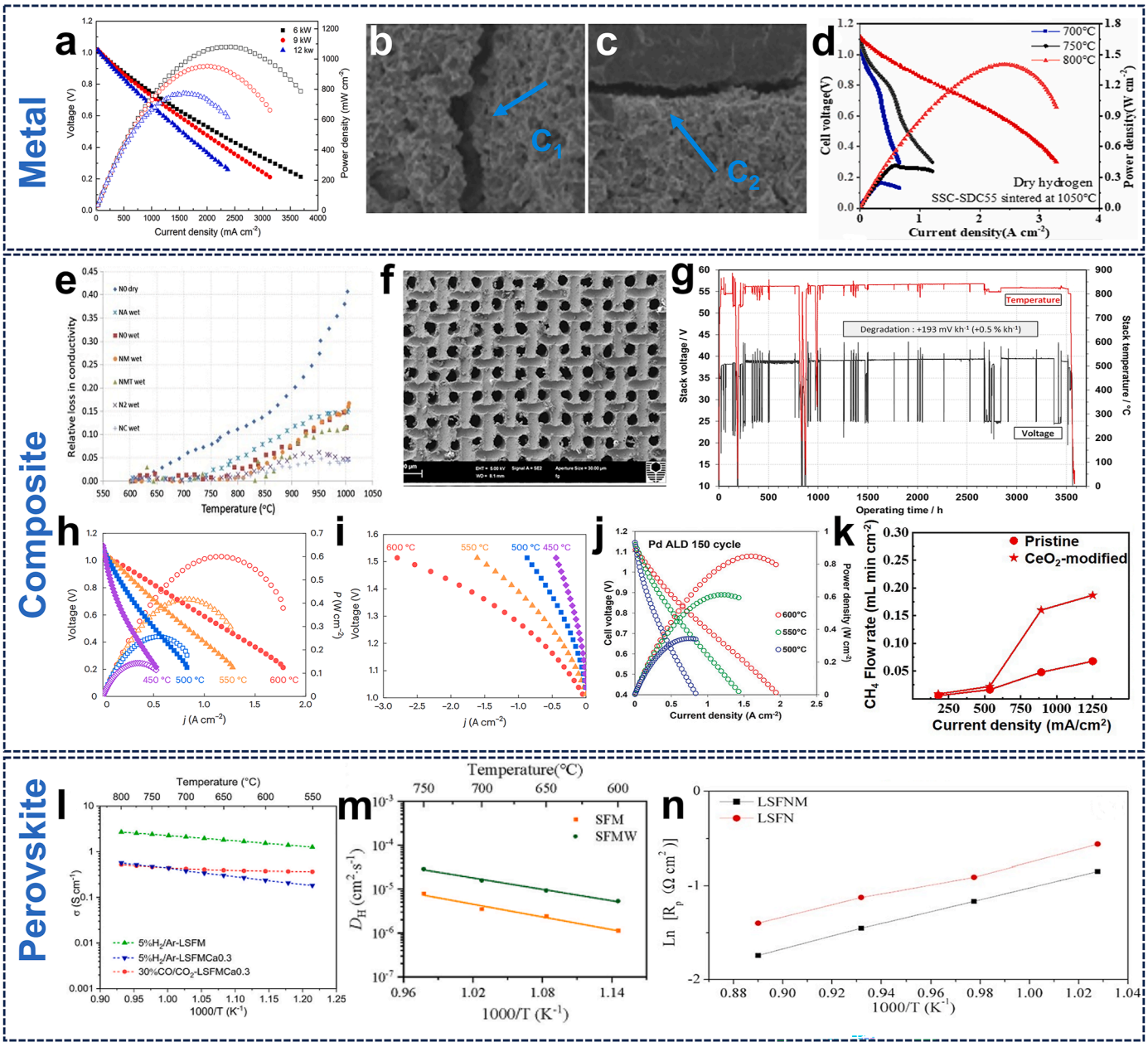

Fig. 9. (a) Current density of Ni electrode at 700 °C. Reproduced with permission [127]. Copyright 2023, Wiley-VCH. (b, c) Cracking and interfacial delamination resulting from TEC mismatch between the nickel-based electrode and the electrolyte. Reproduced with permission [128]. Copyright 2021, Springer Nature. (d) I–V–P curves of Ni–Fe alloy-supported SOFC from 700 to 800 °C. Reproduced with permission [129]. Copyright 2025, Elsevier. (e) Conductivity of Ni-YSZ composite electrode under different atmospheres. Reproduced with permission [135]. Copyright 2012, Elsevier. (f) Surface SEM images of Ni-YSZ microchannel electrodes. Reproduced with permission [138]. Copyright 2016, Elsevier. (g) Long-term behavior of the SOC stack during 3570 h of SOEC operation at 820 °C. Reproduced with permission [143]. Copyright 2020, Wiley-VCH GmbH. (h) I–V–P curves of SOFC with Ni-BZCY from 400 to 600 °C. (i) I–V–P curves of SOEC with Ni-BZCY at 1.3 V from 400 to 600 °C. Reproduced with permission [150]. Copyright 2025, Springer. (j) I–V–P curves of PCFC with Pd ALD Ni-BZCYYb in a temperature range of 500–600 °C. Reproduced with permission [151]. Copyright 2023, Wiley-VCH GmbH. (k) Flow rates of CH_4 under different electrolysis currents for cells with pristine and CeO_2-modified fuel electrodes. Reproduced with permission [152]. Copyright 2023, Royal Society of Chemistry. (l) Electrical conductivities between 550 and 800 °C for LSFM and LSFMCa0.3 in 5% H_2–95% Ar and 30% CO–70% CO_2 atmospheres. Reproduced with permission [157]. Copyright 2022, Elsevier. (m) Diffusion coefficient of protons of SFMW with temperature. Reproduced with permission [159]. Copyright 2024, Elsevier. (n) Dependence of Rp for symmetrical electrodes LSFN and LSFNM with temperature. Reproduced with permission [159]. Copyright 2018, Elsevier.

electrodes designed for oxygen ion conduction have garnered significant attention, with prominent examples including Ni-YSZ and Ni-GDC composites. In such composites, the non-precious metal Ni acts as an electronic conductor and an active catalytic element, whereas YSZ and GDC contribute to the O^{2-} conductivity and modulate the thermal expansion coefficient [131]. In addition, CeO_2-based materials also show good resistance to carbon accumulation, allowing them to use hydrocarbon fuels directly [132]. The interface between nickel and YSZ/GDC is extended at the TPB, providing active sites for electrochemical reduction [133]. The metal–ceramic matrix material has a porous architecture, commonly fulfilling a dual role as the support and functional layers in SOFCs/SOECs. The support layer, which is of sub-millimeter thickness, incorporates large pores designed for gas diffusion, whereas the functional layer, measuring approximately 5–10 μm in thickness, is characterized by an intricate microstructure and an extensive specific surface area that facilitates electrochemical reactions.

The Ni-YSZ support layer is typically synthesized through a solid-state reaction method, which possesses a very high electrical conductivity. The excessive addition of pore-forming agents can suppress the enhancement of electrical conductivity. The composite ratio of Ni to YSZ also affects the material's electrical conductivity, and as the mass percentage of Ni increases, the conductivity significantly improves [134]. Additionally, the Ni-YSZ composite electrode exhibits considerable performance differences under various operating temperatures and atmospheres: its electrical conductivity in dry/wet reducing atmospheres at 600 °C is 1100/1200 S/cm, respectively. However, the electrical conductivity drops sharply in humid atmospheres at temperatures above 800 °C and in dry atmospheres at 850 °C (Fig. 9e) [135]. Regarding variations in the TEC, the increase in pore-forming agents and the ratio of Ni both lead to an increase in the material's TEC. For instance, the addition of a certain mass percentage of pore-forming agents to Ni-YSZ increases its initial TEC value from 11.88×10^{-6} K^{-1} to 12.23×10^{-6} K^{-1} [136]. The Ni-YSZ functional layer can be optimized by controlling the size and distribution of Ni and YSZ particles, as well as the degree of structural distortion, thereby increasing the length of the TPB and the number of active sites for electrochemical reactions [137]. Furthermore, to enhance the electrolysis performance of CO_2 in the Ni-YSZ cathode, a CeO_2 functional layer is introduced into the Ni-YSZ cathode. The uniform catalyst nanoparticle coating formed by the CeO_2 colloid can more firmly adhere to the electrode support, resulting in lower Rp and higher ionic conductivity. For example, Dong et al. developed a hierarchically ordered porous SOEC fuel electrode structure for CO_2 electrolysis, which features a porous functional layer for reducing CO_2 and numerous parallel microchannels conducive to gas diffusion (Fig. 9f) [138]. Costa-Nunes et al., on the other hand, prepared a Cu–CeO_2–Ni-YSZ composite anode for SOFCs operating on CH_4, which demonstrated better anti-coking performance than pure Ni-YSZ anodes [139].

In comparison to YSZ, GDC demonstrates enhanced ionic and electronic conductivity within the cathode operational environment. GDC has the potential to supplant YSZ in the formation of a Ni-GDC cathode, where the proliferation of active sites extends from the TPB to the GDC/gas interface, thus conferring superior electrochemical performance [140]. Thin-film SOFCs, which utilize a NiO-GDC anode, have realized an output power density of 439.5 mW/cm^2 at 500 °C [141]. In practical applications, the addition of a small amount of hematite-type oxide promoters can form new active sites on YSZ, significantly improving anode performance and stability at 700 °C. For instance, Hibino and colleagues studied the direct electrochemical oxidation of methane, ethane, and propane with a Ru–Ni-GDC anode at 600 °C and found that when using dry methane, the maximum power density of the cell could reach 750 mW/cm^2 [142]. Furthermore, the GDC's exceptional oxygen storage capacity positively influences the mitigation of carbon deposition, thereby giving the Ni-GDC cathode exceptional anti-coking characteristics. During a 3370-h durability test, a SOEC equipped with a Ni-GDC cathode achieved a steam conversion rate of 70% at a current density of 0.52 A/cm^2 and a temperature of 820 °C, with a minimal degradation rate of 0.5% per 1000 h (Fig. 9g) [143]. It is noteworthy that GDC and YSZ exhibit chemical incompatibility at temperatures exceeding 1250 °C. Consequently, GDC can be utilized as an alternative to Ni-YSZ in the fabrication of SOFC/SOEC, permitting the cell preparation temperature to be maintained below 1250 °C [144].

Despite the favorable electrochemical performance and chemical stability of Ni-YSZ and Ni-GDC, these materials are susceptible to carbon accumulation when utilized with hydrocarbon fuels and encounter challenges related to thermal expansion coefficient mismatch [145]. To ameliorate these issues, strategies such as surface modification and microstructural optimization can be implemented. By fine-tuning the ratio of Ni to YSZ/GDC or incorporating dopants to enhance the resistance to carbon deposition and improve electrochemical stability, significant advancements can be achieved. Common enhancement techniques also encompass interface modulation and microstructural control. Employing these methodologies can augment the performance and longevity of Ni-YSZ and Ni-GDC as anodes in SOFCs. To circumvent carbon deposition, alternative metal–ceramic electrodes, including Cu-GDC and Ag-GDC, have been proposed for use in SOFCs/SOECs. However, the catalytic activity of Ag and Cu is relatively lower, which restricts their overall performance. Moreover, the low melting points of Ag and Cu can compromise the cathode's stability [146,147].

Beyond the infiltration of the aforementioned ceramic particles into the electrolyte scaffold, it is common practice to alloy metal elements into the electrolyte to augment its performance during electrolysis. The incorporation of Pr into CeO_2 significantly elevates the concentration of surface oxygen vacancies and the electronic conductivity thereof. Consequently, a Cu–$Pr_{0.1}Ce_{0.9}O_{2-\delta}$ anode demonstrates better performance than a Cu-GDC anode. Following a reduction treatment, Ni and Cu, when doped into CeO_2, can precipitate as nanoparticles on the CeO_2 surface, which enhances the catalytic activity and the electrolysis rate [148]. In the context of pure CO_2 electrolysis, single cells employing the $Ce_{0.9}(Cu_{1-x}Ni_x)_{0.1}O_{2-\delta}$ anode have exhibited the highest current density—exceeding 1.6 A/cm^2—at a temperature of 800 °C and a voltage of 1.6 V, which represents a nearly twofold enhancement in performance relative to cells utilizing a CeO_2 anode. This augmentation in electrolysis performance is attributed to the additional electrochemical reduction active sites present at the interfaces of metal nanoparticles/CeO_2/gas [149].

For H–SOFC/SOEC, the concept of the anode/fuel electrode is almost identical to that of oxygen ion conduction, which involves combining metals with proton-conducting electrolytes to construct metal–ceramic composite proton-conducting electrodes. For example, the Ni- BZCY ($BaZr_{0.1}Ce_{0.7}Y_{0.2}O_{3-\delta}$) composite electrode has demonstrated a peak power density of 0.6 W/cm^2 in the SOFC mode at 600 °C (Fig. 9h). In the SOEC mode, operating at 600 °C and 1.3 V, the maximum current density can reach −1.14 A/cm^2 (refer to Fig. 9i) [150]. The incorporation of precious metal elements in composite materials has been shown to enhance electrode performance. For instance, Pd-deposited Ni-BZCYYb ($BaZr_{0.2}Ce_{0.6}Y_{0.1}Yb_{0.1}O_{3-\delta}$) exhibited 1.3 times more power density than a non-deposited counterpart, achieving a power density of 0.85 W/cm^2 (Fig. 9j) [151]. Modifying the catalyst of the fuel electrode has emerged as an effective strategy to enhance SOEC performance. Ye et al. modified Ni-BZCYYb with CeO_2. Under operating conditions of 550 °C and 1250 mA/cm^2, the CeO_2-modified electrode demonstrated a threefold increase in CH_4 selectivity in PCEC compared to the standard electrode (Fig. 9k) [152].

To enhance the efficacy of metal–ceramic anode materials, fluorite electrolyte materials are fashioned into scaffolds and subsequently infiltrated with electronic conductive and electroactive constituents. Recently, porous YSZ scaffolds have been co-sintered with dense YSZ electrolytes and impregnated with solution-dispersed active nanoparticles of SFM to fabricate functional composite fuel electrode materials. Under operating conditions of 800 °C and 1.5 V, SOECs integrated with these composite fuel electrodes have achieved current densities of 1.10 A/cm^2 in pure CO_2 and 1.46 A/cm^2 in a 60% CO_2–40% H_2O mixture, rivaling the performance of SOECs equipped with mechanically mixed Ni-YSZ anodes, which demonstrated a current density of 1.3 A/cm^2 in a 50% CO_2–50% H_2O mixture at 800 °C and 1.5 V [153]. It is important to highlight that the incorporation of active nanoparticles is not confined to single-phase materials; instead, multi-phase components can also be introduced to bolster electrolysis performance. The co-infiltration of $(La_{0.75}Sr_{0.25})_{0.95}(Cr_{0.5}Mn_{0.5})O_{3-\delta}$ (LSCM) and GDC into the YSZ scaffold led to a significant reduction in Rp and enhanced stability due to CO_2 adsorption and CO_3^{2-} dissociation. These improvements are attributed to an increase in TPB and surface oxygen vacancies, as well as the mitigation of nanoparticle coarsening. Similarly, the co-infiltration of LSCF and GDC nanoparticles into the YSZ scaffold was shown to confer equivalent improvements in the performance and stability of SOECs [154].

4.2.3. Perovskite oxide

Perovskite-structured materials have demonstrated considerable potential as anode materials for SOFCs and SOECs. $La_{0.75}Sr_{0.25}Cr_{0.5}Mn_{0.5}O_{3-\delta}$ (LSCM), known for its superior redox stability and catalytic activity, is regarded as a highly promising candidate for anode materials. Comprehensive research has indicated that the electrical conductivity of LSCM escalates with an increase in temperature, and under humid hydrogen atmospheres, doping with Ce can further augment its conductivity. Although Sc doping may negatively influence conductivity, it has been shown to effectively diminish the Rp of the electrode, offering novel insights for electrode performance optimization [155]. In co-electrolysis operations, the Rp of LSCM electrodes tends to rise with the water vapor content, potentially posing challenges for the electrodes' long-term stability. $LaFeO_{3-\delta}$-based materials have exhibited high selectivity for CO_2 and resistance to carbon deposition in SOECs that utilize CO_2-containing fuels [156]. The performance of $LaFeO_{3-\delta}$-based electrodes has been notably enhanced through the incorporation of Sr and Mn, particularly at elevated temperatures, where $La_{0.6}Sr_{0.4}Fe_{0.8}Mn_{0.2}O_{3-\delta}$ has displayed exceptional electrical conductivity (Fig. 9l) [157]. The high current density and CO_2 reduction rate achieved by a LSFM-CuF/LSGM/BLC cell configuration during co-electrolysis further substantiate its application potential as an anode material. $La_{0.2}Sr_{0.8}TiO_{3-\delta}$ (LST) has also shown significant potential as an electrode material in SOECs. The conductivity of LST is substantially influenced by its reduction and oxidation states, and doping with Ca and Fe has been demonstrated to enhance its conductivity and catalytic activity. Specifically, the electrical conductivity of $La_{0.2}Sr_{0.25}Ca_{0.45}Ti_{0.95}Fe_{0.05}O_{3-\delta}$ has been observed to vary with the Ca content, providing a crucial reference for material design [158]. $Sr_2Fe_{1.5}Mo_{0.5}O_{6-\delta}$ (SFM), which has a double perovskite structure, has experienced enhanced fuel electrode performance through strategies such as doping and infiltration [159]. Doping of SFM with W results in greater electrochemical characteristics, with the bulk diffusion coefficient of protons reaching 2.86×10^{-5} cm/s at 750 °C (Fig. 9m) [160]. $SrFe_{0.9}Nb_{0.1}O_{3-\delta}$ when employed as an electrode material in SOFCs has demonstrated an increase in electrical conductivity through the substitution of manganese [161]. The A-site layered double perovskite $(PrBa)_{0.95}(Fe_{0.9}Mo_{0.1})_2O_{5+\delta}$ (PBFM) exhibits elevated electrical conductivity alongside commendable electrochemical performance [162]. The fabrication of the $La_{0.75}Sr_{0.25}Cr_{0.5}Mn_{0.5}O_{3-\delta}$-$Ce_{0.8}La_{0.2}O_{2-\delta}$ (LSCM-LDC) composite anode, realized via the sol-gel process, confers high electrical conductivity and superior electrochemical characteristics. The nanocomposite showed a significant improvement in performance, including up to a fivefold decrease in area-specific resistance and three orders of magnitude greater in-plane conductivity compared to reference single-phase materials. Additionally, remarkable electrochemical activity was observed under an oxidizing atmosphere, and the performance remained stable even after 400 h of thermal aging [163]. Molybdenum (Mo) doping in the perovskite $La_{0.6}Sr_{0.4}Fe_{0.9}Ni_{0.1}O_{3-\delta}$ (LSFNM), utilized as an anode in SOFCs, enhances electrical conductivity and concurrently diminishes Rp (Fig. 9n) [164]. The anode material $La_{0.7}Sr_{0.3}Cr_{0.5}Mn_{0.5}O_{3-\delta}$ for SOFCs, with Ni doping, facilitates an increase in oxygen vacancy formation, thus enhancing electroactivity and the cell's power density [165]. Studies on Sr-doped $LaCrO_3$ chromite (LSCr) as an anode material for SOFCs have highlighted the superior hexacoordinate capacity of Cr under conditions of oxygen deficiency, in addition to the catalytic performance enhancement conferred by cations with a low coordination number on LSCr [166].

4.3. Electrolyte

The electrolyte is a pivotal element in SOFCs and SOECs, fundamentally dictating their operational temperature and electrochemical characteristics. It is imperative for electrolyte materials to exhibit superior ionic conductivity, coupled with minimal electronic conductivity to preclude electrical current leakage. Furthermore, the electrolyte must maintain gas-tight integrity to segregate the electrode atmospheres, necessitating stability under both reductive and oxidative environments, as well as compatibility with electrode materials. The TEC of the electrolyte also must align with that of the electrodes. The efficacy of the electrolyte is contingent upon factors such as temperature, material attributes, and its thickness. Elevated operating temperatures augment the ionic conductivity of the material and stimulate thermally activated electrochemical reactions. In electrolyte-supported SOFCs/SOECs, the material of the electrolyte regulates thermal expansion across the multilayered cell architecture, preventing the delamination of other layers, like electrodes, due to thermal expansion discrepancies. Up to this point, the electrolyte materials employed in SOFCs and SOECs have been nearly identical. Electrolytes that facilitate the conduction of oxygen ions include stable zirconia, doped ceria, and perovskite materials such as $LaGaO_3$ and those with a pyrochlore structure. Conversely, proton-conducting electrolytes predominantly feature perovskite-structured materials like $BaZrO_3$, $BaCeO_3$, as well as heterogeneously structured electrolyte materials, which have garnered interest in recent years.

4.3.1. Oxygen ion-conducting electrolytes

i. Stable zirconia

Stable zirconia (ZrO_2) is renowned for its excellent chemical stability, high mechanical strength, and cost-effectiveness. Additionally, it exhibits high oxygen ion conductivity at elevated temperatures, making it a widely used electrolyte material in SOECs [167]. Pure zirconia undergoes phase transitions from monoclinic to tetragonal to cubic structures at different temperatures, with the cubic phase demonstrating the highest oxygen ion conductivity at high temperatures. To stabilize the cubic phase at lower temperatures and enhance conductivity, Y_2O_3 is commonly used as a dopant. Zirconia doped with 8 mol% Y_2O_3 (8-YSZ) has an oxygen ion conductivity of approximately 0.1 S/cm at 1000 °C (Fig. 10a) [168]. Despite the outstanding performance of YSZ at high temperatures, its conductivity at lower temperatures is compromised due to the low oxygen ion mobility, which restricts its application in low-temperature SOECs. To address this issue, various approaches have been explored to improve the performance of YSZ-based electrolytes. For instance, incorporating the semiconductor ZnO into the YSZ electrolyte layer to form heterojunctions can significantly enhance ionic conductivity at lower operating temperatures (Fig. 10b) [169]. Furthermore, the addition of trace amounts of copper oxide as a sintering aid not only promotes densification of the YSZ electrolyte but also increases the concentration of oxygen vacancies, thereby enhancing conductivity [170]. Scandium-stabilized zirconia (ScSZ) has garnered attention due to its higher oxygen ion conductivity, which is three times that of YSZ at 1000 °C. The primary reason for the increased conductivity is that Sc and Zr have similar ionic radii, which facilitates an increase in oxygen vacancies within the lattice [171]. However, due to its higher cost, lower mechanical strength, and poorer sintering activity, ScSZ has not yet been widely adopted in SOEC applications [172].

Under the operating conditions of SOECs, electrolyte stability is of utmost importance. Studies have demonstrated that a minimum doping level of 8 mol% Y_2O_3 is required to ensure the stability of the electrolyte and to prevent precipitation around the electrodes under the influence of an electric field (Fig. 10c) [173]. Furthermore, the conductivity of 8-YSZ at 800 °C reaches 7.37 S/cm. These properties further substantiate the potential application value of YSZ in SOFCs [174].

ii. Doped CeO_2

Doped CeO_2 electrolyte materials are renowned for their exceptional ionic conductivity, playing a pivotal role in SOFCs and SOECs [175, 176]. The O^{2-} conductivity of these materials is typically enhanced by doping with divalent or trivalent cations such as Y, La, Gd, and Sm [177–179]. Notably, GDC and SDC, whose ionic radii are similar to that of cerium, exhibit higher conductivity compared to YSZ, particularly in the intermediate temperature range (Fig. 10d) [180–182]. Consequently, they have been extensively researched and utilized as electrolyte materials in SOFCs.

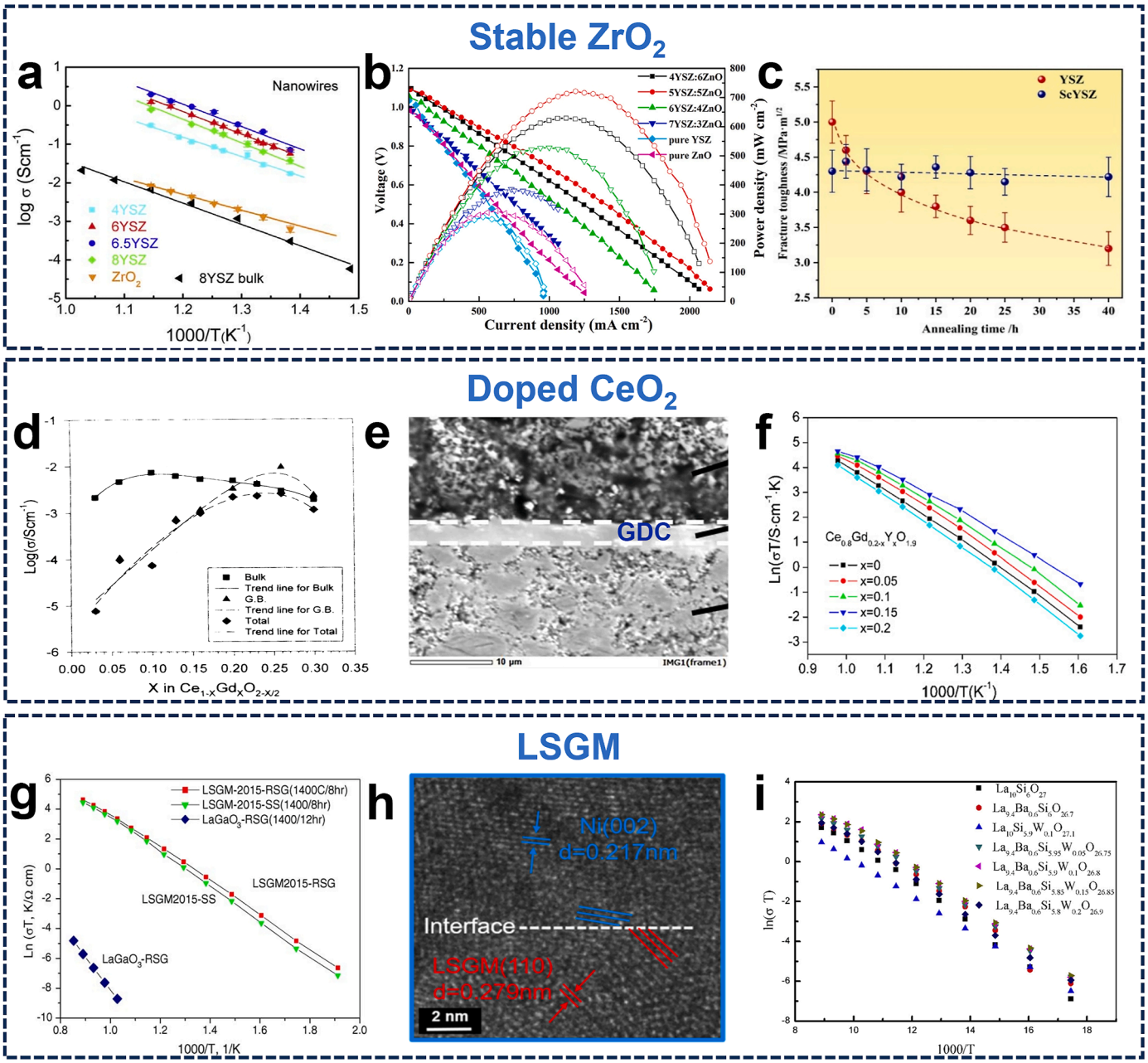

Fig. 10. (a) Ionic conductivity of 8-YSZ. Reproduced with permission [168]. Copyright 2017, Elsevier. (b) Power density of YSZ-ZnO heterojunction electrolytes. Reproduced with permission [169]. Copyright 2023, Elsevier. (c) Thermal stability of YSZ and ScYSZ. Reproduced with permission [173]. Copyright 2022, MDPI. (d) Conductivity of GDC. Reproduced with permission [180]. Copyright 2012, Elsevier. (e) GDC as a barrier layer between the Ni anode and YSZ. Reproduced with permission [184]. Copyright 2019, Elsevier. (f) Conductivity of $Ce_{0.8}Gd_{0.2-x}Y_xO_{1.9}$. Reproduced with permission [186]. Copyright 2008, Elsevier. (g) Conductivity of $La_{1-x}Sr_xGa_{1-y}Mg_yO_{3-\delta}$. Reproduced with permission [187]. Copyright 2006, Elsevier. (h) TEM image of the hetero-interface between Ni and LSGM. Reproduced with permission [188]. Copyright 2024, Elsevier. (i) Conductivity of $La_8B_2Si_6O_{26}$. Reproduced with permission [192]. Copyright 2020, Elsevier.

GDC and SDC, renowned for their elevated conductivity, favorable thermal matching, and chemical compatibility, may exhibit characteristics of MIECs under reducing conditions at low pO_2. This transformation is attributed to the reduction of Ce^{4+} to Ce^{3+}, which can precipitate a deterioration in cell performance. Such deterioration is manifested as diminished OCV and Faradaic efficiency, alongside potential structural disintegration [183]. Furthermore, in comparison to YSZ, ceria-doped electrolyte materials necessitate higher sintering temperatures to achieve densification, consequently amplifying energy expenditure during the fabrication process. Within the context of SOEC applications, the direct utilization of GDC or SDC as the electrolyte, circumventing the need for YSZ to impede electronic conduction, is infrequent. This rarity stems from the chemical incompatibility between the majority of perovskite electrode materials and YSZ electrolytes; hence, GDC or SDC are typically employed as barrier layers to preclude chemical reactions at the electrode–YSZ interface (Fig. 10e) [184]. In the intermediate temperature range of 600–800 °C, GDC surpasses yttria-stabilized zirconia in terms of ionic conductivity, with the conductivity reaching its zenith at approximately 15 mol% dopant concentration (Fig. 10f) [185]. Despite the physical property benefits conferred by ceria-doped electrolytes, 8-YSZ persists as the premier material option on account of its economic viability. A plethora of tests executed in various research institutions have substantiated the enduring stability of SOFCs employing 8-YSZ electrolytes.

iii. Doped $LaGaO_3$ with perovskite structure

The perovskite oxide $LaGaO_3$, once doped with Sr and Mg to form the compound $La_{1-x}Sr_xGa_{1-y}Mg_yO_{3-\delta}$ (LSGM), manifests superior O^{2-} conductivity relative to YSZ, especially within the intermediate temperature range of 400–800 °C. This attribute renders LSGM a prospective candidate for the electrolyte in intermediate-temperature SOECs (Fig. 10g) [186]. Notably, the compositions $La_{0.85}Sr_{0.15}Ga_{0.8}Mg_{0.2}O_{3-\delta}$ and $La_{0.8}Sr_{0.2}Ga_{0.85}Mg_{0.15}O_{3-\delta}$ achieve an oxygen ion conductivity of 0.15 S/cm at 800 °C, a value comparable to that of YSZ at 1000 °C [186,187]. However, the chemical incompatibility of LSGM with nickel-based electrode materials at elevated temperatures, coupled with the propensity for Sr segregation under high current densities, can result in diminished cell performance (Fig. 10h) [188]. Moreover, the mechanical robustness of LSGM does not surpass that of conventional YSZ and GDC electrolyte materials, presenting challenges for its practical utilization [189]. In parallel, lanthanum-based apatite structures, exemplified by $La_8B_2Si_6O_{26}$ (B =Mg, Ca, Sr, Ba), have attracted considerable interest due to their elevated conductivity at moderate temperatures. Lanthanum silicate apatite, possessing a hexagonal structure (*P*63/m), demonstrates superior oxide ion conductivity compared to YSZ within the intermediate temperature spectrum (Fig. 10i) [190,191]. This conductivity can be further enhanced by the strategic doping of cations at the La and B sites. Despite the many advantageous properties of apatite materials, though, their synthesis necessitates high-temperature processes, thereby escalating the costs and complexity associated with manufacturing [192,193].

4.3.2. Proton-conducting electrolytes

In recent years, there has been significant research interest in Ba-based perovskite proton-conducting electrolyte materials [194,195]. Section 2.2.2 delved into the mechanism of proton conduction, highlighting the significance of the B-site cations within the [BO_6] octahedra as the crucial component for facilitating proton conduction. The proton conduction performance of a material can be directly affected by B-site cations through their influence on the material's electronic structure and crystal cell parameters. Researchers have thus focused on exploring strategies of B-site doping to enhance the proton conductivity of Ba-based perovskite electrolyte materials. Taking the most representative Ba-based perovskite electrolytes, $BaZrO_3$ and $BaCeO_3$, as an example, $BaZrO_3$ is characterized by a stable crystal structure and robust chemical stability, yet it exhibits inferior sintering activity and relatively low proton conductivity [196]. To augment its conductivity, researchers have employed strategies such as B-site ion doping and the utilization of sintering aids. For instance, ytterbium-doped $BaZrO_3$ (BZY), when sintered at 1500 °C with the incorporation of CuO as a sintering aid, realizes an ionic conductivity of 5.5×10^{-3} S/cm at 600 °C (Fig. 11a) [197]. Conversely, $BaCeO_3$ demonstrates superior proton conductivity, reaching $\geq 10^{-2}$ S/cm at 600 °C. Nevertheless, its chemical stability in the presence of H_2O and CO_2 is compromised, with a propensity to form secondary phases such as $BaCO_3$ and CeO_2, which restricts its application in H–SOFCs. The incorporation of rare earth elements can enhance the material's oxygen vacancy concentration, thereby improving proton conduction [198]. For example, yttrium-doped $BaCe_{0.9}Y_{0.1}O_{2.95}$ (BCY) electrolyte material exhibits a proton conductivity of 0.6 S/m at 500 °C, escalating to 1.38 S/m at 700 °C [199]. Gawel et al. observed that $BaCe_{0.85}Y_{0.15}O_{3-\delta}$ (BCY15) is susceptible to decomposition in environments containing CO_2–H_2O or N_2–H_2O; however, its stability can be markedly enhanced through the formation of composites with $Ce_{0.85}Y_{0.15}O_{2-\delta}$ (YDC15) [200]. Capitalizing on the high stability of $BaZrO_3$ and the high proton conductivity of $BaCeO_3$, researchers have developed solid-solution materials of $BaZrO_3$–$BaCeO_3$, further augmenting proton conduction properties via additional ion doping [201]. For instance, yttrium-doped $Ba_{0.98}Ce_{0.6}Zr_{0.2}Y_{0.2}O_{3-\delta}$ manifests an OCV of approximately 1.13 eV in a humid fuel atmosphere (H_2/N_2–2.76%H_2O) at 600 °C, coupled with a maximum power density of 493 mW/cm^2 [202]. The $BaZr_{0.55}Ce_{0.2}Y_{0.25}O_{3-\delta}$ (BCZY-Y25) achieves a conductivity of 9.4×10^{-3} S/cm in a humid atmosphere at 600 °C (Fig. 11b). [203]. Kim et al. reported that the electrolyte material $BaZr_{0.5}Ce_{0.3}Y_{0.1}Yb_{0.1}O_{3-\delta}$ (BZ5C3YYb) attains pure proton conductivity in a humid 5% H_2/Ar atmosphere at 600 °C, while retaining excellent chemical stability even when subjected to a CO_2 atmosphere (Fig. 11c) [204].

The manipulation of oxygen defect formation energy and a proton migration energy barrier within the electrolyte through A-site doping will be instrumental in enhancing proton conductivity. Dudek et al. performed A-site Ca ion substitution on $BaCeO_3$ and $BaZrO_3$, yielding samples $(Ba_{0.95}Ca_{0.05})(Zr_{0.9}Y_{0.1})O_3$ and $(Ba_{0.95}Ca_{0.05})(Ce_{0.9}Y_{0.1})O_3$ that demonstrated proton conductivities of 1.24×10^{-3} S/cm and 8.21×10^{-3} S/cm, respectively, under humid 5% H_2/Ar conditions at 600 °C (Fig. 11d) [205]. Murali et al. explored the proton conductivity of $Ba_{1-x}Sr_xCe_{0.65}Zr_{0.25}Pr_{0.1}O_{3-\delta}$ at 500 °C, identifying the optimal sample $Ba_{0.92}Sr_{0.08}Ce_{0.65}Zr_{0.25}Pr_{0.1}O_{3-\delta}$, which has a proton conductivity of 3.41×10^{-6} S/cm under humid conditions (Fig. 11e) [206]. In a study by Luo et al., it was observed that the introduction of 1% Ca doping in $Ba_{0.99}Ca_{0.01}Ce_{0.8}Gd_{0.2}O_{3-\delta}$ led to a significant enhancement in conductivity. Specifically, the conductivity increased from 1.1×10^{-3} S/cm to 2.8×10^{-3} S/cm at 400 °C, and from 1.7×10^{-2} S/cm to 3.1×10^{-3} S/cm at 800 °C (Fig. 11f) [207]. Certain perovskite materials with mixed ionic conduction properties are also considered potential candidates for proton-conducting electrolytes. For example, $SrTiO_3$ exhibits ionic conductivities of 0.16, 0.21, and 0.24 S/cm at 450, 500, and 550 °C, respectively, alongside maximum power densities of 0.3, 0.44, and 0.62 W/cm^2 under humid air [208]. Kato et al. reported that $La_{0.675}Sr_{0.325}Sc_{0.99}Al_{0.01}O_3$, a material with mixed ionic conduction, achieved a power density of 158 mW/cm^2 under humid conditions at 1000 °C, and below 700 °C, the material predominantly displayed proton conduction [209]. It is noteworthy that the novel perovskite electrolyte $SmNiO_3$, engineered by leveraging the Mott transition induced by strong electron correlation to inhibit electronic conduction, attained a maximum output power of 225 mW/cm^2 and an OCV of 1.03 V at 500 °C, offering a high-performance material design strategy for low-temperature SOFCs [210].

Beyond single perovskite-structured materials, the potential of certain double perovskite-structured materials as proton-conducting electrolytes is increasingly gaining the attention of researchers. Shi et al. focused on the double perovskite-structured $RbBiNb_2O_7$ and used first-principles calculations to delve into its theoretical potential as a proton-conducting electrolyte material. The findings indicated that $RbBiNb_2O_7$ demonstrates superior proton transport behavior on the a–b plane, outperforming numerous perovskite oxides, including $CaTiO_3$, $CaZrO_3$, and $SrZrO_3$. The diffusion of protons between the Rb–O and Nb–O layers is impeded by a relatively high energy barrier of 0.86 eV, while the pronounced octahedral tilting in $RbBiNb_2O_7$ facilitates proton transport (Fig. 11g) [211].

$Ba_2In_2O_5$ is a unique material that exhibits a brownmillerite structure at room temperature but transitions to a cubic perovskite phase with enhanced proton conductivity at elevated temperatures. Leveraging this property, Abubakar et al. stabilized the cubic phase of $Ba_2In_2O_5$ at room temperature through doping with boron and silicon, thereby enabling B/Si-doped $Ba_2In_2O_5$ to primarily conduct protons at medium and low temperatures. Specifically, $Ba_2In_{1.4}B_{0.4}Si_{0.2}O_{5.1}$ achieved an electrical conductivity of 1.05×10^{-3} S/cm at 400 °C. Concurrently, the incorporation of B enhanced the material's density and grain morphology (Fig. 11h) [212]. The doping of Sr and Fe can also modulate the proton conduction properties of $Ba_2In_2O_5$. Wang et al. reported on the perovskite material $BaSrInFeO_5$ (BSIF), which predominantly exhibits proton conduction within the temperature range of 500–600 °C; a H–SOFC using the BSIF generates a relatively high peak power density of 1192 mW/cm^2 at 700 °C, which is superior to many cells in previous reports (Fig. 11i) [213].

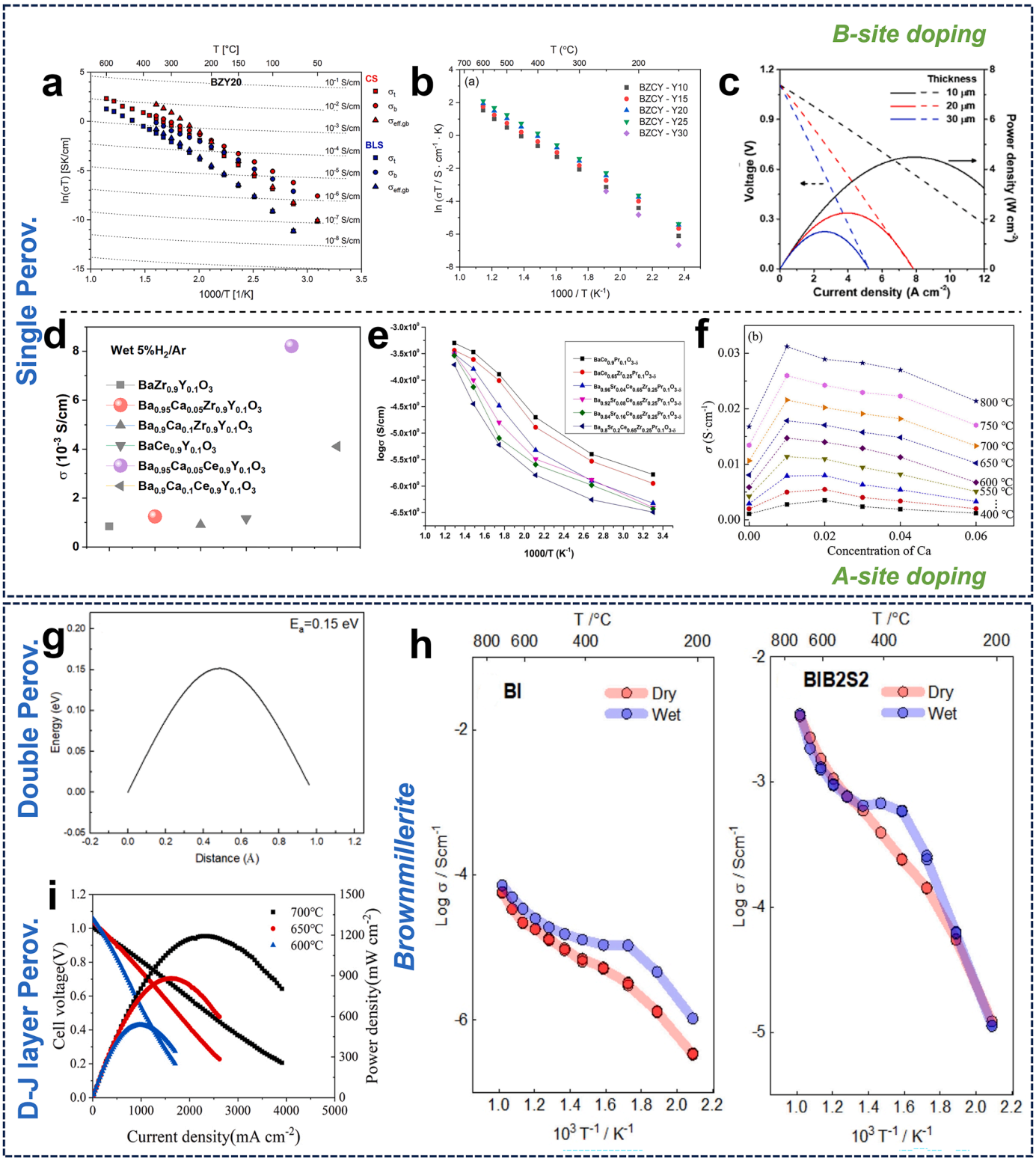

Fig. 11. (a) Conductivity of $BaZrO_3$. Reproduced with permission [197]. Copyright 2025, Elsevier. (b) Power density of BZCY. Reproduced with permission [203]. Copyright 2025, Royal Society of Chemistry. (c) I–V–P curves of BZ5C3YYb in a humid 5% H_2/Ar atmosphere at 600 °C. Reproduced with permission [204]. Copyright 2023, Elsevier. (d) Conductivity of $Ba_{1-x}Ca_x$ $(Ce/Zr)_{0.9}Y_{0.1}O_3$ (x = 0, 0.05, 0.1) in the wet 5% H_2/Ar at 600 °C. (e) Conductivity of $Ba_{1-x}Sr_xCe_{0.65}Zr_{0.25}Pr_{0.1}O_{3-\delta}$ ($0 \le x \le 0.2$) in wet air. Reproduced with permission [206]. Copyright 2018, Elsevier. (f) Conductivity of $Ba_{1-x}Ca_x$ $Ce_{0.8}Gd_{0.2}O_{3-\delta}$ with concentration of Ca in the temperature range 400–800 °C. Reproduced with permission [207]. Copyright 2023, Elsevier. (g) Power density of $RbBiNb_2O_7$. Reproduced with permission [211]. Copyright 2021, MDPI. (h) Conductivity of $Ba_2In_2O_5$ and $Ba_2In_{1.4}B_{0.4}Si_{0.2}O_{5.1}$ in dry/wet atmosphere. Reproduced with permission [212]. Copyright 2023, Elsevier. (i) I–V and I–P curves of $BaSrInFeO_5$. Reproduced with permission [213]. Copyright 2023, Elsevier.

4.4. Connectors and other components

In SOFC and SOEC technologies, interconnects play a critical role. They not only provide electrical connections between individual cells but also ensure the effective separation of oxygen, fuel gases, and steam during the electrolysis process. Ideal interconnect materials should possess excellent electrical conductivity, structural stability under high-temperature conditions, and a TEC compatible with other cell components. Based on these criteria, iron-chromium alloys, such as Crofer22 APU, SUS304, and SUS430, have become mainstream choices for manufacturing interconnects. However, these materials are prone to the formation of chromium oxide (CrO) scales during prolonged high-temperature operation, which can lead to performance degradation and may also cause material embrittlement and fracture. The interaction between chromium oxide scales and electrode materials also can negatively impact electrode performance. To address these issues, researchers are actively developing a new generation of interconnect materials, aimed at enhancing durability, reducing manufacturing costs, and optimizing compatibility with electrode materials.

4.4.1. Iron-based alloys

Iron-based alloys have garnered significant research attention owing to their superior ductility, machinability, and economic viability. The Fe/Cr/Mn and Fe/Cr/W alloy systems are exemplary, characterized by relatively low TEC and a minimum chromium content exceeding 17%, which facilitates the formation of a corrosion-resistant Cr_2O_3 oxide layer. The issue of chromium element volatilization warrants attention, though [214]. Elemental doping has been demonstrated to promote the growth of the metal oxide layer in alloys such as Fe–Cr doped with rare earth element Y and transition metal element X (X = Co, Mn, Ni, Cu). However, while doping elements enhance oxide layer growth, they also lead to a swift increase in alloy weight, rendering these materials unsuitable for interconnect applications [215]. The incorporation of Y can mitigate the oxidation kinetics of iron alloys, resulting in reduced weight gain for alloys Fe/Cr/Y, which positions them as potential candidates for connector materials. Doping Fe/Cr alloy with multiple elements, such as Co, Ni, Mn, and Al, can lead to the formation of various oxide layers, including spinel layers, on top of the Cr_2O_3 layer. This can effectively decrease resistance and greatly enhance corrosion resistance. It has been determined that the oxidation-induced surface changes correlate with each element's interactive tendencies with the environment, guided by thermodynamics. Initial oxidation stages follow atomic size and redox potential, with the latter becoming dominant over time, causing composition inversion [216]. To bolster the performance of interconnect materials, it is imperative to conduct further research on the thermal expansion and chemical stability of Fe/Cr alloys. This will be covered in more detail in a later section.

4.4.2. Cr-based alloys

Cr-based alloys have garnered considerable attention for their superior resistance to oxidation and corrosion, positioning them as prime candidates for interconnect materials in SOFCs and SOECs. The Cr_2O_3 oxide layer, which forms at elevated temperatures, exhibits a relatively low resistivity, and its propensity for spallation occurs at temperatures significantly higher than those encountered in the operation of these cells, thus underscoring its potential benefits in high-temperature settings. Moreover, the thermal expansion coefficient of these alloys aligns closely with that of electrolytes such as YSZ, mitigating thermal stress and bolstering structural integrity [217]. However, the electronic conductivity of Cr_2O_3 may be adversely affected at high temperatures due to accelerated growth, particularly when the diffusion rate of chromium surpasses that of aluminum at temperatures exceeding 1000K [218]. Consequently, a delicate balance must be achieved between enhancing conductivity and curbing the proliferation of the oxide layer. Commercial chromium alloys designed for high-temperature applications often incorporate trace amounts of rare earth elements, including Y, Ce, La, and Zr, which effectively fortify the alloy's resistance to oxidation by retarding the kinetics of the oxidation process [219]. Despite the commercial advances of chromium-based alloys for SOFC interconnects, their utility is predominantly confined to mid-temperature operations (below 700 °C), with lingering concerns about their long-term stability. At temperatures surpassing 800 °C, the oxidation resistance, even in the presence of active elements, is inadequate for dependable service [220]. Additionally, volatile hexavalent Cr^{6+} forms within the SOFC environment; this is a byproduct of chromium-based alloys that can severely compromise the cell's electrochemical performance [221]. To dampen chromium's volatility and make it more chemically compatible with other SOFC components, the application of doped $LaCrO_3$ perovskite coatings is a promising strategy. Research has demonstrated that the deposition of a strontium-doped $LaCrO_3$ ceramic protective layer on a $Cr/5Fe/Y_2O_3$ interconnect substantially diminishes the chromium evaporation rate at 950 °C [222].

4.5. Property degradation and resolution strategies

4.5.1. Degradation and corrosion

The materials and operating conditions of solid oxide electrochemical systems are highly intricate, and many well-established grid integration and industrial applications are hindered by various degradation processes; these can manifest in the form of solid–solid, solid–gas, or solid–liquid reactions [223]. The lifespan and reliability challenges for SOFCs and SOECs primarily revolve around the volatility of electrode materials and their vulnerability to poisoning in different gaseous environments. Volatility, an inherent characteristic of the materials themselves, poses an intrinsic challenge, while susceptibility to poisoning stems from external factors such as the operating environment. To optimize the overall performance of solid-state electrochemical devices, it is therefore imperative to develop materials with resilience to poisoning and minimal volatility.

i. Electrode volatilization and interface degradation

The volatility of electrode materials is primarily due to the high vapor pressure of certain components within these materials under high-temperature operating conditions, which facilitates their evaporation at elevated temperatures. For instance, in SOFCs and SOECs, commonly used nickel-based electrode materials are prone to nickel vaporization at high temperatures. This volatilization loosens the electrode structure, thereby reducing the electrode's electronic conductivity and catalytic activity. Consequently, the Rp of the cell increases, resulting in decreased output power and reduced electrolysis efficiency (Fig. 12a) [224]. For example, although GDC can effectively block vapor diffusion when used as an interlayer, lanthanum and strontium from lanthanum strontium manganite (LSM) and lanthanum strontium cobalt ferrite (LSCF) can still diffuse through grain boundaries and deposit at the YSZ/GDC interface. This deposition forms compounds such as $La_2Zr_2O_7$ or $SrZrO_3$, which increase local surface resistance, leading to localized joule heating and non-uniform temperature distribution, and ultimately damaging the interface (Fig. 12b) [225]. These surface chemical changes not only affect the performance of the electrode itself but also lead to the degradation of the electrode–electrolyte interface, resulting in a continuous decline in cell performance.

Moreover, the volatility of electrode materials can cause the accumulation of specific cations on the surface, thereby altering the physicochemical and catalytic properties of the oxide materials [226–228]. This segregation phenomenon is primarily related to cations such as Sr and Ba, which tend to accumulate on the electrode surface in high-temperature oxidizing atmospheres, forming oxides like SrO or BaO. The formation of these oxides not only alters the chemical composition and structure of the electrode surface but also may affect the electrode's catalytic activity and electrochemical performance (Fig. 12c) [229]. The microscopic mechanisms of segregation involve multiple factors, including elastic energy, electrostatic interactions, and lattice strain. In perovskite-structured oxides, the size differences

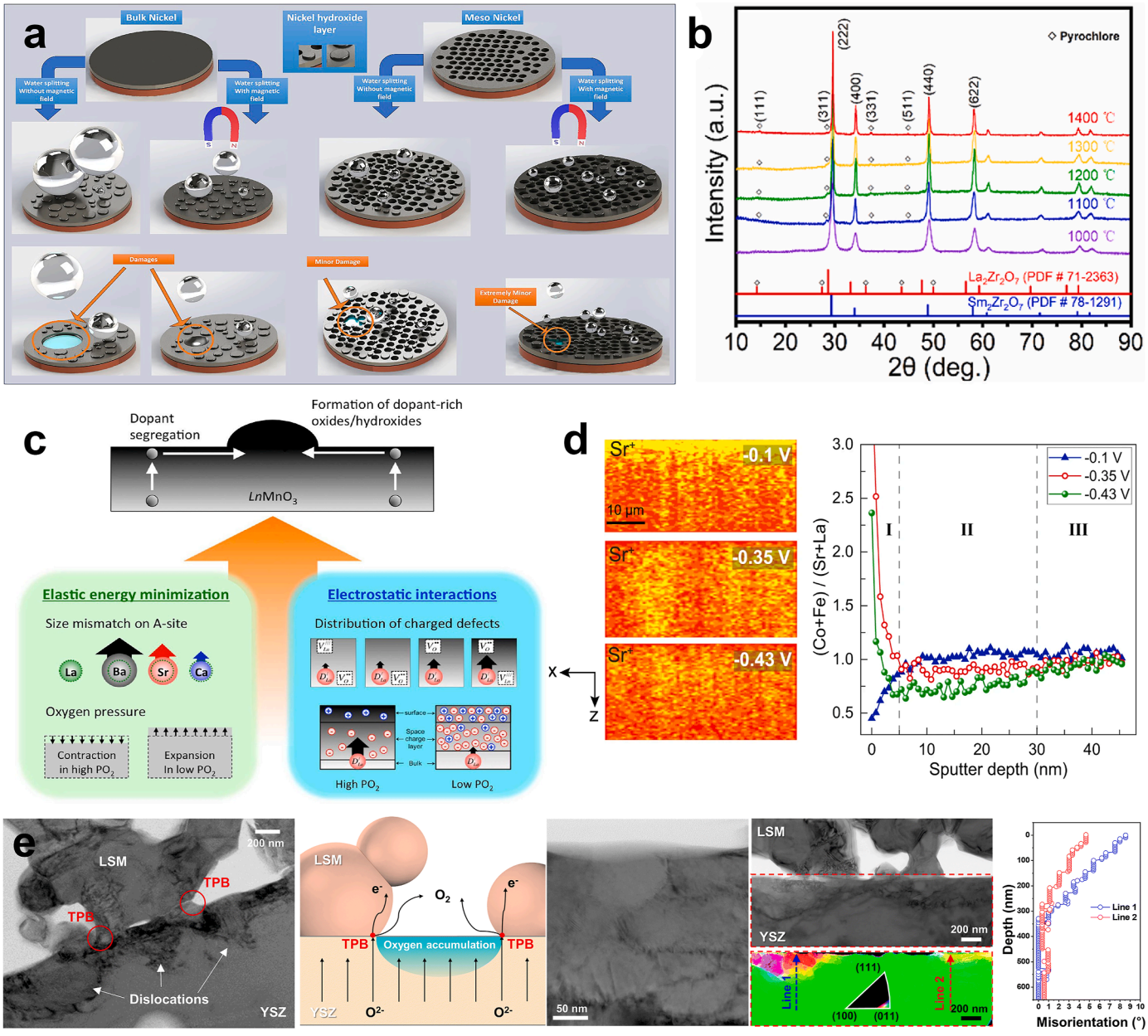

Fig. 12. (a) Schematic representation of the types of damage observed on electrode surfaces. Reproduced with permission [224]. Copyright 2025, Springer. (b) $La_2Zr_2O_7$ formed by the deposition of La on the grain boundaries. Reproduced with permission [225]. (c) Illustration of the mechanisms that drive dopant segregation to the surface of $LnMnO_3$. Reproduced with permission [229]. Copyright 2013, American Chemical Society. (d) Reconstructed cross-sectional images of −0.1, −0.35, and −0.43 V electrodes using Bi_3^+ as primary and Ar^+ as sputtering ion guns. Depth profile of (Co + Fe)/(La + Sr) for −0.1, −0.35, and −0.43 V LSCF-SDC electrodes. Reproduced with permission [230]. Copyright 2023, Springer. Copyright 2023, Elsevier. (e) Interfacial defects formed by the injection of excess oxygen ions. Reproduced with permission [234]. Copyright 2024, Royal Society of Chemistry.

between A-site cations (such as Sr) and B-site cations (such as Co or Fe) can lead to lattice distortion, resulting in elastic energy. To reduce this elastic energy, cations tend to migrate toward the surface, forming a surface reconstruction to Ruddlesden–Popper phases, thereby minimizing lattice distortion (Fig. 12d) [230]. Electrostatic interactions also play an important role in cation segregation. In oxidizing atmospheres, the concentration of oxygen vacancies at the electrode surface is relatively high; these positively charged oxygen vacancies can attract negatively charged inert cations, promoting their migration to the surface [231]. Moisture on the electrode surface at high temperatures can further promote the segregation of inert cations. The presence of moisture increases the concentration of oxygen vacancies, enhancing electrostatic interactions and facilitating the migration of cations like Sr to the surface [232]. The evolution mechanisms of the electrode surface in high-temperature, water-containing, oxidizing atmospheres are a complex, dynamic process that involves surface reconstruction, phase separation, and morphological changes.

Another major factor contributing to the degradation of the electrode–electrolyte interface, as mentioned in Section 4, is TEC mismatching. The difference in TEC between the electrode and the electrolyte causes varying degrees of deformation under thermal stress during the operation of SOFCs or SOECs, thereby leading to interface degradation [233]. Chemical reactions at the interface can also cause changes in the interfacial structure. For example, the high external voltage and electrode polarization in SOECs make the interface more susceptible to delamination failure. Studies have

shown that when electrolyte materials such as SDC operate under electrolysis conditions, the chemical stress at the interface significantly increases with changes in temperature, oxygen partial pressure, and external voltage. Specifically, an increase in voltage, a decrease in oxygen partial pressure, and an increase in temperature all reduce the electrolyte's mechanical stability, with oxygen partial pressure and temperature exhibiting an exponential relationship with the maximum tensile stress (Fig. 12e) [234]. This interface degradation can lead to a rapid decline in cell performance and may even cause cell failure.

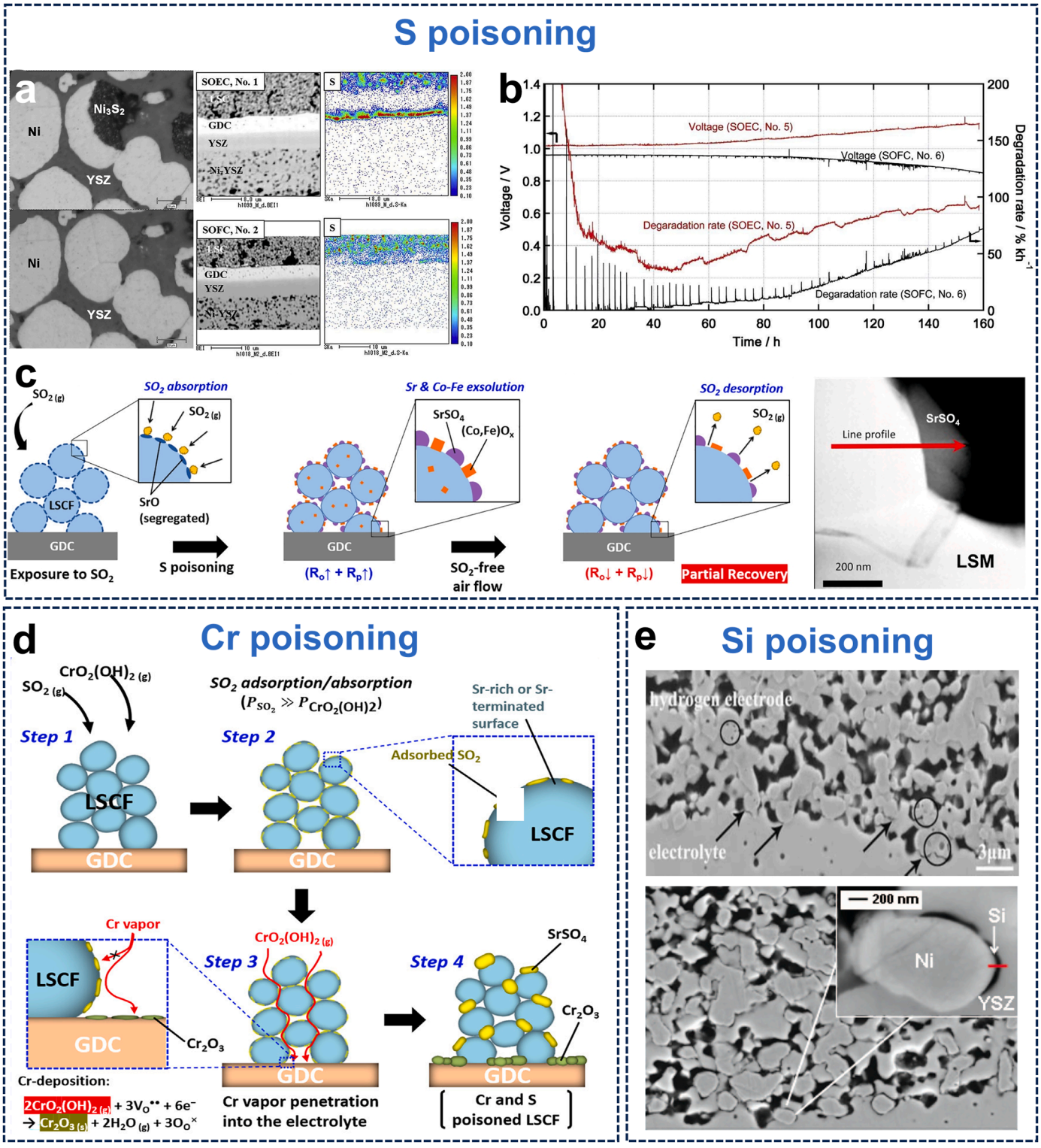

Fig. 13. (a) S poisoning in Ni-YSZ. Reproduced with permission [236]. Copyright 2011, Royal Society of Chemistry. (b) Comparison of S deposition in SOFC and SOEC models. Reproduced with permission [237]. Copyright 2017, Elsevier. (c) $SrSO_4$ generation process in LSM and LSCF. Reproduced with permission [238]. Copyright 2021, Frontiers. (d) Mechanism of Cr deposition in LSCF/YSZ cells. Reproduced with permission [229]. Copyright 2020, Elsevier. (e) Si deposition at the Ni-YSZ electrode/YSZ electrolyte interface. Reproduced with permission [245]. Copyright 2007, IOPscience.

ii. Sulfur poisoning

The poisonous effect of sulfur on the electrodes of SOFCs and SOECs is a critical challenge that necessitates mitigation, given that sulfur compounds can markedly diminish the performance of electrode materials. Sulfur is present in various forms within these systems, including H_2S, S, and SO_4^{2-}, each of which is capable of reacting with electrode materials and causing performance deterioration. The interaction of sulfur with electrode materials results in the formation of sulfides, which diminish catalytic activity and electrical conductivity [235]. For example, under sulfur-containing conditions, the anode of a Ni-YSZ cell experiences degradation as H_2S reacts with Ni to form Ni–S, thereby reducing the effective TPB for ionic transfer, decreasing the electrochemically active surface area, and augmenting the Rp (Fig. 13a) [236]. The chemical adsorption of sulfur and eventual saturation of the electrode surface leads to a reduction in active sites, while the oxidation of nickel intensifies polarization, resulting in decreased cell voltage and catalyst efficacy. The performance degradation of SOECs and SOFCs under operating conditions of 973 K and 0.3 A/cm^2 has been examined, revealing that sulfur primarily permeates at the interface of the CGO electrolyte and LSC oxygen electrode in SOECs, whereas in SOFCs, sulfur is distributed throughout the LSC. Elevated oxygen partial pressures may expedite sulfur permeation. The implementation of activated carbon air filtration has been demonstrated to reduce the degradation rate of SOECs to 0.6% k/h. Furthermore, experimental outcomes utilizing air with 0.2 ppm SO_2 have indicated that sulfur poisoning exerts a more pronounced impact in SOEC mode than in SOFC mode (Fig. 13b) [237]. Sulfur in the form of SO_2 as an air-borne contaminant can also influence the cathode air stream. On the air electrode, sulfur dioxide impurities in the oxidant react with strontium to form $SrSO_4$ deposition layers on the electrode surface, impeding oxygen diffusion (Fig. 13c) [238].

iii. Chromium poisoning

The deleterious impact of Cr on the electrodes of SOFCs and SOECs is predominantly expressed through its reactivity with electrode materials, resulting in the genesis of novel phases that diminish the electrodes' catalytic and conductive attributes, potentially impeding the diffusion pathways essential for oxygen. Cr primarily emanates from alloy constituents within the cell components and system, and under operational conditions, it may be emitted as volatile Cr compounds that can infiltrate the TPB and accumulate on the electrode surface. Cr accumulation on the electrodes can engender a spectrum of adverse effects, including reduced catalytic activity, greater electrode resistance, and modifications to the chemical and physical characteristics of the electrode materials, consequently compromising the cell's performance and durability [239].

Among SOFCs/SOECs, cathode/air electrode materials are exceptionally vulnerable to the pernicious influence of Cr poisoning. Substances containing Cr, liberated from stack and system materials such as Fe–Cr ferritic stainless steel, can migrate to the cathode and engage in reactions with the cathode materials to produce new phases; and these emergent phases can attenuate the electrode's catalytic and conductive capabilities and may even obstruct the diffusion pathways for oxygen. The toxicological effects of Cr are intricately linked to temperature, humidity, and current density; increased humidity corresponds to heightened volatilization of Cr in the guise of $CrO_2(OH)_2$. Under specific conditions, the thermodynamic favorability of reaction enthalpy substantiates the formation of this entity, affirming that such volatile Cr species are the principal perpetrators of electrode intoxication. At the TPB, $CrO_2(OH)_2$ undergoes reduction, yielding solid Cr_2O_3, which diminishes the active area of the boundary and impedes oxygen reduction [240]. In systems containing strontium and manganese, such as LSM, the presence of gaseous Cr can induce the crystallization and proliferation of solid $SrCrO_4$ and spinel-type (Cr, Mn)$_3O_4$ phases. These phases precipitate within the active boundary regions, thereby diminishing the electrochemical reactivity [241]. In LSCF electrodes, Cr will undergo different deposition processes, depending on whether S is present. When only Cr is present, it typically manifests as $SrCrO_4$ on the LSCF surface, similar to LSM. However, the presence of both Cr and S alters the deposition process, with Cr species primarily accumulating at the LSCF/GDC interface through electrochemical reduction, suggesting there is no interaction between Cr and SrO on the LSCF surface. Thermodynamic analysis supports this observation, indicating that SrO on the LSCF surface can absorb SO_2, consequently losing its ability to effectively trap Cr and allowing Cr vapor to penetrate through the LSCF and reach the LSCF/GDC interface for deposition (Fig. 13d) [242]. Within the context of Pr-Ba-Co based materials, exemplified by $PrBaCo_2O_{5+\delta}$ (PBCO), electrode polarization can precipitate the insulating BaO phase on the electrode surface. The polarization current facilitates the preferential segregation of Ba at the electrode surface, and subsequent contact with Cr-bearing stainless steel interconnects results in a pronounced decline in electrode performance, characterized by a significant escalation in Rp and overpotential. In the context of Ni–Cr alloy electro-deposition, the initial deposition of Ni serves as a catalytic agent, prompting a positive shift in the initial reduction potential of Cr ions. This corroborates the finding that chromium's deposition mechanism is not attributable to electrochemical reduction but is intricately linked to surface segregation phenomena [243].

iv. Silicon poisoning

The toxicological impact of Si on the electrodes of SOFCs and SOECs is a growing area of investigation. Silicon predominantly stems from contaminants within cell components or materials and is frequently attributed to the volatilization of glass sealants employed for sealing cell assemblies. Throughout the cell's operational cycle, Si may be emitted and translocate to the electrodes as siliconaceous compounds. Within the confines of SOFCs/SOECs, electrode materials are susceptible to reactions with silicon, culminating in the formation of silicates or other silicon compounds. These compounds have the propensity to blanket active sites and obstruct gaseous diffusion, consequently diminishing the catalytic activity and electrical conductivity of the electrodes [244]. Additionally, Si accumulation can induce structural alterations in the electrode materials, thereby compromising the electrodes' mechanical stability and longevity. The silicidation effect on electrodes is contingent upon a multitude of factors, including the concentration of Si, the operational temperature of the cell, and the compositional and microstructural attributes of the electrode materials. In an effort to ameliorate the silicidation effects, researchers have endeavored to engineer electrodes that exhibit resilience to Si poisoning, refine cell fabrication methodologies, and enhance cell sealing and material selection protocols. Within the context of SOFC and SOEC systems, volatile Si may precipitate as SiO_2 on the catalytically active sites situated at the TPB between the Ni-YSZ electrode and the YSZ electrolyte. The generation of SiO_2 and H_2O within the cell compartment from $Si(OH)_4$ solutions, with the subsequent reduction of H_2O to H_2, is most pronounced in the vicinity of the hydrogen electrode and the electrolyte. This leads to a concentrated deposition of SiO_2 at the electrode/electrolyte interface, obstructing catalytic sites. The deposition of SiO_2 also causes non-ohmic polarization to escalate and increases the fuel electrode's ohmic losses (Fig. 13e) [245].

v. Carbonation

Coking predominantly transpires when hydrocarbons serve as fuels, precipitating fuel decomposition and deposition on the electrode's active sites. This obstruction impedes direct interactions between the fuel gas and electrode material, culminating in lower electrode catalytic activity and, by extension, lower overall performance of the fuel cell or electrolyzer. With the accrual of carbonaceous deposits, the effective reactive surface area of the electrode shrinks, leading to a marked diminution in current and power density. Concurrently, discrepancies in the TEC between the carbonaceous deposit and the electrode material may induce thermal stresses under high-temperature operational parameters, potentially precipitating fractures or delamination in the electrode material and exacerbating the deterioration of cell performance and longevity. Jiang et al., in their investigation of Sn-doped Ni/ScCeSZ as a SOFC anode material, observed that at an operational

temperature of 750 °C, the cell could attain a peak power density of 0.997 W/cm^2. However, a pronounced escalation in carbon deposition on the anode without Sn doping stopped operation after 11 h of exposure to biogas at 750 °C due to severe carbon deposition [246]. The primary factors influencing carbon deposition include temperature, pressure, and gas composition. The C–H–O phase diagram provides a clear visualization of the areas where carbon generation and stability occur under varying conditions of temperature, pressure, and gas composition [247]. It is observed that at a constant pressure, carbon deposition increases with decreasing temperature. Furthermore, a higher oxygen content in the gas composition at the same temperature results in a greater propensity for carbon deposition, as depicted in Fig. 14a. Changes in pressure also impact the formation and elimination of carbon. In environments of high pressure, the collision frequency between gas molecules rises, potentially enhancing carbon generation reactions. Conversely, in low-pressure situations, the tendency for gas molecules to escape increases, which is less favorable for stable carbon deposition, as demonstrated in Fig. 14b. With respect to different gas compositions, a higher hydrogen content typically inhibits carbon generation, since hydrogen can react with carbon to produce gaseous substances like methane. Conversely, a greater CO content may facilitate carbon formation through disproportionation reactions. Introducing an appropriate quantity of water vapor or oxygen into the fuel gas can modify the local gas composition, shifting the system away from the stable carbon region and reducing the likelihood of carbonization, as shown in Fig. 14c.

Due to variations in the chemical potential trends of oxygen at different interfaces in the SOFC and SOEC modes, carbon deposition exhibits distinct behaviors in these two modes, as depicted in Fig. 14d. In Fig. 14e, under OCV conditions, the oxygen chemical potential at the electrode–electrolyte interface ($\mu_{O_2}^{FE|EL}$) is approximately equivalent to that in the electrode ($\mu_{O_2}^{FE}$), resulting in no carbon deposition at this juncture. When $\mu_{O_2}^{FE|EL} > \mu_{O_2}^{FE}$, the reduction potential at the interface in the SOFC mode is lower than within the bulk of the electrode, rendering the Boudouard reaction in the electrode's bulk phase more favorable than carbon deposition at the FE|EL interface. Conversely, in the SOEC mode, $\mu_{O_2}^{FE|EL} < \mu_{O_2}^{FE}$, indicating that the reduction potential at the interface exceeds that within the bulk of the electrode. This leads to a substantial thermodynamic driving force at the FE|EL interface, significantly exceeding that within the electrode's bulk phase. This observation suggests that carbon deposition is more prone to occur in the SOEC mode, as the reduction conditions at the interface facilitate carbon formation [248]. There are other factors leading to the deposition of carbon on electrodes—for instance, during the electrolysis of a 5% CO and 95% CO_2 mixture at 600 °C, GDC fuel electrodes exhibited no coking under OCV conditions, yet coking manifested after the application of a 0.1 A current for 5 h, indicating that the presence of current catalyzes the coking process (Fig. 14f) [249]. Moreover, coking predominantly occurs at the electrode–electrolyte interface, in contrast to the Ni surface [250]. Ni-based fuel electrodes characterized by low porosity (20–25%) are more prone to coking than those with higher porosity (30–40%), a phenomenon potentially correlated with the efficacy of gas transport and conversion [251].

vi. Corrosion of interconnects

Another challenge in the context of SOFCs and SOECs is finding interconnect materials with significant stability. Interconnects are subjected to a dual-atmosphere environment comprising air and hydrogen or carbon-based fuels. Their essential function is to efficiently segregate these gases and channel them to the appropriate electrodes, ensuring the cell's proper functioning. To preserve the cell's long-term performance and longevity, interconnect materials must maintain stability within this atmosphere. However, at intermediate temperatures, interconnect materials are prone to corrosion due to the oxidation of iron. Specifically, when one side of the interconnect alloy is exposed to an air environment, the corrosion process accelerates, with visible signs of corrosion often manifesting within 50 h (Fig. 14g) [252]. In contrast, under a singular atmosphere (either air or a hydrogen-based fuel), the extent and severity of the interconnect material's oxidation are significantly reduced, underscoring the intensified corrosive effects under dual-atmosphere conditions. The permeation of hydrogen through metals and its migration across the metal and oxide film facilitate the growth of iron oxide, leading to the formation of flaky structures that propagate outward. Under carbon-based atmospheres, processes such as carburization, carbon deposition, or metal dusting may ensue, which can embrittle the material and diminish the chromium content within the alloy. This reduction impairs the alloy's ability to form a protective oxide layer, thereby accelerating corrosion. Particularly between 400 and 650 °C, the significant fluctuations in oxygen partial pressure and carbon activity during thermal cycling exacerbate metal dusting corrosion. Alnegren et al. investigated the dual-atmosphere corrosion behavior of AISI 441 alloy at 600 °C, 700 °C, and 800 °C, observing that dual-atmosphere corrosion was most pronounced at 600 °C. At elevated temperatures, the increased formation of Cr/$MnCr_2O_4$ spinel scale layers and a reduction in the excessive growth of iron oxide were noted. This "inverse temperature effect", wherein more severe corrosion occurs at lower temperatures, is not yet comprehensively understood [253].

4.5.2. Resolution strategies

i. Composite electrode suppresses attenuation and interface degradation

The essence of the composite electrode strategy lies in enhancing the electrochemical performance and stability of electrodes by utilizing synergistic interactions among multiple components. For example, to combat the volatility and performance deterioration of nickel-based anode materials at elevated temperatures, introducing a nanoscale interlayer into the nickel-based anode can effectively suppress nickel volatilization and particle coarsening (Fig. 15a) [254]. This coating not only prevents the diffusion of nickel atoms but also maintains the electrode's high catalytic activity, leading to significantly greater stability and longevity.

Researchers have also utilized composite electrode structures to enhance the stability of the electrode–electrolyte interface. As an illustration, Liu et al. introduced a cathode-supported SOFC using $La_{0.8}Sr_{0.2}MnO_{3-\delta}$-YSZ composite material, achieving a peak power density of 253 mW cm^{-2} at 800 °C (Fig. 15b) [255]. However, the composite electrode may experience decreases in specific surface area and electrochemical activity due to particle coarsening during high-temperature sintering. Fan et al. verified that a 7% Mo-doped modified $SFM_{0.07}$+GDC composite electrode demonstrates exceptional stability. This was observed during accelerated thermal cycling at a rapid heating/cooling rate of 30 °C/min between 700 and 400 °C. The electrode also operated consistently and without interruption at a fixed voltage of 0.7 V for over 250 h (Fig. 15c) [256]. These results were achieved by integrating the principles of Smith acidity and average metal–oxygen bond energy.

In SOECs, the deterioration of the electrode–electrolyte interface is particularly evident. Song et al. incorporated a layer of GDC between the LSM-YSZ composite cathode and the YSZ electrolyte. This approach effectively reduced interfacial degradation. Through experiments, it was observed that after 100 h of operation at 800 °C, the interfacial polarization impedance of the electrode with this optimized interface only increased by approximately 0.005 Ω cm^2 (Fig. 15d). This optimization technique not only improved the electrochemical performance of the electrode but also significantly prolonged the cell's life [257].

ii. Reduction of electrode poisoning

Researchers have adopted a multitude of strategies to address the challenge of sulfur poisoning. One such strategy involves the application of sulfur adsorbents or catalysts to diminish the adsorption and accumulation of sulfur on electrode materials. Marina et al. observed a

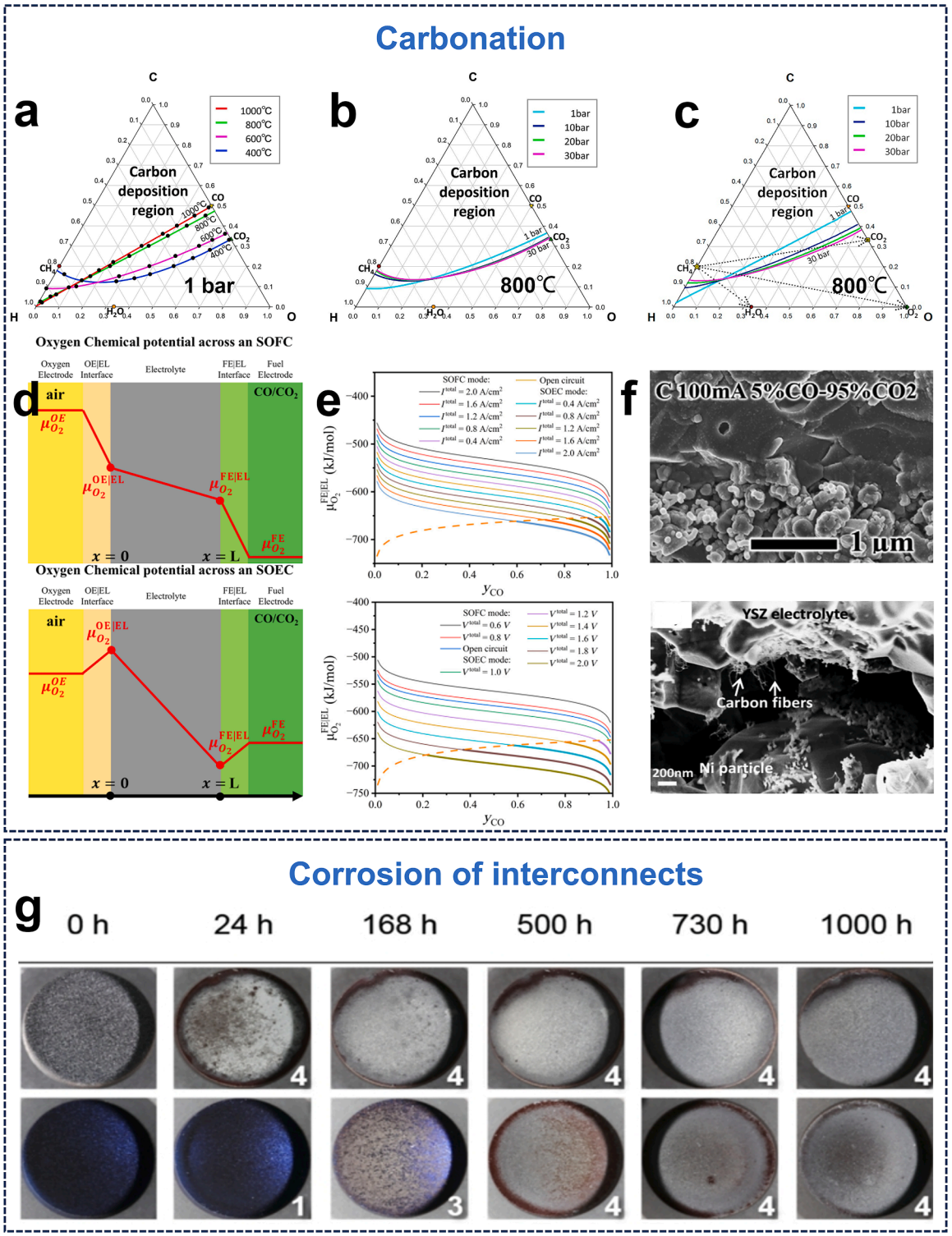

Fig. 14. C–H–O phase diagram of the carbon deposition region as a function of (a) temperature, (b) barometric pressure, and (c) gas composition. Reproduced with permission [247]. Copyright 2017, Elsevier. (d) Schematics of oxygen chemical potential distribution (red line) across the cross-section from oxygen electrode to fuel electrode in the SOFC and SOEC modes. (e) Oxygen chemical potential $\mu_{O_2}^{FE|EL}$ of RSOC working at different fuel compositions and working loads, in constant current density or constant voltage mode. The orange dashed line shows the threshold of $\mu_{O_2}^{FE|EL}$ for the forward-moving CO electrochemical reduction reaction. The region where carbon deposition is thermodynamically favored is bolded. Reproduced with permission [248]. Copyright 2025, APS Journals. (f) C deposition on the Ni–YSZ interface. Reproduced with permission [249]. Copyright 2013, Elsevier. (g) Corrosion processes of metallic connecting parts in an air atmosphere. Reproduced with permission [252]. Copyright 2018, Elsevier.

marked decrease in sulfur adsorption by incorporating Sb and Sn into Ni-YSZ anodes. This reduction may be ascribed to the occupancy of active sites on the electrode surface by these elements, which might weaken the sulfur's adsorptive capacity, enhance its oxidative conversion, or facilitate the formation of secondary phases. Notably, the segregation of Sn at surface grain boundaries may act as a direct barrier to sulfur adsorption. The TPB model of the Ni/YSZ system with a Sn adatom on the Ni can form a physical barrier and effectively prevent sulfur diffusion to the O vacancy at the interface. The diffusion barriers are as large as 1.41 eV, therefore eliminating sulfur poisoning at the TPB and showing promise for alleviating sulfur poisoning (Fig. 16a) [258].

In the context of chromium poisoning, the resistance of electrode materials to chromium-induced degradation can be significantly bolstered by the incorporation of a minor amount of Nb^{5+} at the B-site of the perovskite structure. Such doping effectively curbs the precipitation of metallic strontium on the material's surface. Additionally, the application of protective coatings on metallic interconnect surfaces, including active element oxides, rare earth perovskites, or spinel coatings, can mitigate chromium oxidation and volatilization, thus inhibiting its detrimental impact on the cathode. Specifically, doping modifications, such as the incorporation of Cu and Fe, have been shown to enhance the electrical conductivity and oxidative resistance of Mn–Co spinel coatings. Couturier et al. conducted extensive testing for oxidation resistance and Cr evaporation, as well as long-term SOEC operation in a short stack scale for up to 5000 h at 700–800 °C. The effectiveness of a dense CeCo-based coating produced by PVD HiPIMS was confirmed on AISI441 steel, which is commonly used as an interconnect in SOC technology. Through SEM/EDX characterizations conducted after the tests, it was determined that the thickness of the poorly conductive Cr oxide layer was reduced on CeO_2/Co- and CeO_2/Co/CeO_2-coated samples after 5000 h at 800 °C, with no spallation observed, unlike on bare steel. As a result, the coating minimized the loss of steel thickness (Fig. 16b) [259]. Spinel oxide coatings have demonstrated superior efficacy in inhibiting chromium evaporation. Yang et al. applied a spinel $Mn_{1.5}Co_{1.5}O_4$ coating to ferritic stainless steel via a slurry coating technique and conducted extensive testing, confirming that the coating effectively precludes chromium ingress into the LSCM cathode (Figs. 16c, d) [260]. In an effort to further ameliorate chromium poisoning within the electrode, Aphale et al. engineered a chromium getter in the form of SrO–NiO or Sr_xNi_yO solid solution traps. These traps maintain stability at temperatures below 900 °C and capture gaseous chromium species through targeted chemical reactions (Fig. 16e) [261]. The utilization of such materials offers a cost-effective and potent solution for mitigating chromium poisoning.

The scientific rationale underpinning the getter's function can be extrapolated to lessen the impact of other toxicants, including sulfur or silicon. This approach may represent the most efficacious strategy for retarding silica deposition, as alternative methods have not been extensively investigated, and the sole substitute involves replacing the framework $NaAlSi_3O_8$ sealant employed on the hydrogen electrode.

iii. Corrosion inhibition strategies for metal components

Within the realm of dual-gas-layer corrosion inhibition, the preferential growth of the oxide layer has emerged as a potent strategy. Cr_2O_3 and $(Cr,Mn)_3O_4$ spinel are notably advantageous for curbing the corrosion of metallic interconnects. During the pre-oxidation treatment of ferritic stainless steel, a compact passive chromium layer can be established by manipulating the oxygen partial pressure (pO_2), and this layer mitigates intergranular corrosion effectively when subjected to a dual-gas-layer environment. Diminishing pO_2 fosters the development of chromia and $(Cr, Mn)_3O_4$ spinel; in contrast to untreated samples, pre-oxidized samples demonstrate enhanced ability to curb the excessive proliferation of iron oxide. The formation of a dense passive film of chromium is regarded as an efficacious way to mitigate corrosion. The similar sizes of chromium and iron lead to comparable relative migration rates within their respective oxides; however, chromium's function within the passive film is more pronounced. Research indicates that the diffusion of iron within chromia is thermodynamically impeded, and chromia diminishes the oxygen potential at the metal–oxide interface, thus preventing iron oxidation. The diffusion coefficient of bulk cations fluctuates with variations in oxygen partial pressure, adhering to the relationship $(pO_2)^{3/16}$. Oxygen partial pressure is the principal impetus for the growth of iron oxide scales.

The application of doped spinel oxide coatings is an efficacious strategy for attenuating the corrosion of metallic components. Zhu et al. conducted a study on three Ni-doped variants: $CuNi_{0.2}Mn_{1.8}O_4$, $CuNi_{0.4}Mn_{1.6}O_4$, and $CuNi_{0.6}Mn_{1.4}O_4$. The findings indicated that Ni doping enhances phase stability, increases electrical conductivity, and reduces chromium diffusion in $CuMn_2O_4$ spinel. Among the three variants, $CuNi_{0.2}Mn_{1.8}O_4$ demonstrated superior performance, exhibiting the highest conductivity (94–103 S/cm) within the temperature range of 700–800 °C. This variant also formed the densest coating and served as the most effective diffusion barrier against chromium migration (Fig. 16f) [262]. These spinel coatings have demonstrated the capacity to diminish the parabolic oxidation rate at elevated test temperatures (ranging from 800 to 900 °C), yet their corrosion-inhibiting efficacy is attenuated as the temperature decreases. The long-term impact of these coatings for enhancing corrosion resistance remains inconclusive, as certain regions of Crofer22APU with applied coatings exhibit markedly inferior corrosion resistance compared to the uncoated alloy (Fig. 16g) [263].

The modification of coatings with active elements constitutes an effective approach to bolster the corrosion resistance of metallic materials. These active elements, predominantly rare earth elements such as Zr, La, Ce, and Y, exhibit a heightened oxygen affinity and larger ionic radii in comparison to Cr. This endows them with the ability to better augment the oxide film's conductivity. This phenomenon is recognized as the reactive element effect (REE), the precise mechanisms of which remain unclear. As delineated by Pint, potential mechanisms underlying the REE encompass the enhancement of chemical bonding at the oxide–alloy interface, as well as alterations in oxygen vacancy-assisted diffusion that may arise from the incorporation of active elements. Furthermore, several hypotheses posit that the introduction of active elements might influence the growth kinetics, microstructural attributes, and adhesive properties between the oxide film and the alloy substrate. However, these hypotheses have yet to coalesce into a cohesive theoretical framework, necessitating further work to unravel the explicit mechanisms underpinning the REE.

4.6. Key materials exploration with new strategies

4.6.1. High-entropy doping strategies

High-entropy materials (HEMs), including high-entropy oxides (HEOs), are cutting-edge materials that combine multiple elements to create synergistic effects. The high-entropy doping strategy, an innovative material design approach, has in recent years proven to be crucial in the advancement of key materials for SOFCs/SOECs. In the realm of high-entropy oxide research, several pivotal factors—including size mismatch and lattice distortion, the diversity of acidity and redox properties, variations in oxidation states, and the contributions of configurational entropy—significantly influence the microstructural and macroscopic properties of these materials.

Size mismatch and lattice distortion exert substantial effects on the crystalline structure and physicochemical characteristics of HEOs. The inherent differences in ionic radii among various cations induce localized stress within the lattice, resulting in lattice distortion. This distortion may present as anisotropic alterations in lattice constants or irregular displacements of atomic positions, subsequently modifying the crystalline structure and potentially leading to the formation of novel phases. Such new phases may exhibit distinctive physicochemical properties, including superconductivity, magnetism, or electrical conductivity. Moreover, lattice distortion can influence the mechanical properties of materials, such as hardness and strength;

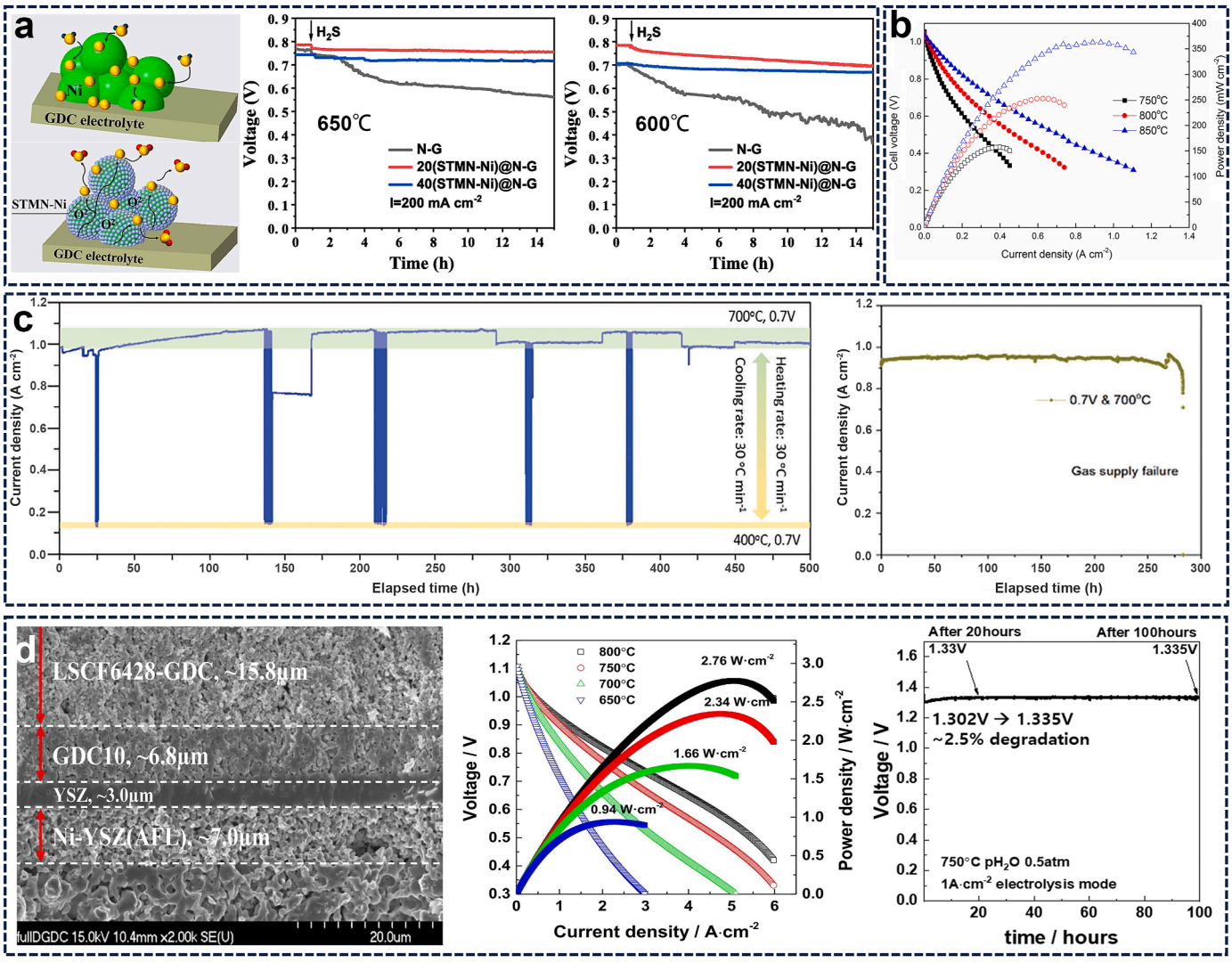

Fig. 15. (a) Enhanced stability of Ni-based anode coating with nano $SrTi_{0.5}Mo_{0.2}Ni_{0.3-x}O_{3-\delta}$-Ni interlayer. Reproduced with permission [254]. Copyright 2018, Springer. (b) Voltage and power density versus current density. Reproduced with permission [255]. Copyright 2023, Elsevier. (c) Long-term stability measurement of cell at 0.7 V and 700 °C. Reproduced with permission [256]. Copyright 2024, Wiley-VCH. (d) Power density and stability after the introduction of a GDC coating. Reproduced with permission [257]. Copyright 2024, Elsevier.

while the stresses within the lattice may enhance hardness, they can concurrently diminish ductility. Lattice distortion may also affect a material's thermal and chemical stability, as the internal stresses and distortions can compromise thermal stability and accelerate degradation under high-temperature or chemically corrosive conditions [271].

The introduction of diverse cations engenders a variety of acidity and redox properties, which play a critical role in modulating the surface chemistry and catalytic performance of HEOs. Different cations possess unique electronic structures and chemical reactivities, exhibiting distinct Lewis acid (electron acceptor) and Lewis base (electron donor) characteristics that affect the charge distribution and electron density on the material's surface [268]. Such alterations can enhance the adsorption capacity for reactants while facilitating the desorption of products, as the bonding strength between product molecules and active surface sites may be diminished. The differential redox properties of various cations also impact the material's electronic structure, particularly the position of the d-band center, which is essential for optimizing the electronic characteristics of active sites. The d-band center's position is closely linked to catalytic activity, as it governs the strength of interactions between reactants and the catalyst surface. By fine-tuning the d-band center, it is possible to optimize the adsorption and activation of reactants, thereby enhancing catalytic activity [269].

In HEOs, the incorporation of different elements is frequently accompanied by variations in oxidation states, a characteristic that significantly affects the electronic structure and catalytic capabilities of the materials [266]. Typically composed of five or more principal elements, high-entropy oxides can exhibit multiple stable oxidation states. This feature of multiple oxidation states can complicate the internal electronic structure of the material, subsequently influencing its conductivity, magnetism, and catalytic activity. Elements in different oxidation states can provide a diverse array of active sites, which may display heightened activity or selectivity for specific chemical reactions [267]. For instance, in electrocatalytic processes, transition metals in varying oxidation states can optimize the generation rates of hydrogen and oxygen during water splitting. The presence of different oxidation states can also enhance catalytic activity by modulating the position of the d-band center, as this position is closely correlated with the interaction strength between reactants and catalyst surfaces, thus impacting the kinetics and efficiency of catalytic reactions.

The contribution of configurational entropy in HEOs is multifaceted, arising from the random distribution of multiple elements within the

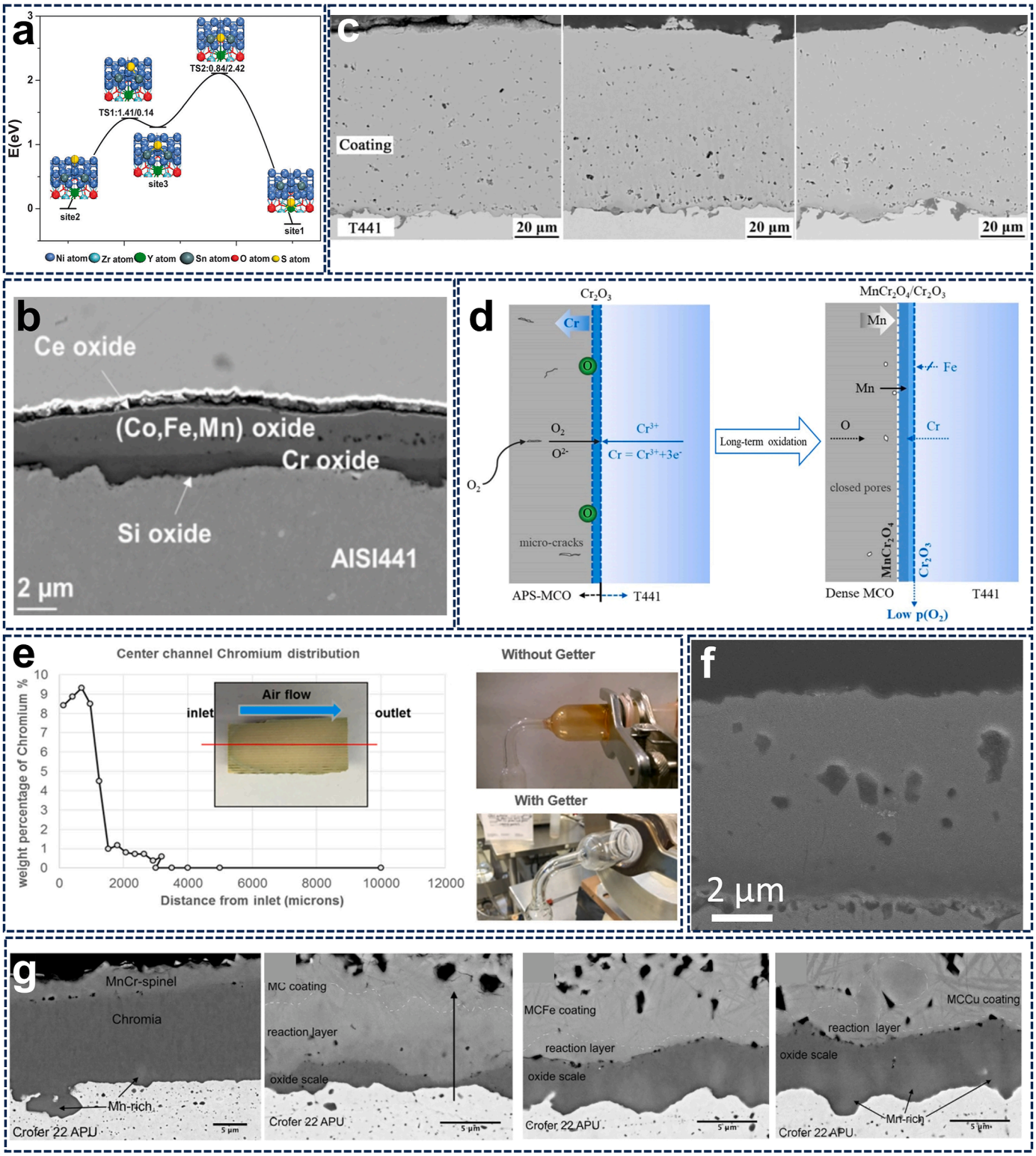

Fig. 16. (a) Sn-doped Ni-YSZ mitigated S deposition. Reproduced with permission [258]. Copyright 2014, Royal Society of Chemistry. (b) After 5000 h at 800 °C, the presence of the CeO_2/Co- and CeO_2/Co/CeO_2 coatings significantly reduced the loss of steel thickness. Reproduced with permission [259]. Copyright 2024, Springer. (c, d) Mn–Co–O coating against Cr poisoning. Reproduced with permission [260]. Copyright 2020, Elsevier. (e) Reduction of Cr poisoning by Cr trapping agents. Reproduced with permission [261]. Copyright 2018, Springer. (f) SEM images of cross-sectional microstructure of $CuNi_{0.2}Mn_{1.8}O_4$. Reproduced with permission [262]. Copyright 2025, Springer. (g) SEM images of Cr deposition in Crofer22APU under different oxide coatings. Reproduced with permission [263]. Copyright 2017, Elsevier.

material, which increases the microscopic disorder and, consequently, the configurational entropy. The augmentation of configurational entropy is commonly associated with a material's thermodynamic stability, as high-entropy states typically lower the free energy of the system, rendering the material thermodynamically more stable. This enhanced stability assists in preserving the structural integrity of the material under elevated temperatures or extreme conditions, mitigating the risks of phase separation and lattice defects. An increase in configurational entropy may also lead to improved kinetic stability—for instance, by elevating the diffusion barriers for elements, thereby inhibiting elemental migration and phase transitions—which is critical for enhancing a material's long-term stability and durability. It can improve ion transport and charge transfer by modifying ion diffusion pathways and adjusting electronic structures, ultimately enhancing the overall performance and reliability of SOFC/SOEC systems [264]. This method introduces fresh ideas and techniques for producing high-performance, durable key materials for SOFCs/SOECs, offering significant scientific value and promising practical applications.

i. High-entropy electrolytes

High-entropy doping oxides targeting the fluorite structure have proven to be highly effective for enhancing the performance of oxygen ion-conducting electrolytes. Gild et al. prepared 11 types of high-entropy fluorite-structured electrolyte materials containing five elements, doped through high-energy ball milling, spark plasma sintering, and air annealing processes:

$(Hf_{0.25}Zr_{0.25}Ce_{0.25}Y_{0.25})O_{2-\delta}$

$(Hf_{0.25}Zr_{0.25}Ce_{0.25})(Y_{0.125}Yb_{0.125})O_{2-\delta}$

$(Hf_{0.2}Zr_{0.2}Ce_{0.2})(Y_{0.2}Yb_{0.2})O_{2-\delta}$

$(Hf_{0.25}Zr_{0.25}Ce_{0.25})(Y_{0.125}Ca_{0.125})O_{2-\delta}$

$(Hf_{0.25}Zr_{0.25}Ce_{0.25})(Y_{0.125}Gd_{0.125})O_{2-\delta}$

$(Hf_{0.2}Zr_{0.2}Ce_{0.2})(Y_{0.2}Gd_{0.2})O_{2-\delta}$

$(Hf_{0.25}Zr_{0.25}Ce_{0.25})(Yb_{0.125}Gd_{0.125})O_{2-\delta}$

$(Hf_{0.2}Zr_{0.2}Ce_{0.2})(Yb_{0.2}Gd_{0.2})O_{2-\delta}$

These successfully form pure-phase fluorite-structured solid solutions with high configurational entropy in the cation sub-lattice. Except for the two high-entropy fluorite oxides (HEFOs) containing Yb and Gd, most HEFOs can be sintered to high relative densities. However, compared with 8YSZ materials, these HEFOs have lower ionic conductivity, which may be due to the high phonon scattering caused by multiple cations and strained lattices (Fig. 17a) [265]. Bonnet et al. used spark plasma sintering to synthesize a high-entropy oxide electrolyte of $(Hf_{1/3}Ce_{1/3}Zr_{1/3})_{1-x}(Y_{1/2}Gd_{1/2})_xO_{2-\delta}$ under reducing conditions, achieving an ionic conductivity of 2.9 mS/cm in an oxygen atmosphere at 800 °C (Fig. 17b) [266]. In-depth research on high-entropy design has led to the introduction of Pr, which has shown the potential to enhance the electrical performance of ZrO_2-based HEOs. Zhang et al. realized $(Zr_{0.2}Hf_{0.2}Pr_{0.2}La_{0.2})Y_{0.2}O_{2-\delta}$ with a configurational entropy (Sconf) value of 1.61 R. The formation of a cubic fluorite structure was confirmed by Raman spectroscopy, and the maximum conductivity at 600 °C was 5.35×10^{-4} S/cm (Fig. 17c), which is the highest value among the studied ZrO_2-based HEOs [267].

Yapryntsev et al. achieved HEOs using CeGdDySmPrO, CeGdYSmPrO, and CeGdLaSmPrO. These powder materials retained their single-phase structure even after undergoing a calcination process at 900 °C, maintaining a body-centered cubic structure (space group *Ia-3*). Among the materials synthesized, the CeGdLaSmPrO ceramic exhibited an impressive electrical conductivity of 0.001 S/cm at 510 °C (Fig. 17d). The activation energy (*Ea*) values for this ceramic were found to be lower than 0.52–0.53 eV in both the low-temperature range (170–400 °C) and the medium-temperature range (450–560 °C), which is significantly lower than that of common high-entropy fluorite-structured materials [268]. Dabrowa et al. obtained $(CeGdNdPrSmMo_{0.5})O_2$ and $(CeGdLaNdPrMo_{0.5})O_2$ with fluorite structures using a solid reaction. At 600 °C, the conductivities of $CeGdNdSmPrO_2$, $CeGdLaNdPrO_2$, and the two samples doped with 0.5% Mo^{6+} were 2.3, 2.1, 1.0, and 1.3 mS/cm, respectively. The authors suggested that by stabilizing the oxygen vacancy disordered *Fm-3m* crystal structure, Mo^{6+} doping helps to enhance ionic conductivity (Fig. 17e) [269].

The pyrochlore-structured zirconates, another mainstream type of oxygen ion-conducting electrolyte material, have also seen a growing trend toward high-entropy design. Wang et al. prepared a series of high-entropy zirconates, namely $(Nd_{0.2}Sm_{0.2}Gd_{0.2}Dy_{0.2}Y_{0.2})_2Zr_2O_7$, $(Nd_{0.2}Sm_{0.2}Eu_{0.2}Dy_{0.2}Y_{0.2})_2Zr_2O_7$, $(Nd_{0.2}Sm_{0.2}Eu_{0.2}Gd_{0.2}Er_{0.2})_2Zr_2O_7$, $(Nd_{0.2}Sm_{0.2}Eu_{0.2}Gd_{0.2}Y_{0.2})_2Zr_2O_7$, and $(Nd_{0.2}Sm_{0.2}Eu_{0.2}Gd_{0.2}Dy_{0.2})_2Zr_2O_7$, using a high-temperature solid reaction with a sintering temperature of 1700 °C [270]. As illustrated in Fig. 17f, the degree of order in the HEOs grew as the cation radius ratio r (A^{3+})/r (B^{4+}) increased, maintaining a structured pyrochlore configuration. The conductivity also rose in tandem with the r (A^{3+})/r (B^{4+}) ratio, reaching its peak in the $(Nd_{0.2}Sm_{0.2}Eu_{0.2}Gd_{0.2}Dy_{0.2})_2Zr_2O_7$ sample, which displayed a conductivity value of 7.1×10^{-3} S/cm at 800 °C.

In addition to oxygen ion-conducting electrolyte materials, there has been progress in the development of HEOs for use in proton-conducting electrolytes. These HEOs have shown promising characteristics such as high proton conductivity and strong structural stability. Gazda et al. synthesized various high-entropy perovskite proton-conducting electrolyte materials using a solid reaction. This included compositions such as $BaZr_{0.2}Sn_{0.2}Ti_{0.2}Hf_{0.2}Ce_{0.2}O_3$, $BaZr_{0.2}Sn_{0.2}Ti_{0.2}Hf_{0.2}Y_{0.2}O_3$, $BaZr_{1/7}Sn_{1/7}Ti_{1/7}Hf_{1/7}Ce_{1/7}Nb_{1/7}Y_{1/7}O_3$, and $BaZr_{0.15}Sn_{0.15}Ti_{0.15}Hf_{0.15}Ce_{0.15}Nb_{0.15}Y_{0.1}O_3$. The final sintering temperatures ranged from 1400 to 1500 °C, depending on the specific material composition. Among these materials, $BaZr_{0.2}Sn_{0.2}Ti_{0.2}Hf_{0.2}Y_{0.2}O_3$ displayed the highest water absorption rate (Fig. 17g) and had a proton defect concentration of 1.6×10^{-2} mol/mol. The $BaZr_{0.2}Sn_{0.2}Ti_{0.2}Hf_{0.2}Y_{0.2}O_3$ sample demonstrated the highest conductivity values, with 4.4×10^{-4} S/cm in dry air and 6.0×10^{-4} S/cm in humid air at 600 °C [271]. Guo et al. successfully synthesized a perovskite-structured high-entropy oxide named $BaSn_{0.16}Zr_{0.24}Ce_{0.35}Y_{0.1}Yb_{0.1}Dy_{0.05}O_{3-\delta}$ (BSZCYYbD) using a sol-gel method, with a sintering temperature of 1500 °C. The proton conductivity of BSZCYYbD exhibited exceptional performance, measuring 8.3 mS/cm at 600 °C, surpassing the 6.1 and 7.4 mS/cm conductivities of the traditional $Ba_{1.03}Ce_{0.6}Zr_{0.2}Yb_{0.2}O_{3-\delta}$ and $BaCe_{0.6}Zr_{0.3}Y_{0.1}O_{3-\delta}$ electrolytes, respectively (refer to Fig. 17h) [272]. Furthermore, an anode-supported cell utilizing BSZCYYbD as the electrolyte (45 μm thick) and $BaCo_{0.4}Fe_{0.4}Zr_{0.4}Y_{0.1}O_{3-\delta}$ (BCFZY) as the cathode achieved an impressive maximum power density of 318 mW/cm^2 at 600 °C. The cell displayed an OCV of 1.06 V at 600 °C and 1.08 V at 500 °C, demonstrating the potential for high performance in various operating conditions. This research lays the groundwork for the advancement of high-entropy proton-conducting electrolytes. Xiang et al. developed a high-entropy perovskite oxide, $Ba_{1.05}Ce_{0.45}Zr_{0.1}Y_{0.1}Yb_{0.1}Pr_{0.10}Gd_{0.15}O_{3-\delta}$, via A-site stoichiometric compensation and oxygen vacancy engineering. Its conductivity reached 8.95 mS/cm at 600 °C in wet air (3% H_2O). The electrolyte showed great electrochemical performance and stability in wet air: a single cell with this electrolyte achieved a peak power density of 397 mW/cm^2 at 600 °C. The sintering temperature of the electrolyte was 1400 °C, 150–200 °C lower than other reported high-entropy electrolytes, helping to reduce Ba volatilization loss [273]. He et al. designed and synthesized a novel perovskite electrolyte material, BZCYYbHG ($BaZr_{0.06}Ce_{0.7}Y_{0.06}Yb_{0.06}Hf_{0.06}Gd_{0.06}O_{3-\delta}$). It shows a conductivity of 0.012 S/cm at 500 °C and 0.006 S/cm at 400 °C. Under high humidity (30% H_2O), its conductivity degradation rate is only 5.5%, much lower than BZCYYb's (19.6%). It delivers peak power densities of 1.39 W/cm^2 in H_2, 1.12 W/cm^2 in NH_3, and 0.7 W/cm^2 in wet CH_4 at 550 °C. It also achieves stable reversible cycling for 330 h at 550 °C. This material is the highest-performing high-entropy protonic electrolyte reported so far, showing the great potential of high-entropy strategies for protonic ceramic cells [274].

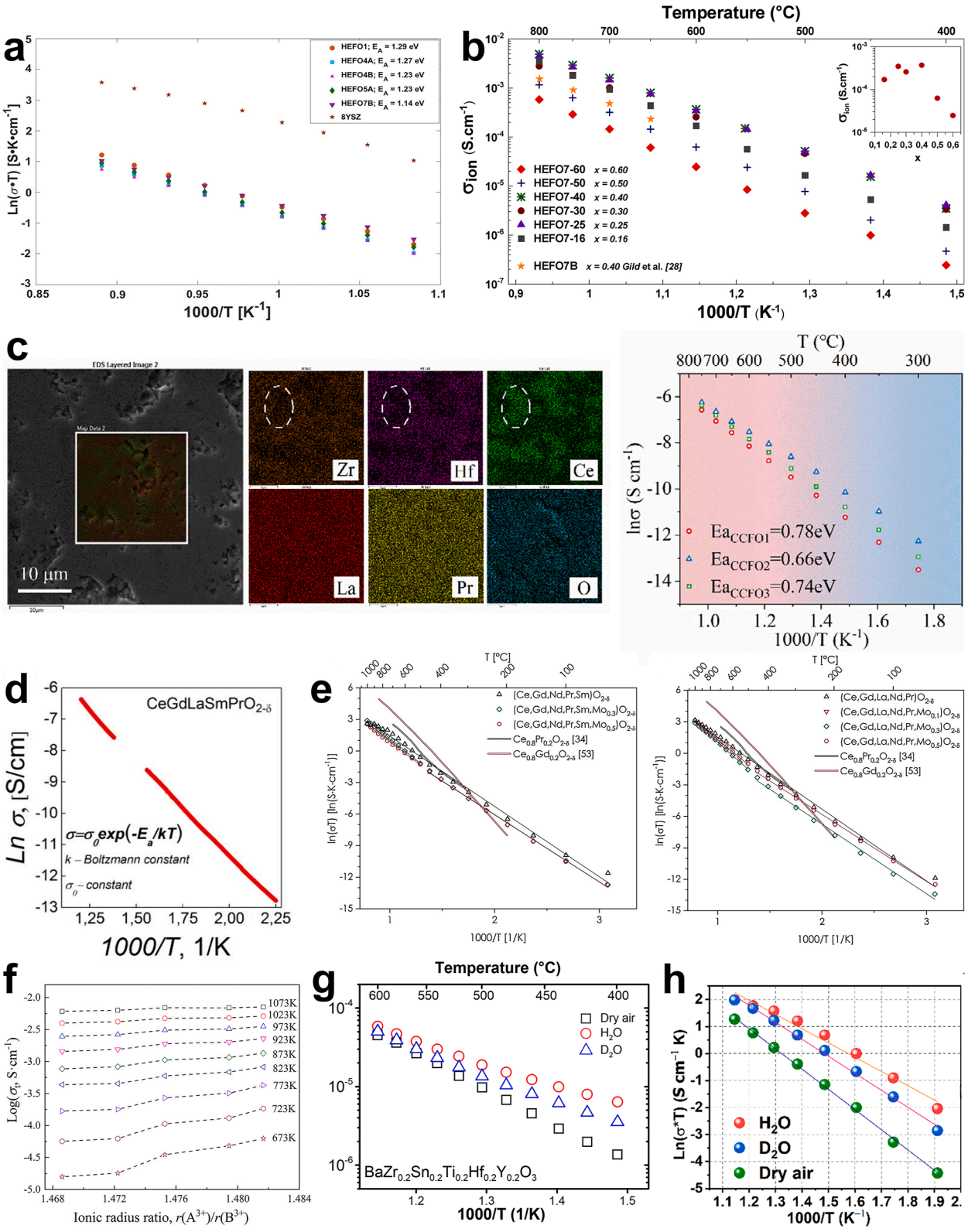

Fig. 17. (a) Electrical conductivity of a series of high-entropy materials. Reproduced with permission [265]. Copyright 2018, Elsevier. (b) Arrhenius plots of the ionic conductivity of $(Hf_{1/3}Ce_{1/3}Zr_{1/3})_{1-x}$ $(Y_{1/2}Gd_{1/2})_xO_{2-\delta}$. Reproduced with permission [266]. Copyright 2021, Elsevier. (c) EDS mapping and Ionic conductivity of $(Zr_{0.2}Hf_{0.2}Pr_{0.2}La_{0.2})Y_{0.2}O_{2-\delta}$ at 600 °C. Reproduced with permission [267]. Copyright 2022, Elsevier. (d) Conductivity of CeGdLaSmPrO at 510 °C. Reproduced with permission [268]. Copyright 2022, Elsevier. (e) Conductivity of $CeGdLaSmPrO_{2-\delta}$ at 600 °C. Reproduced with permission [269]. Copyright 2022, Wiley-VCH. (f) Relationship between electrical conductivity and the ratio of the radius of the cation. Reproduced with permission [270]. Copyright 2022, Elsevier. (g) Total electrical conductivity of HEOs in dry and humidified air [271]. Copyright 2020, American Chemical Society. (h) Proton conductivity of BSZCYYbD. Reproduced with permission [272]. Copyright 2022, American Chemical Society.

ii. High-entropy electrodes

In high-entropy strategies, a mixture of different cations with distinctive acid-base properties coexists, leading to a synergistic effect that enhances the material's surface chemistry. This results in improved reactant adsorption and product desorption, ultimately leading to superior catalytic performance compared to single-component materials. The high-entropy approach also introduces elements with varied valence states, which in turn alters the electronic structure and d-band center position of the material. This optimization of electronic properties at active sites further boosts catalytic activity. These advantages are particularly beneficial for electrode materials, and there has been significant progress in high-entropy doping strategies for such applications. Significant advancements have been achieved in the exploration of high-entropy doping strategies for electrode materials. Han et al. successfully incorporated five transition metal species (Mn, Fe, Co, Ni, and Cu) into the B-site of $LaMnO_{3-\delta}$ (LMO) perovskite materials, resulting in the development of $LaMn_{0.2}Fe_{0.2}Co_{0.2}Ni_{0.2}Cu_{0.2}O_{3-\delta}$ (HE-LMO). This new material demonstrated significantly improved electrochemical performance compared to traditional LMO cathodes. Solid oxide fuel cells (SOFCs) equipped with HE-LMO as the cathode were able to achieve a power density of 550.89 mW/cm^2 at 800 °C, which is almost 1.5 times higher than that of cells using LMO cathodes. Furthermore, the electrode's Rp decreased from 0.532 to 0.212 Ω cm^2 at 800 °C, and from 4.051 to 1.538 Ω cm^2 at 650 °C (Fig. 18a), indicating the enhanced performance of HE-LMO cathodes at different operating temperatures [275].

The high-entropy strategy also effectively inhibits the segregation of Sr elements. Yang et al. incorporated five components into the A-site of LSM perovskite, resulting in the development of $La_{0.2}Pr_{0.2}Nd_{0.2}Sm_{0.2}Sr_{0.2}MnO_{3-\delta}$ (HE-LSM) cathode material. The strong dispersion of A-site cations created a chaotic stress field around Sr, thereby restricting its transport rate (refer to Fig. 18b). Extended stability assessments demonstrated that HE-LSM exhibited remarkable stability under operational conditions, showcasing its potential as a cathode material for mitigating cation segregation in SOFCs. Furthermore, $La_{0.2}Pr_{0.2}Nd_{0.2}Sm_{0.2}Ba_{0.1}Sr_{0.1}Co_{0.1}Fe_{0.6}Ni_{0.1}Cu_{0.1}O_{3-\delta}$ (HEP) was introduced as a promising cathode option for intermediate-temperature SOFCs. HEP adopted a single-phase cubic perovskite structure and seamlessly aligned with the neighboring $Gd_{0.2}Ce_{0.8}O_{2-\delta}$ (GDC) electrolyte. It displayed a high conductivity of 635.15 S/cm at 800 °C, with a Rp of merely 0.3 Ω cm^2, in contrast to the 0.6 Ω cm^2 recorded for the conventional $La_{0.8}Sr_{0.2}FeO_{2-\delta}$ (LSF) cathode [276].

Zhao et al. utilized A_2BO_4-type high-entropy oxides in the development of SOFC cathodes, leading to the synthesis of $(La_{0.2}Pr_{0.2}Nd_{0.2}Sm_{0.2}Gd_{0.2})_2CuO_4$ high-entropy oxide (HE-LCO). This HE-LCO material exhibited a tetragonal structure with a thermal expansion coefficient of 12.19×10^{-6} K^{-1} between 100 and 800 °C, making it a good match with the GDC electrolyte. The HE-LCO material demonstrated an area-specific resistance of 0.52 Ω cm^2 and a peak power density of 528 mW cm^{-2} at 700 °C in an anode-supported fuel cell (Fig. 18c) [277]. Elsewhere, Xu et al. were the first to explore the use of a high-entropy ceramic oxide as a cathode material for proton-conducting SOFCs. They successfully synthesized $Fe_{0.6}Mn_{0.6}Co_{0.6}Ni_{0.6}Cr_{0.6}O_4$ (FMCNC) high-entropy spinel oxide. *In situ* chemical stability tests demonstrated that FMCNC exhibited excellent stability against CO_2. The H–SOFC with the FMCNC cathode achieved a peak power density (PPD) of 1052 mW cm^{-2} at 700 °C. The presence of high-entropy material enhanced protonation capability and shifted the O p-band center closer to the Fermi level, giving the cathode better catalytic activity and significantly reducing the Rp to 0.057 Ω cm^2 at 700 °C (Fig. 18d) [278].

4.6.2. Machine learning accelerates materials development

Machine learning (ML) is the utilization of computational algorithms and statistical models to analyze vast datasets, identify meaningful patterns, and make projections or improvements based on the knowledge thereby acquired. This process aids in directing experimental work towards the most promising areas. Recent years have seen notable progress in the development of high-performance SOFC/SOEC materials, including enhancements in ion/proton conductivity, catalytic activity, and durability. However, these advancements have predominantly been achieved through traditional material development approaches that rely on empirical experimentation and trial-and-error methods, which are slow, costly, and often limited in scope. Consequently, the integration of machine learning in materials science has attracted considerable interest, offering innovative solutions to expedite the discovery, design, and enhancement of materials with specific desired properties.

ML has made significant progress in predicting material ion/proton conductivity, hydration enthalpy, and entropy. In a study by Yamazaki et al., gradient boosting regression (GBR) was used to predict hydration quantities, equilibrium constants of hydration (K-hyd), hydration entropy (S-hyd), and hydration enthalpy (H-hyd). The researchers focused on mappings with higher confidence within the interpolation domain of these descriptors to ensure that the predictions for $SrSnO_3$ aligned with the measured data. They also developed data-driven structure–property maps that were transformed into a space of top-ranking descriptors (Fig. 19a), facilitating the selection of candidate materials for practical development [279,280]. Additionally, Wang et al. built five ML models—support vector machine (SVM) with a radial basis function kernel, k-nearest neighbors (KNN), Gaussian process regression (GPR) with a radial basis function kernel, random forest (RF), and gradient boosting machine (GBM)—to predict the hydration quantities of P–SOFC cathode oxides [281]. The study identified a high-precision and interpretable model using the RF algorithm to predict the hydration quantities of 3200 perovskite-structured oxides at temperatures ranging from 500 to 700 °C. In Fig. 19b, $(La_{0.7}Ca_{0.3})(Co_{0.8}Ni_{0.2})O_3$ was singled out, and its hydration quantity was further analyzed using the RF model to assess variations with temperature and water pressure.

ML has shown promise for predicting and assisting in the screening of catalytic activity and durability for SOFC/SOEC electrode materials. The catalytic activity of SOFC cathodes can be quantitatively described by measuring the Rp, which is inversely proportional to catalytic activity. Lower Rp corresponds to higher catalytic activity and more efficient oxygen reduction kinetics. Ni et al. utilized a ML-driven approach to expedite the discovery of efficient oxygen reduction electrodes for solid oxide fuel cells. The model successfully predicted the Rp of 6871 oxides, with the best fit among eight different regression models. The importance scores of each descriptor revealed that the A-site/B-site ion Lewis acid (AISA/BISA) had the most significant impact on model predictions (Fig. 19c), suggesting a strong correlation with intrinsic ORR activity. While some deviations were observed, lower AISA and higher BISA values were found to lead to better ORR activity within a certain range [282]. Moon et al. overcame the challenge posed by the scarcity of experimental data for training models, which had previously hindered the application of ML for studying the most promising perovskite-based catalysts. They developed a high-precision model using a small dataset, supplemented with information-rich structural characterization data and coupled with closed-loop experiments (Fig. 20). The screened oxides demonstrated exceptional OER overpotentials compared to the known four metal perovskite oxides [283].

4.6.3. Nano-fabrication technology

i. Nanostructured catalysts

Electrode reactions in solid oxide fuel cells involve complex multiphase processes, including gas diffusion, adsorption, and charge transfer. Nanocatalysts, with their high specific surface area, provide more active sites. In the ORR at the SOFC cathode, nanocatalysts increase the number of sites for oxygen molecule adsorption and activation, enhancing the reaction rate. When nanoparticle size is reduced to 1–10 nm, the particle's electronic structure changes, with more evident electron energy level splitting and quantization. This makes electrons in nanocatalysts more easily excited and transferred, facilitating fuel

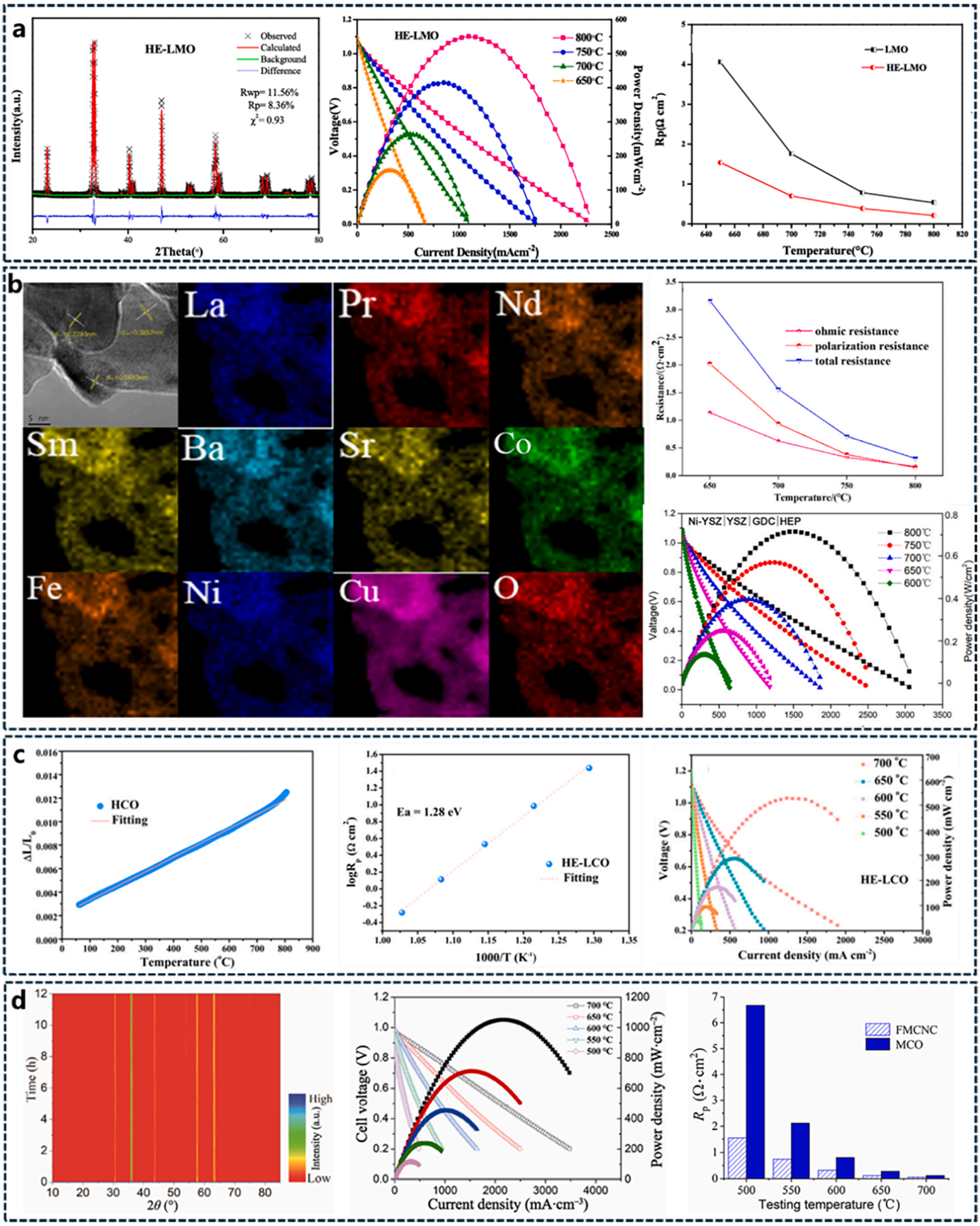

Fig. 18. I–V, I–P, and the corresponding Rp of a single cell with (a) HE-LMO. Reproduced with permission [275]. Copyright 2021, Elsevier; and (b) HEP. Reproduced with permission [276]. Copyright 2021, Elsevier. (c) Thermal expansion curves, Arrhenius plot of ASRs, and I–V, I–P curves for HE-LCO cathodes. Reproduced with permission [277]. Copyright 2022, Elsevier. (d) I–V, I–P curves for FMCNC cell, and a comparison of the *R*p values of cells using FMCNC and MCO cathodes at different temperatures. Reproduced with permission [278]. Copyright 2022, Springer.

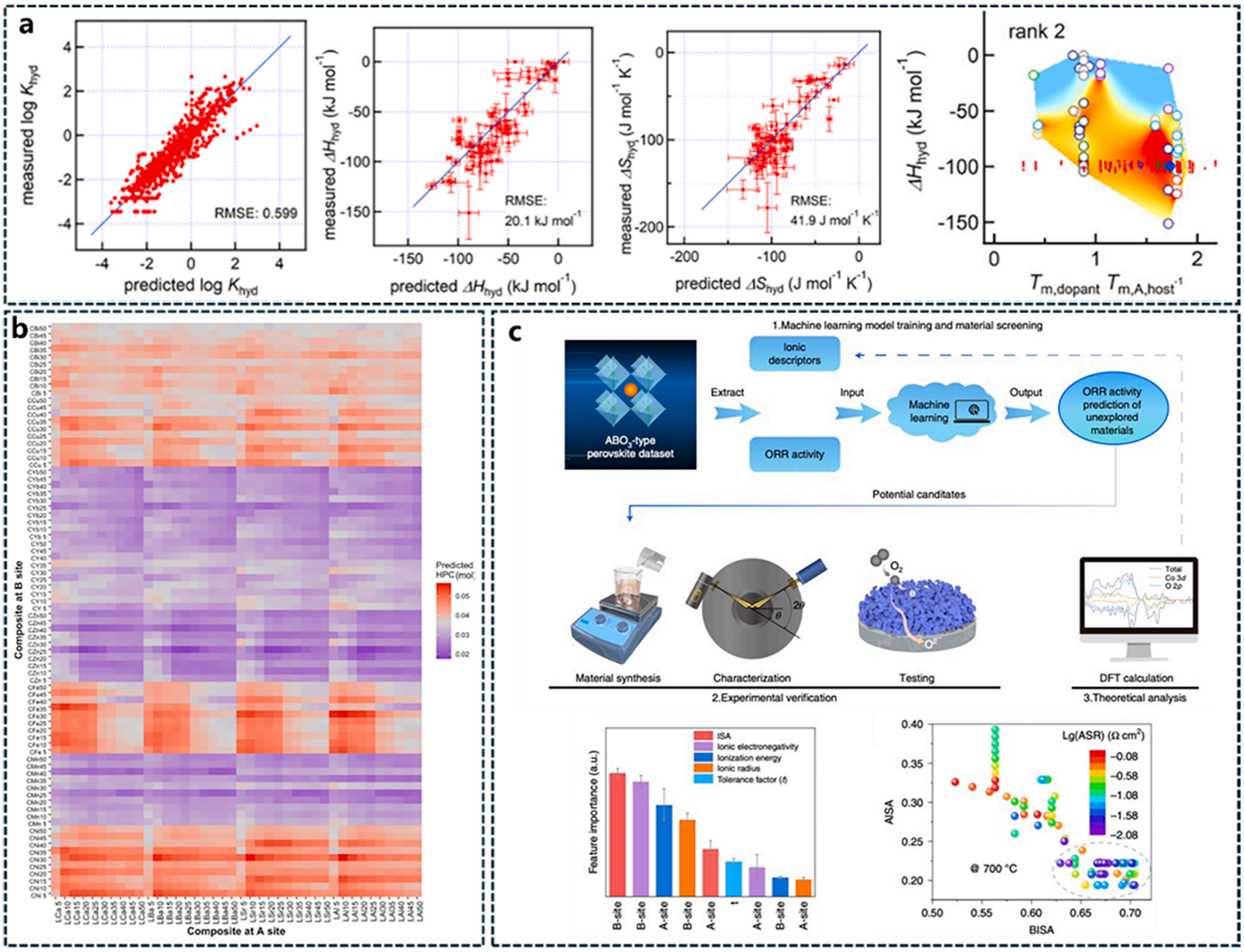

Fig. 19. (a) Structure–property maps of hydration in descriptor spaces for selecting candidate materials from 8, 613 hypothetical compounds. Reproduced with permission [280]. Copyright 2021, American Chemical Society. (b) Reproduced with permission [281]. Copyright 2022, Springer. (c) Machine-learning model training and material screening, experimental verification, and DFT analysis. Reproduced with permission [282]. Copyright 2022, Springer.

molecule adsorption and dissociation, and promoting the reaction. Nanocatalysts can more effectively promote complete hydrocarbon oxidation rather than the incomplete cracking that produces carbon deposits. This broadens the fuel adaptability of SOFCs.

Namgung et al. used a glycine-assisted CTAB infiltration technique to prepare uniformly distributed SSC ($Sm_{0.5}Sr_{0.5}CoO_{3-\delta}$) nanocatalysts on an LSCF6428 cathode substrate. The infiltrated cell had a lower polarization resistance (0.24 Ω cm^2 at 700 °C). DRT analysis showed that infiltrating SSC nanoparticles significantly accelerated the oxygen reduction kinetics, indicating effective activation polarization reduction. The LSCF6428 cathode with SSC nanoparticles achieved a power density of 1.57 W/cm^2 at 700 °C, a significant improvement over the non-infiltrated sample (Figs. 21a, b) [284]. Jeon et al. infiltrated Ag nanocatalysts into an LSCF cathode and deposited a ZrO_2 layer via ALD. The Ag nanocatalysts were uniformly dispersed on LSCF with a particle size of 8.2 ± 1.5 nm. The ZrO_2 layer, about 5.1 ± 0.3 nm thick, covered the Ag nanocatalysts and LSCF surface, preventing Ag nanoparticle aggregation and Sr-ion surface enrichment. The LSCF/Ag/ZrO_2 cathode had an electrode resistance of 0.085 Ω cm^2 at 650 °C (Figs. 21c, d) [285]. Zhang et al. introduced *in situ* precipitated CoRu bimetallic nanoparticles/PBMCRO5 catalyst layers on the anode. The Ru and Co sites activate H_2O and CH_4, respectively, lowering the methane steam-reforming energy barrier. This promotes methane-to-H_2/CO conversion and curbs carbon-deposit formation. At 800 °C, methane conversion reached 21.7%. At 700 °C, the cell with the PBMCRO5 catalyst-coated anode had a peak power density of 617 mW/cm^2 on H_2 and 522 mW/cm^2 on CH_4. Long-term testing showed it stably ran for over 150 h at 700 °C and 0.65 V, with just 1.4% power-density decay (Figs. 21e, f) [286]. Chen et al. used ALD to deposit dual nanocatalysts (Pt and CoO_x) on a LSM/YSZ cathode. The Pt layer was about 3 nm thick and the CoO_x layer about 2 nm. Synergy between the Pt and $(MnCo)O_x$ boosted the ORR activity. CoO_x stabilized the Pt nanoparticles and worked with Pt to promote oxygen reduction, achieving a peak power density of 1.245 W/cm^2 (Figs. 21g, h) [287].

ii. Novel synthesis methods

Traditional material synthesis methods (e.g., sol-gel, co-precipitation, hydrothermal) increasingly fail to meet SOFC requirements for electrochemical activity, stability, and mechanical properties, so novel methods like atomic layer deposition (ALD), pulsed laser deposition (PLD), and electrospinning are emerging. These methods allow precise control over material size and morphology, enable accurate doping and compositing, and facilitate the preparation of nanoscale electrodes and electrolytes with exact regulation. Choi et al. first deposited a nanoscale $La_{0.6}Sr_{0.4}CoO_{3-\delta}$ (LSC) film on a LSCF cathode surface via ALD. The oxygen p-band center of the ALD-LSC film is closer to the Fermi level,

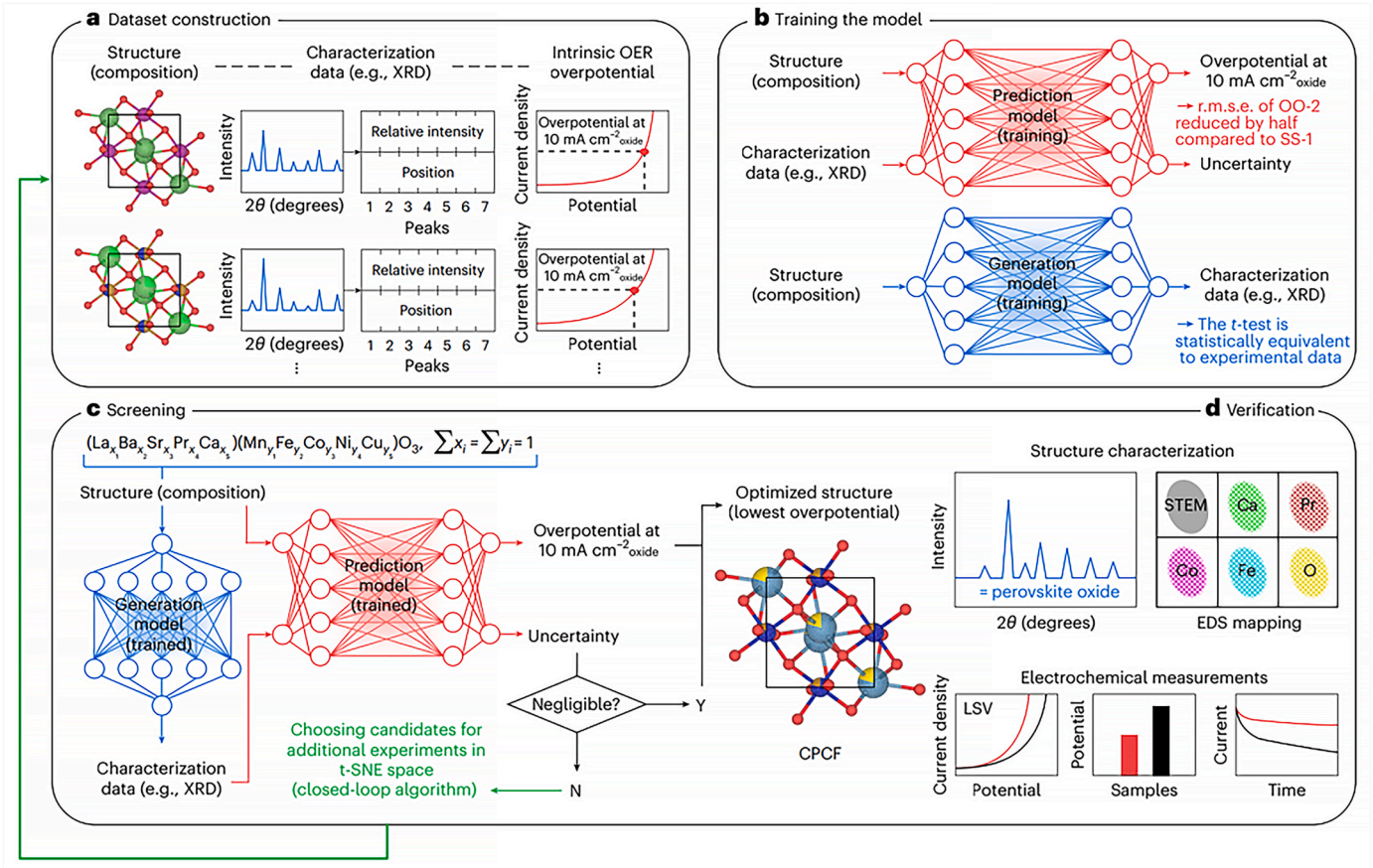

Fig. 20. Overview of the algorithms used and the workflow of the closed-loop protocol. (a) The initial data set of 30 ABO_3 four-metal perovskite oxides provides compositional and structural data in graph format, characterization data, and OER performance data. The red (prediction) model and the blue (generation) model are individually trained (b) and combined (c) for the screening process. Additional experiments are carried out if the prediction uncertainty is not negligible. Reproduced with permission [283]. Copyright 2023, Springer.

lowering the oxygen removal energy and boosting the ORR activity. Consequently, the ALD-LSC cell's maximum power density at 600 °C was 1.8 times higher than that of an untreated LSCF cathode (Fig. 22a) [288]. Li and colleagues applied PEALD to deposit an ultrathin Ru film on a LSCF oxygen electrode. This enhanced the electrode's performance and durability. At 700 °C, it achieved a peak power density of 803 mW/cm^2 in fuel cell mode and a current density of 656 mA/cm^2 in electrolysis mode. The deposited Ru reacted with surface Sr species to form $SrRuO_3$, suppressing Sr segregation and maintaining good OER performance (Fig. 22b) [289]. Develos-Bagarinao et al. used PLD to combine a nanoporous $La_{0.6}Sr_{0.4}CoO_{3-\delta}$ (LSC) cathode with an ordered, self-assembled LSCF and GDC nanocomposite cathode layer. This increased the cathode–electrolyte interfacial area density and lowered the polarization resistance. At 650 and 700 °C, the optimized cathode achieved high current densities of about 2.2 and 4.7 A/cm^2, equivalent to power densities of about 1.5 and 3.3 W/cm^2 (Fig. 22c) [290]. Kong et al. used electrospinning to make $CuCo_2O_4$−ESB composite fibers with a 1D structure. The composite-fiber cathode had a polarization resistance of 0.021 Ω cm^2 at 800 °C, 3.3 times lower than that of $CuCo_2O_4$−ESB nanoparticle-based composite cathodes. In the fiber structure, the $CuCo_2O_4$ and ESB nanoparticles were evenly distributed. This extended the TPB length and increased the ORR active sites (Fig. 22d) [291].

5. Systems integration of SOFCs/SOECs

SOFC power systems are comprised of two integral components: the cell stack and the balance of plant (BOP) support systems. The cell stack, an assembly of numerous cell units connected in series, generates a defined level of power. The BOP, which includes gas supply systems, exhaust gas recovery systems, control systems, and power conversion systems, collectively ensures a stable and optimal operating environment for the cell stack. The gas supply system regulates the flow of fuel and air and pre-processes the fuel to guarantee a clean and consistent supply. The exhaust gas recovery system is tasked with the collection of unreacted fuel and the preheating of incoming cold air and fuel, thereby improving the system's thermal efficiency. The control system provides oversight of the entire system to ensure secure operation and swift load response, while also optimizing temperature management. The power conversion system transforms the direct current from the cell stack into a form suitable for general usage and manages the energy. The overarching design objective of the SOFC system is to realize high efficiency, stability, and longevity in operation (Fig. 23a) [292–295]. The integration of SOEC systems involves a stack composed of multiple SOEC units and an array of BOP components. The BOP primarily consists of pumps, heat exchangers, and steam generators that operate in unison to enhance system efficiency. Water is initially preheated in the heat exchanger, subsequently transformed into steam in the steam generator, and further superheated via an electric heater to satisfy the requirements for the electrolysis reaction. To reduce electricity consumption and augment efficiency, the effluent hydrogen and oxygen gases are utilized to preheat the steam. The availability of external steam allows for further system optimization. The SOEC system is engineered to facilitate efficient and sustainable hydrogen production while minimizing energy expenditure and environmental impact (Fig. 23b) [296–298]. In recent years, companies including Bloom Energy in the United States and Mitsubishi in Japan have developed various SOFC and SOEC systems. For instance, the AlwaysON microgrid SOFC system designed to withstand power grid outages and extreme weather disturbances, the 250 kW

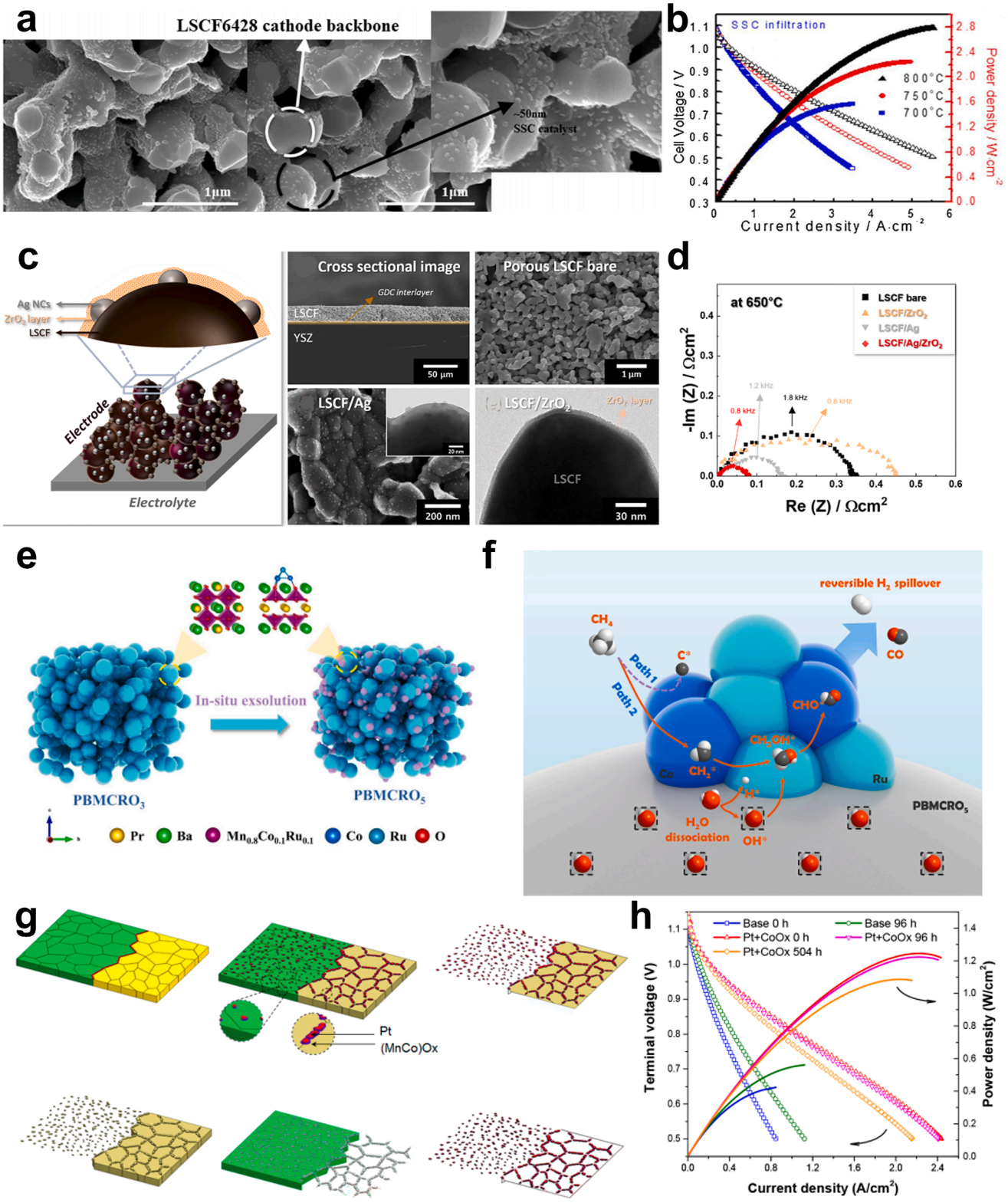

Fig. 21. (a) FESEM micrograph showing SSC nanoparticles infiltrating a porous LSCF6428 support, at different magnifications of the optimized sample. (b) I–V curves of full cells after infiltration of SSC nanoparticles via the glycine method at 700, 750, and 800 °C. Reproduced with permission [284]. Copyright 2020, Elsevier. (c) The overall morphology of fabricated LSCF electrode: schematic illustration of fabricated LSCF-based electrode; cross-sectional SEM images of screen-printed LSCF; top-view SEM images of porous bare LSCF; SEM and inset TEM image of LSCF/Ag; and TEM image of LSCF/ZrO_2. (d) Resistance evaluation of surface-treated LSCF symmetrical cells. Reproduced with permission [285]. Copyright 2023, Elsevier. (e) Schematic illustration of structural evolution of exsolved PBMCRO5 electrocatalyst. (f) Schematic diagram of the proposed reaction pathway of MSR on the CoRu/PBMCRO5 catalyst, where * represents the adsorbed species. Reproduced with permission [286]. Copyright 2024, Elsevier. (g) Schematic of TPB interface of pristine LSM/YSZ with Pt/Mn–Co–O distribution changes (h) Terminal voltage as a function of current density for the cells with LSM at 750 °C. Reproduced with permission [287]. Copyright 2019, American Chemical Society.

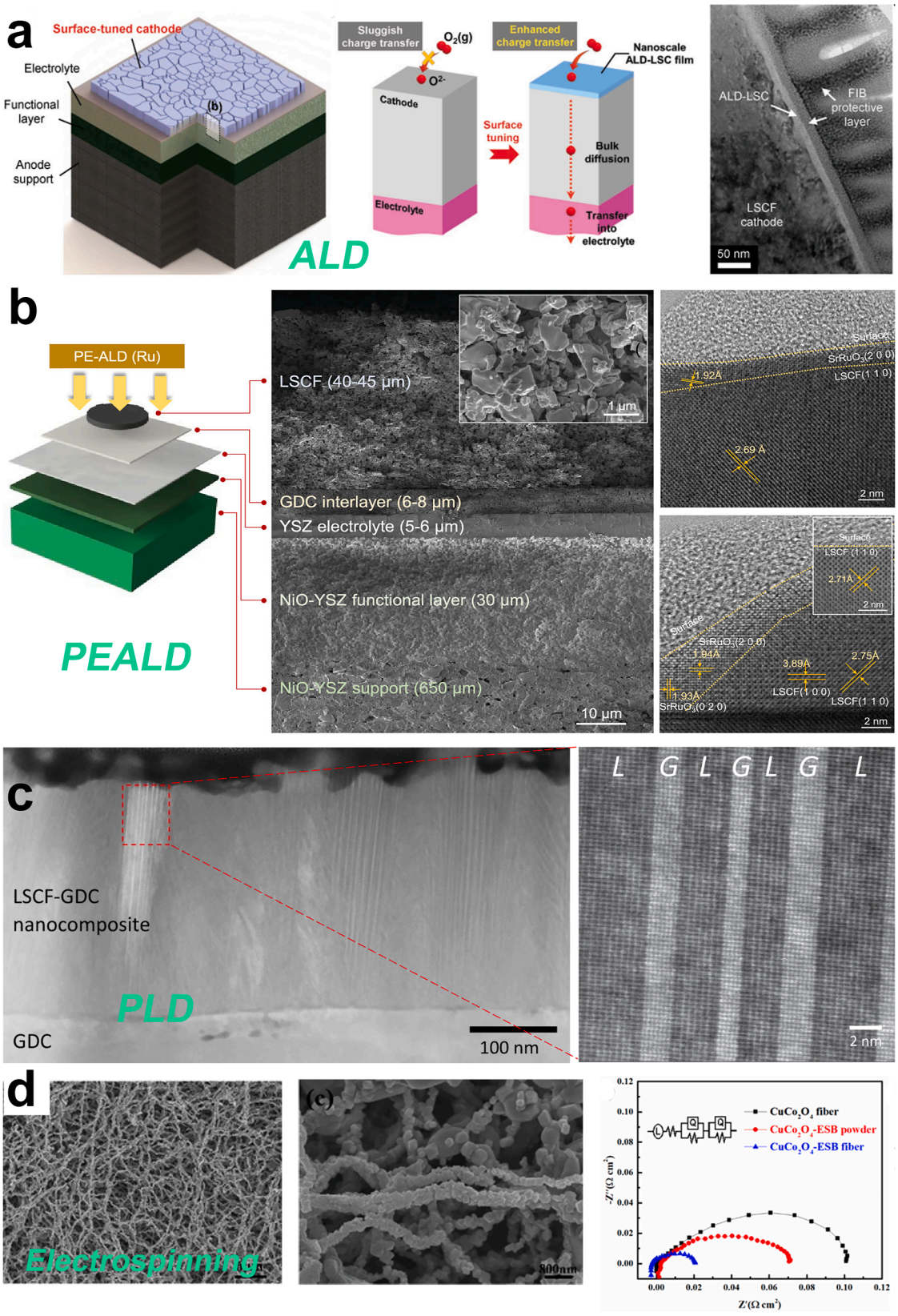

Fig. 22. Strategies: (a) ALD to prepare LSC cathode films. Reproduced with permission [288]. Copyright 2018, Wiley VCH GmbH. (b) PEALD to prepare Ru films on LSCF. Reproduced with permission [289]. Copyright 2023, American Chemical Society. (c) ALD to prepare nanocomposite cathode layers of LSCF and GDC. Reproduced with permission [290]. Copyright 2021, Springer. (d) Electrospinning to prepare a composite-fiber cathode of $CuCo_2O_4$-ESB. Reproduced with permission [291]. Copyright 2020, American Chemical Society.

SOFC-GT system operational in Japan since 2017, and the 1 MW SOFC system used in shore power applications since 2018 have all highlighted the immense potential of SOFC/SOEC in the energy sector (Fig. 23c) [299–301].

In the process of integrating SOFC and SOEC systems, several critical factors can significantly influence the performance of the integrated system:

i. The high-temperature operating conditions can induce thermal expansion and stress in materials, potentially leading to damage or performance degradation for cell components. ii. The uniform distribution of fuel and oxidant is essential for optimal cell performance. Research in fluid dynamics is necessary to optimize gas flow paths, minimize resistance to flow, and ensure that reactive gases effectively reach the electrode surfaces. iii. Discrepancies in the TECs of different cell components during temperature fluctuations can create mechanical stresses at interfaces, possibly resulting in material cracking or delamination. iv. System integration necessitates the harmonized functioning of various components to achieve the efficient conversion of electrical energy and storage of chemical energy. This encompasses cell stack assembly, electrical interconnections, gas sealing, and thermal management. v. The effective utilization of the high-quality waste heat generated by the system is crucial.

Thermal management is a critical aspect that significantly influences the performance and longevity of high-temperature SOFC/SOEC systems. Operating at elevated temperatures can result in thermal stress, material degradation, and uneven temperature distribution, all of which can hinder system performance. For example, the heat generated by SOFC systems through electrochemical reactions and ohmic losses must be effectively dissipated to avoid excessive temperature gradients. Failure to do so can lead to thermal stress and material degradation over time [302]. To tackle these challenges, researchers have put forward various thermal management strategies. One approach involves optimizing the design of gas channels and fluid flow to enhance heat dissipation. By fine-tuning the channel design, Zeng et al. were able to reduce the axial temperature gradient from 24 to 14 K/10 mm (Fig. 24a) [303]. Cascade control methods have also proven effective in managing thermal issues. By implementing a secondary loop, cascade control can swiftly suppress major disturbances, thereby improving system response time and stability. In SOFC thermal management systems, incorporating disturbances from the reformer into the secondary loop can effectively mitigate disruptions caused by fluctuations in air flow. Simulation results show that cascade control can reduce regulation times for stack temperature and inlet air temperature to 200 s and 0 s, respectively, with no overshoot, surpassing single-loop control methods (Fig. 24b) [304]. Furthermore, temperature management systems utilizing LabVIEW have demonstrated effective thermal control. Equipped with high-precision temperature sensors and rapid feedback mechanisms, these systems can swiftly raise the SOFC stack temperature to the desired level and keep it within the optimal range of 750–800 °C (Fig. 24c) [305]. This not only reduces preheating time but also boosts system efficiency and reliability.

Fluid dynamics plays a crucial role in SOFC/SOEC systems, as it directly impacts the transport efficiency of fuel and oxidant, thus influencing the overall performance of the system. In SOEC systems, precise control of gas flow and distribution is necessary to ensure uniform reactant concentrations on the electrode surfaces. In SOFCs, optimizing the flow of fuel and air is essential to avoid local fuel depletion or oxygen deficiency. The impact of various channel structures, including rectangular, triangular, and trapezoidal, on heat and mass transfer has been examined. Results indicated that for the trapezoidal channel design, the highest heat transfer efficiency was achieved when the thickness of the porous region equaled 90% of the channel height (Fig. 24d) [306]. Additionally, when compared to the conventional rectangular channel, the trapezoidal structure displayed superior heat transfer capabilities (Fig. 24e) [307]. The triangular channel configuration proved effective in directing fluid flow and reducing fluid residence time within the channel, consequently enhancing heat transfer efficiency (Fig. 24f) [308]. Computational fluid dynamics (CFD) simulations can be used to optimize fluid flow paths and reduce pressure losses.

6. Summary and outlook

Significant strides have been made in recent years within the realms of materials science, mechanistic understanding, and device engineering pertinent to SOFCs and SOECs, augmenting our comprehension of these technologies and broadening the spectrum of their practical applications. Nonetheless, to surmount the intricate challenges inherent to SOFCs and SOECs, more profound and systematic investigations through theoretical and empirical research are imperative. The following areas are replete with opportunities and significance, necessitating further exploration and resolution.

i. Nanofabrication and homogenization of ceramic materials for SOFCs/SOECs

The nanofabrication and homogenization of ceramic materials play a pivotal role in enhancing the performance of both materials and SOFC/SOEC cells. Nanotechnology has the potential to augment the specific surface area of materials, amplify the quantity of reactive sites, and consequently boost the efficacy of electrochemical reactions. Furthermore, electrolytes crafted from uniformly dispersed nanoscale powders manifest denser microstructures and more uniform grain boundary conditions, which are particularly beneficial for ionic conduction, notably for protons. However, the nanofabrication and homogenization of ceramic materials also engender a plethora of challenges, including the sintering and agglomeration of nanoparticles at elevated temperatures, and pronounced variations in size distribution and compositional homogeneity among particles, all of which can precipitate a deterioration in the electrochemical and ionic conduction properties of the materials. Consequently, it is imperative to delve into more sophisticated nanomaterial synthesis techniques to actualize the uniform nanoscale homogenization of ceramic materials. Investigating low-temperature rapid sintering methodologies is crucial to mitigate the sintering and agglomeration of nanoparticles at high temperatures and to preserve the heightened reactivity of the nanostructure.

ii. Electrochemical mechanisms of SOFCs and SOECs, with a focus on transport mechanisms in electrolytes under protonic conduction

Despite substantial research on the electrochemical mechanisms of SOFCs and SOECs, the transport mechanisms in electrolytes under protonic conduction are not fully understood. Protonic conduction cells, such as H–SOFCs and H-SOECs, demonstrate enhanced conductivity and performance at lower temperatures due to their distinct transport mechanisms. However, the transport pathways and interactions of protons with electrolyte materials, as the smallest elemental ions, cannot be entirely elucidated by the mechanisms applicable to oxygen ion conduction. Additionally, the experimental observation and characterization of protons present significant challenges, warranting the utilization of theoretical simulation methods to investigate the transport pathways and energetic states of protons within electrolyte materials. The development and application of advanced *in situ* characterization techniques are essential for achieving real-time observations of proton transport in electrolytes.

iii. Exploration of co-electrolysis of H_2O and CO_2 and synthesis gas production under the SOEC mode

SOECs have shown significant potential in the co-electrolysis of H_2O and CO_2 to produce synthesis gas, yet challenges remain regarding the reaction mechanisms, control of product distribution, and stability of electrode materials during this process. In-depth research is required to understand the co-electrolysis reaction mechanisms of H_2O and CO_2 at the electrode and to reveal the patterns of product distribution under various operating conditions. Research is needed to develop electrode materials with high catalytic activity, stability, and selectivity, tailored to the characteristics of co-electrolysis reactions.

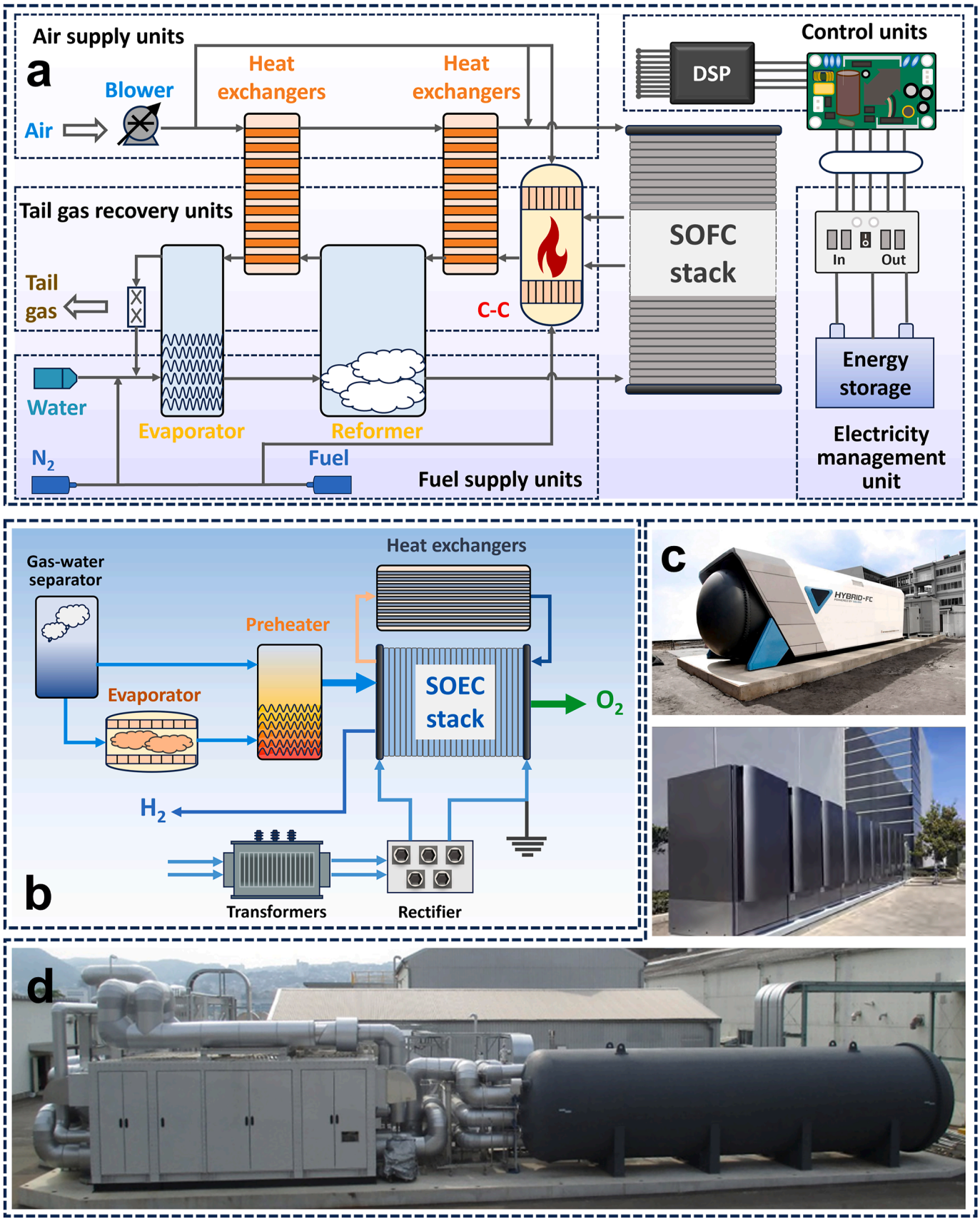

Fig. 23. (a, b) Topology of the SOFC and SOEC systems. (c) Photos of various SOFC/SOEC systems. Reproduced with permission [299]. Microgrid Solution Overview. Reproduced with permission [301]. Copyright 2018, Mitsubishi Heavy Industries Technical Review.

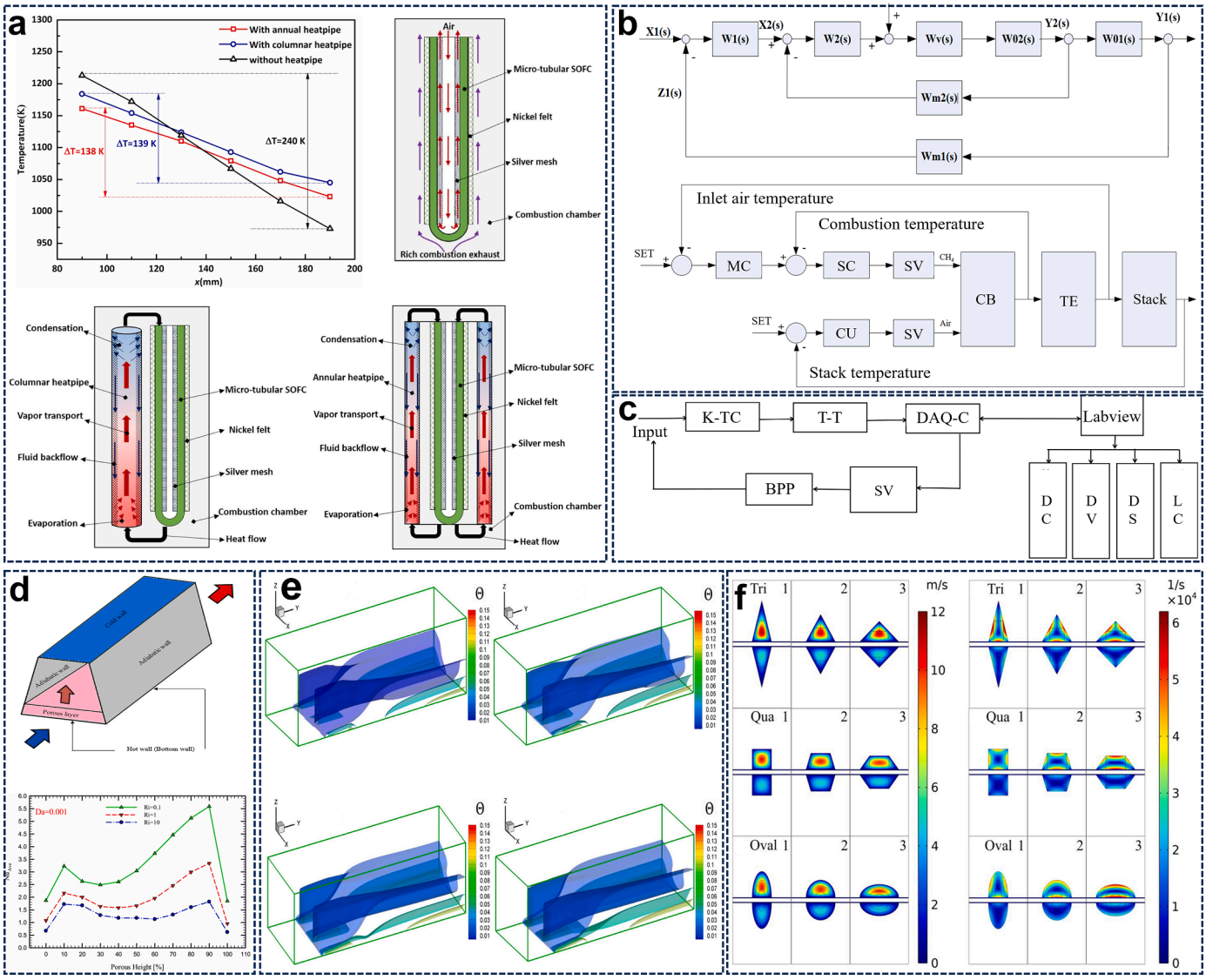

Fig. 24. (a) Schematic diagram of different flame-assisted fuel cells with heat pipe. Reproduced with permission [303]. Copyright 2017, IOPscience. (b) SOFC thermal management system cascade control structure. Reproduced with permission [304]. Copyright 2011, Hanspub. (c) Flow chart of temperature management with LabVIEW. Reproduced with permission [305]. Copyright 2011, Hanspub. (d) Trapezoidal flow channel thickness and heat transfer efficiency. Reproduced with permission [306]. Copyright 2021, Elsevier. (e) Rectangular flow channel thermal field simulation. Reproduced with permission [307]. Copyright 2025, Elsevier. (f) Velocity and vorticity magnitude distribution at cross-section. Reproduced with permission [308]. Copyright 2024, Elsevier.

iv. Establishment of standardized testing conditions and performance evaluation criteria

There is currently an absence of uniform standards in the performance testing conditions and evaluation criteria for SOFCs and SOECs, which hinders the direct comparison of results across different studies. It is imperative to collect and organize performance testing data from various research teams and establish a public performance database to facilitate data sharing and comparative analysis among researchers. Collaborating with international standardization organizations to formulate unified performance testing procedures and evaluation standards for SOFCs and SOECs is crucial for advancing the field.

v. Transition from laboratory research to practical applications

The prospective applications of SOFCs and SOECs are exceedingly broad. Thus far, the majority of research on SOFCs and SOECs has been confined to the laboratory development phase, with the transition to practical applications encountering numerous challenges, including cost, durability, and scalability. The high efficiency, cleanliness, and stability of SOFCs/SOECs make them promising candidates for a wide array of uses, ranging from distributed power generation to propulsion systems in heavy-duty transport vehicles, as well as in maritime, aerospace, and space applications.

vi. Integration and optimization of SOFC and SOEC systems

The integration and optimization of SOFC and SOEC systems are pivotal for achieving high efficiency, stability, and longevity in operation. However, numerous challenges persist for system integration, such as thermal management, fluid dynamics, mechanical stress control, and compatibility among the various components. Furthermore, the optimization of system design often necessitates trade-offs among several performance metrics, including power density, thermal efficiency, and cost. It is imperative to develop an efficient thermal management system to ensure stable operation of the fuel cell stack while effectively harnessing waste heat to enhance the overall thermal efficiency of the system. By integrating numerical simulation with experimental validation, the flow paths and distribution of gases and steam within the system can be optimized to improve reaction efficiency and product separation. Investigating the thermal expansion behavior of materials at high temperatures and designing appropriate cell structures and interconnect materials can mitigate the impact of thermal stress on system

stability. Dividing the system into multiple independent modules facilitates maintenance, upgrades, and scalability while reducing system complexity and cost. Leveraging the Internet of Things, big data, and artificial intelligence technologies can lead to intelligent monitoring, predictive maintenance, and remote system operation, thereby enhancing system reliability and operational efficiency.

CRediT authorship contribution statement

Peng Feng: Writing – review & editing, Writing – original draft, Investigation, Funding acquisition, Formal analysis, Data curation. **Kuan Yang:** Writing – original draft, Formal analysis, Data curation. **Xuanyou Liu:** Writing – original draft, Formal analysis. **Jiujun Zhang:** Writing – review & editing, Supervision, Formal analysis. **Zhi-Peng Li:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

There are no conflicts to declare.

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